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From Morse Theory on Finite Dimensions to Floer Theory for Minimal Surfaces

by

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Abstract

In this thesis, we outline the methodology for determining the CW-decomposition of Sobolev spaces through the Morse theory of the energy functional. We examine the resulting information about the harmonic maps, including stability and indices as determined by the Jacobi operator. Additionally, we address obstacles to CW decomposition and review recent research aimed at overcoming challenges with these spaces. As the main challenge is that the CW-decomposition cannot be determined through Morse theory, further analysis of the Jacobi operator may be pursued by adopting a Floer-theoretic perspective on harmonic maps from the torus to Kähler manifolds. This approach follows the theory of Oliver Fabert and Ronen Brilleslijper on polysymplectic geometry and the CRPS action functional, which generalizes symplectic geometry. With this, we aim to formulate a relation between the Morse index and the Conley-Zehnder index for minimal surfaces, as was done for geodesics through the symplectic action functional.

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1. Introduction

Morse theory is a mathematical discipline connecting geometry, topology, and dynamics. When it is first introduced, it is usually in the case of functions from a finite-dimensional manifold to $\mathbb{R}$. These functions can provide a helpful description of a manifold through its critical points. Indeed, it is possible to determine the homotopy type of a manifold and how it transforms with respect to the critical points.

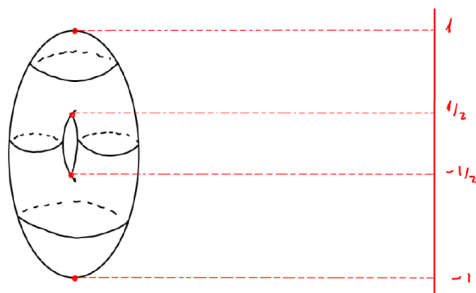


Figure 1.1.: Height function for the torus.

A straightforward example is the height function on the torus $h : \mathbb{T}^2 \rightarrow \mathbb{R}$ as in Figure 1.1. This function has four critical points: a minimum at the value -1 , two saddle points at the values $\pm\frac{1}{2}$, and a maximum at the value 1 . Via these critical points, we can reconstruct the torus. This reconstruction from the minimum to the maximum, as seen in the figure below, is reminiscent of a 3D printer, and it is called the Handle-Body decomposition. Via the Handle body decomposition, we are able to describe the topology

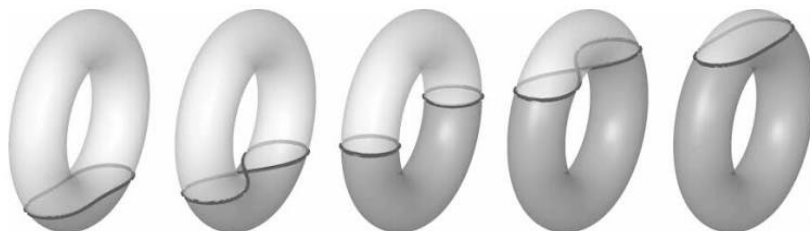


Figure 1.2.: Handle-body decomposition of the torus via the height function.

of the torus. Indeed, it can be noted that the subset $h^{-1}\left(\left(-\infty, -\frac{1}{2}\right)\right)$ is homotopic to a point – as one can deform a disk into a point. What is more interesting is that the subset $h^{-1}\left(\left(-\infty, \frac{1}{2}\right)\right)$ is homotopic to the subset $h^{-1}\left(\left(-\infty, -\frac{1}{2}\right)\right)$ with a 1-cell attached and that the torus is homotopic to the subset $h^{-1}\left(\left(-\infty, 1\right)\right)$ with a 2-cell attached.

Interestingly, at first, Morse theory was not constructed with functions in mind. In fact, Morse theory was originally developed in the 1920's by Marston Morse to study geodesics on a sphere. Geodesics are often described as the shortest paths between two points. In a Euclidean space, geodesics are straight lines. While on a sphere, geodesics are arches; this is why, when on a plane, one does not take the 'straight' course but the 'curved' one, see Figure 1.3 It turns out that there exist infinitely many geodesics between antipodal points of a sphere, as well as other surfaces. Geodesics can be seen



Figure 1.3.: Geodesics as flight routes.

to be elements of the space of all paths of a manifold that are critical points of the length functional on this Path space. As was the case with functions, one can apply the same methodology to the length functional to describe the topology of the path space. However, as the length functional is invariant under reparametrization, it does not allow for unique solutions. To solve this, we can use the energy functional. It turns out that geodesics are also the critical points of the energy functional. With this, it is possible to construct a bridge between topological and geometric objects. For example, Cartan



Figure 1.4.: Some funky closed geodesics on the torus.

showed that every free homotopy class of loops in a compact Riemannian manifold has a geodesic. With this theorem, Preissman showed that every nontrivial abelian subgroup of the fundamental group of a compact Riemannian manifold with negative curvature is infinite cyclic. The latter theorem connects topology with geometry as it excludes metrics of negative curvature on a considerable class of manifolds.

It is, therefore, interesting to study if the same holds for higher-dimensional analogues of geodesics: volume-minimizing submanifolds. A real-world example of an area-

minimizing surface is a soap film: via the surface tension, the liquid is pulled into the smallest possible area. In this case, one would apply Morse theory on the spaces of maps

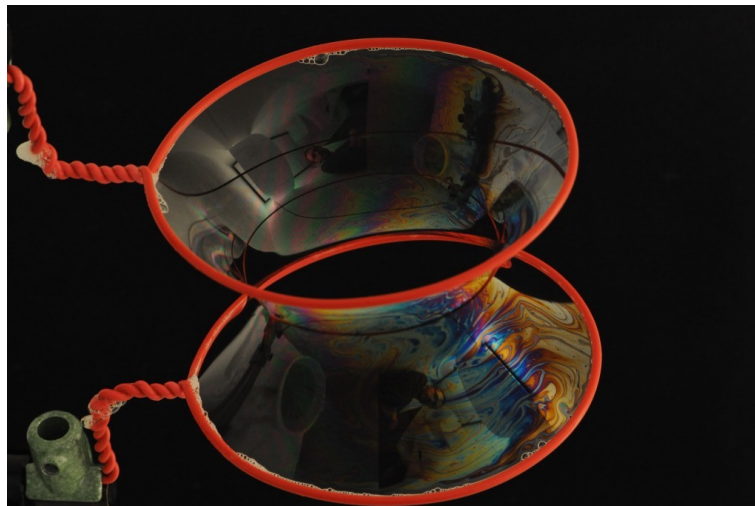


Figure 1.5.: Soap films as minimal surfaces.

between manifolds. As will be seen in this thesis, and as was true in the case of geodesics, it is preferred to use the energy functional. It turns out that there exists a relation between minimal submanifolds – the critical points of the volume functional – and harmonic maps – the critical points of the energy functional. Once again, Morse theory allows us to construct a bridge between topological and geometric objects. Thanks to this, Morse theory has been applied and generalized to different areas of mathematics. Some of them are the Morse-Witten complex and Morse-Smale systems in dynamical systems, and Floer homology in symplectic geometry; the latter of which will play a refreshing role in this thesis.

This thesis is divided into three parts and is organized as follows. The first part, titled ‘The One-Dimensional Case: Geodesics and the Path Space’, considers the classic case of Morse theory through geodesics. It allows us to familiarize ourselves with the methodology and provides a blueprint for the higher-dimensional analogues. The second part, titled ‘Generalizing Geodesics to Higher Dimensions: Minimal Sub-manifolds and Harmonic Maps’, considers the higher-dimensional generalization of geodesics. It explores the analogies of geometric objects and operators found in the first part, as well as discusses the limitations of using Morse theory in higher dimensions. Finally, the third part, titled ‘Moving Forward: Hamiltonian Formalism’, examines a different method specifically for minimal surfaces. This method aims to describe the topology of Sobolev spaces via Floer theory to counteract the failings of Morse theory.

Part I.

The One-Dimensional Case: Geodesics and the Path Space

2. Good Old Morse Theory

This chapter is heavily inspired by *Morse Theory* [22].

Before talking about geodesics and harmonic maps, let us refresh some Morse theory and Morse homology; the latter will be of interest in the last third of this thesis.

2.1. Morse Theory on zero-dimensions

Definition 2.1. Let $f \in C^\infty(M, \mathbb{R})$. A point $p \in M$ is called a *critical point* of f if the induced map $df_p : T_p M \rightarrow T_{f(p)} \mathbb{R}$ is zero.

If $(x^1, \dots, x^n)$ are local coordinates in a neighborhood U of p , then

$$\frac{\partial f}{\partial x^1}(p) = \dots = \frac{\partial f}{\partial x^n}(p) = 0.$$

The real number $f(p)$ is called a *critical value* of f .

If $a \in \mathbb{R}$ is not a critical value of f , by the *Implicit Function Theorem* the set

$$M^a := \{x \in M : f(x) \leq a\} = f^{-1}([-\infty, a]) \quad (2.1)$$

is a smooth manifold with boundary :

$$\partial M^a = f^{-1}(a). \quad (2.2)$$

Definition 2.2. A critical point p is called *non-degenerate* if and only if the Hessian matrix $\frac{\partial^2 f}{\partial x^i \partial x^j}(p)$ is non-singular.

Definition 2.3. If p is a critical point of f , the *Hessian* $H_p(f)$ of f at p is the symmetric bilinear functional, $H_p(f) : T_p M \times T_p M \rightarrow \mathbb{R}$.

Definition 2.4. The *Morse index* of the Hessian $H_p(f)$ is the maximal dimension of a subspace of $T_p M$ on which H_p is negative definite. The *nullity* is the dimension of the *nullspace* $\{v \in T_p M : H_p(f)(v, w) = 0 \ \forall w \in T_p M\}$.

It is then clear that

Proposition 2.5. A critical point p is a non-degenerate critical point of f if and only if $H_p(f)$ on $T_p M$ has nullity equal to 0.

Remark 2.1.1. The index of $H_p(f)$ on $T_p M$ will be referred to as the index of f at p and denoted as $\mu_M(p)$.

Lemma 2.6. *Let f be a smooth function in a convex neighborhood V of 0 in $\mathbb{R}^n$, with $f(0) = 0$. Then*

$$f(x_1, \dots, x_n) = \sum_{i=1}^n x_i g_i(x_1, \dots, x_n)$$

for some suitable smooth functions g_i defined in V with $g_i(0) = \frac{\partial f}{\partial x^i}(0)$.

Lemma 2.7 (Morse Lemma). *Let p be a non-degenerate critical point for f . Then there is a local coordinate system $(y_1, \dots, y_n)$ in a neighborhood U of p with $y^i(p) = 0$ for all i and such that the identity*

$$f = f(p) - (y^1)^2 - \dots - (y^\lambda)^2 + (y^{\lambda+1})^2 + \dots + (y^n)^2$$

holds through U , where λ is the index of f at p .

Corollary 2.7.1. *Non-degenerate critical points are isolated.*

From the above, we say that a smooth function $f : M \rightarrow \mathbb{R}$ is a *Morse function* if all its critical points are non-degenerate.

2.2. CW-structure via Morse functions

Theorem 2.8. *Let f be a smooth real-valued function on a manifold M . Let $a < b$ and suppose that the set $f^{-1}([a, b]) := \{p \in M : a \leq f(p) \leq b\}$ is compact and contains no critical points of f . Then M^a is diffeomorphic to M^b . Furthermore, M^a is a deformation retract of M^b , so that the inclusion map $M^a \hookrightarrow M^b$ is a homotopy equivalence.*

The idea is the ‘push’ M^a down to M^b along the orthogonal projection of the hypersurface $f^{-1}(c)$.

Proof. Choose a Riemannian metric g on M , and let $\langle X, Y \rangle_g$ denote the inner product of two tangent vectors. The gradient $\text{grad } f$ vanishes at the critical points of f . If $c : \mathbb{R} \rightarrow M$ is a curve with velocity vector $\frac{dc}{dt}$, then

$$\left\langle \frac{dc}{dt}, \text{grad } f \right\rangle_g = \frac{d(f \circ c)}{dt} \quad (2.3)$$

Let $\rho : M \rightarrow \mathbb{R}$ be a smooth function such that on $f^{-1}[a, b]$ it is

$$\frac{1}{\langle \text{grad } f, \text{grad } f \rangle_g}$$

and 0 outside a compact neighborhood of $f^{-1}[a, b]$. Then the vector field X , defined by

$$X_q = \rho(q)(\text{grad } f)_q$$

generates a 1-parameter group of diffeomorphism (by [22][Lemma 2.4]) $\varphi_t : M \rightarrow M$. For fixed $q \in M$ consider the function $t \mapsto f(\varphi_t(q))$. If $\varphi_t(q)$ lies in the set $f^{-1}([a, b])$, then

$$\begin{aligned} \frac{d(f(\varphi_t(q)))}{dt} &= \left\langle \frac{d\varphi_t(q)}{dt}, \text{grad } f \right\rangle_q \\ &= \langle X, \text{grad } f \rangle_g = 1 \end{aligned}$$

thus the correspondence $t \mapsto f(\varphi_t(q))$ is linear with derivative 1 as long as $a \leq f \circ \varphi_t(q) \leq b$.

Now consider the diffeomorphism $\varphi_{b-a} : M \rightarrow M$, this carries M^a diffeomorphically onto M^b . This proves the first half of the theorem.

Define a 1-parameter family of maps $r_t : M^b \rightarrow M^b$ be

$$r_t(q) = \begin{cases} q & f(q) \leq a \\ \varphi_{t(a-f(q))}(q) & a \leq f(q) \leq b. \end{cases}$$

Then r_0 is the identity and r_1 is a retraction from M^b to M^a . Hence M^a is a deformation retract of M^b $\square$

Theorem 2.9. *Let $f : M \rightarrow \mathbb{R}$ a smooth function and p a non-degenerate critical point with index λ . Setting $f(p) = c$, suppose that $f^{-1}([c - \epsilon, c + \epsilon])$ is compact and contains no critical point of f other than p , for some $\epsilon > 0$. Then, for all sufficiently small ϵ , the set $M^{c+\epsilon}$ has the homotopy type of $M^{c-\epsilon}$ with a λ -cell attached.*

The proof is very interesting so we refer to [22][proof of Theorem 3.2].

Corollary 2.9.1. *Let $f : M \rightarrow \mathbb{R}$ be a smooth function. Suppose that there are k non-degenerate critical points $p_1, \dots, p_k$ with indices $\lambda_1, \dots, \lambda_k$ in $f^{-1}(c)$. Suppose that $f^{-1}([c - \epsilon, c + \epsilon])$ is compact and contains no other critical point of f , for some $\epsilon > 0$. Then for all sufficiently small ϵ , the set $M^{c+\epsilon}$ has the homotopy type of $M^{c-\epsilon} \cup e^{\lambda_1} \cup \dots \cup e^{\lambda_k}$.*

Theorem 2.10. *If f is a differentiable function on a manifold M with no degenerate critical points, and if each M^a is compact, then M has the homotopy type of a CW-complex with one cell of dimension λ for each critical point of index λ .*

Proof. Let $c_1 < c_2 < \dots$ be the critical values of $f : M \rightarrow \mathbb{R}$. The sequence $\{c_i\}$ has no cluster points since each M^a is compact. The set M^a is vacuous for $a < c_1$. Suppose $a \neq c_1, c_2, \dots$ and that M^a is of the homotopy type of a CW-complex. Let c be the smallest $c_i > a$. $M^{c+\epsilon}$ has the homotopy type of $M^{c-\epsilon} \cup_{\varphi_1} e^{\lambda_1} \cup \dots \cup_{\varphi_j} e^{\lambda_j}$ for certain attaching maps $\varphi_1, \dots, \varphi_j$ when ϵ is small enough and there is a homotopy equivalence $h : M^{c-\epsilon} \rightarrow M^a$. We have assumed that there is a homotopy equivalence $h' : M^a \rightarrow K$, where K is a CW complex. Then each $h' \circ h \circ \varphi_i$ is homotopic by cellular approximation to a map ψ_i from the sphere e^{λ_i} to the $(\lambda_i - 1)$ -skeleton of K . Then $K \cup_{\psi_1} e^{\lambda_1} \cup_{\psi_2} \dots \cup_{\psi_j} e^{\lambda_j}$ is a CW-complex. If M is compact, this completes the proof. If M is not compact, but all critical points lie in one of the compact sets M^a , then a similar proof to that of Theorem 2.8 (*which we did not show*), will give that the set M^a is a deformation retract

of M , so that the proof is again complete. If there are infinitely many critical points, then the above construction gives an infinite sequence of homotopy equivalences

$$\begin{array}{ccccccc} M^{a_1} & \subset & M^{a_2} & \subset & M^{a_3} & \subset & \dots \\ \downarrow & & \downarrow & & \downarrow & & \\ K_1 & \subset & K_2 & \subset & K_3 & \subset & \dots \end{array}$$

each extending the previous one. Let K denote the union of the K_i in the direct limit and let $g : M \rightarrow K$ be the limit map. Then g induces isomorphisms of homotopy groups in all dimensions. Since both M and K are dominated by CW-complex, g is a homotopy equivalence. $\square$

2.3. Morse inequalities

Definition 2.11. Let S be a function from certain pairs of spaces to $\mathbb{Z}$. S is *sub-additive* if whenever $X \supset Y \supset Z$ we have

$$S(X, Z) \leq S(X, Y) + S(Y, Z).$$

If equality holds, then S is called *additive*.

We will focus on two sub-additive functions for CW-complexes that relate to the relative homology of (X, Y) $H_i(X, Y)$:

1.

$$R_i(X, Y) = i\text{th Betti number of } (X, Y) = \text{rank}_{\mathbb{F}}(H_i(X, Y, \mathbb{F})) \quad (2.4)$$

for any pair (X, Y) such that the rank is finite. This is sub-additive.

2. The Euler Characteristic $\chi(X, Y)$ is additive

$$\chi(X, Y) = \sum_i (-1)^i R_i(X, Y). \quad (2.5)$$

Lemma 2.12. Let S be sub-additive and let $X_0 \subset \dots \subset X_n$. Then

$$S(X_n, X_0) \leq \sum_{i=1}^n S(X_i, X_{i-1}).$$

If S is additive then equality holds.

Proof by induction on n . For $n = 1$, equality holds trivially and the case $n = 2$ is the definition of sub-additivity. If the result holds for $n - 1$, then $S(X_{n-1}, X_0) \leq \sum_{i=1}^{n-1} S(X_i, X_{i-1})$. Thus

$$\begin{aligned} S(X_n, X_0) &\leq S(X_n, X_{n-1}) + S(X_{n-1}, X_0) \\ &\leq S(X_n, X_{n-1}) + \sum_{i=1}^{n-1} S(X_i, X_{i-1}) \\ &= \sum_{i=1}^n S(X_i, X_{i-1}). \end{aligned}$$

$\square$

Remark 2.3.1. We define $S(X, \emptyset) = S(X)$. Taking $X_0 = \emptyset$ in the above lemma we have $S(X_n) \leq \sum_{i=1}^n S(X_i, X_{i-1})$, with equality if S is additive.

Let M be a compact manifold and $f \in C^\infty(M, \mathbb{R})$ with isolated, non-degenerate critical points. Let $a_1 < \dots < a_k$ be such that M^{a_i} contains i critical points, and $M^{a_k} = M$. Then $H_*(M^{a_i}, M^{a_{i-1}}) = H_*(M^{a_{i-1}} \cup e^{\lambda_i}, M^{a_{i-1}})$, where λ_i is the index of the critical point a_i . By Excision

$$\begin{aligned} H_*(M^{a_i}, M^{a_{i-1}}, \mathbb{F}) &= H_*(e^{\lambda_i}, \partial e^{\lambda_i}, \mathbb{F}) \\ &= \begin{cases} \mathbb{F} & * = \lambda_i \\ 0 & \text{else} \end{cases} \end{aligned}$$

Remark 2.3.2 (on Excision). [13][Theorem 2.2]: Given subspaces $A, B \subset X$, whose interiors cover X then the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces isomorphisms $H_n(B, A \cap B) \xrightarrow{\sim} H_n(X, A)$ for all n . So in our case, $X = M^{a_{i-1}} \cup e^{\lambda_i}$, $A = M^{a_{i-1}}$, $B = e^{\lambda_i}$ and $A \cap B = \partial e^{\lambda_i}$.

The sequence $\emptyset = M^{a_0} \subset M^{a_1} \subset \dots \subset M^{a_k} = M$ with the sub-additive function $S = R_\lambda$ we have

$$R_\lambda(M) \leq \sum_{i=1}^k R_\lambda(M^{a_i}, M^{a_{i-1}}) = C_\lambda \quad (2.6)$$

where C_λ is the number of critical points of index λ . Applying the above to the Euler Characteristic one has

$$\chi(M) = \sum_{i=1}^k \chi(M^{a_i}, M^{a_{i-1}}) = C_0 - C_1 + C_2 \dots \pm C_n, \quad (2.7)$$

and we have proved

Theorem 2.13. *If C_λ denotes the number of critical points of index λ on the compact manifold M , then*

1. $R_\lambda(M) \leq C_\lambda$
2. $\sum (-1)^\lambda R_\lambda(M) = \sum (-1)^\lambda C_\lambda$

Slightly sharper inequalities can be proven via the following argument

Lemma 2.14. *The function*

$$S_\lambda(X, Y) = R_\lambda(X, Y) - R_{\lambda-1}(X, Y) + R_{\lambda-2}(X, Y) - \dots \pm R_0(X, Y)$$

is sub-additive.

Applying S_λ to $\emptyset = M^{a_0} \subset M^{a_1} \subset \dots \subset M^{a_k} = M$ we obtain the *Morse Inequalities*

$$S_\lambda(M) \leq \sum_{i=1}^k S_\lambda(M^{a_i}, M^{a_{i-1}}) = C_\lambda - C_{\lambda-1} + \dots \pm C_0 \quad (2.8)$$

$$\text{or} \quad (2.9)$$

$$R_\lambda(M) - R_{\lambda-1}(M) + \dots \pm R_0(M) = C_\lambda - C_{\lambda-1} + \dots \pm C_0 \quad (2.10)$$

One can use the Morse Inequalities to show

Corollary 2.14.1. *If $C_{\lambda+1} = C_{\lambda-1} = 0$, then $R_\lambda(M) = C_\lambda$ and $R_{\lambda+1}(M) = R_{\lambda-1}(M) = 0$.*

2.4. Morse Homology

As the last section showed, one can use the tools from Morse Theory to construct a Homology: *the Morse homology*. The set of critical points will serve the same role that cells and simplices serve in Cellular and Simplicial Homology; in Morse homology, one uses the Morse index and the stable and unstable manifolds to define the abelian groups and boundary morphisms crucial in the construction of a homology theory.

The Stable and Unstable Manifolds

Definition 2.15. Let $X \in \mathfrak{X}(N)$ be a complete vector field with flow φ^t . For $p \in N$ such that $X(p) = 0$ we define the *Stable* and *Unstable* sets, respectively:

$$W^s(p) := \left\{ y \in N : \lim_{t \rightarrow +\infty} \varphi^t(y) = p \right\} \quad (2.11)$$

$$W^u(p) := \left\{ y \in N : \lim_{t \rightarrow -\infty} \varphi^t(y) = p \right\} \quad (2.12)$$

respectively. Furthermore, for $p, q \in N$ such that $X(p) = 0 = X(q)$, we define

$$\mathcal{M}(p, q) = \left\{ y \in N : \lim_{t \rightarrow -\infty} \varphi^t(y) = p \text{ and } \lim_{t \rightarrow +\infty} \varphi^t(y) = q \right\} = W^u(p) \cap W^s(q) \quad (2.13)$$

There exists a natural free $\mathbb{R}$ -action on the $\mathcal{M}(p, q)$ given by the flow φ^t :

$$\mathbb{R} \times \mathcal{M}(p, q) \rightarrow \mathcal{M}(p, q) \quad (2.14)$$

$$(t, y) \mapsto \varphi^t(y) \quad (2.15)$$

thus it helps to work with the quotient, $\widetilde{\mathcal{M}}(p, q) = \mathcal{M}(p, q)/\mathbb{R}$, instead.

Lemma 2.16. Let $f : (N, g) \rightarrow \mathbb{R}$ be a Morse function and φ^t be the flow of the negative gradient $-\nabla f$. Then for every critical point $p \in N$ of f The stable and unstable sets are finite-dimensional manifolds with

$$\dim W^s(p) = \mu_M(p) \quad \text{and} \quad \dim W^u(p) = \dim N - \mu_M(p).$$

Definition 2.17. Let N be a closed, smooth n -manifold. A pair (f, g) is said to be *Morse-Smale* if under the Riemannian metric g , for all critical points p, q of a morse function f , $W^s(p)$ and $W^u(q)$ intersect transversally.

Remark 2.4.1. When (f, g) is Palais-Smale, for any $p, q \in \text{Crit}(f)$ $\mathcal{M}(p, q)$ is a smooth manifold of dimension

$$\dim \mathcal{M}(p, q) = \mu(p) - \mu_M(q)$$

The Morse Complex

Let N be a compact manifold with (f, g) a Morse-Smale pair. We define the *Morse complex* $\{C_k\}_{k \in \mathbb{N}}$ as the group whose elements are finite formal sums of non-degenerate critical points with morse index k :

$$C_k(f) := \left\{ \sum_{i=1}^m a_i p_i : m \in \mathbb{N}, a_i \in \mathbb{Z}_2, p_i \in \text{Crit}(f), \mu_M(p_i) = k \right\} \quad (2.16)$$

and we define the boundary operator $\partial_k : \text{Crit}_{k+1}(f) \rightarrow \text{Crit}_k(f)$ as the linear extension of the map

$$\partial_k(p_i) = \sum_{\substack{q \in \text{Crit}(f) \\ \mu_k(q)=k}} \# \widetilde{\mathcal{M}}(p, q) q \quad (2.17)$$

since for $p, q \in \text{Crit}(f)$ such that $\mu_M(p) - \mu_M(q) = 1$, $\dim \widetilde{\mathcal{M}}(p, q) = 0$ and it has a finitely many points.

Definition 2.18. *Morse homology* for a compact manifold N is

$$MH_k(M) = \frac{\ker \partial_k}{\text{im } \partial_{k+1}}$$

and it is independent of the choice of Palais-Smale pair.

Theorem 2.19. *The Morse homology is isomorphic to the singular homology of N .*

3. The Art of Geodesics: Length-Minimizing Curves

This chapter references *Introduction to Riemannian manifolds*[15], *Morse Theory*[22] and “Mastermath Differential Notes Fall 2024”[9].

We begin this thesis by tracing back to the beginning. Before analyzing harmonic maps and the energy functional, we must first consider the precursor to all of this: the *geodesics*. Specifically, geodesics as the shortest curves within a manifold. We will show that the critical points of the length functional are indeed geodesics using concepts from the field of *Calculus of Variations*.

3.1. Length of a Curve

We will first introduce some definitions that will play an important role in understanding length-minimizing curves.

Definition 3.1. A smooth curve $\gamma : I \rightarrow M$ is said to be *regular* if $\gamma'(t) \neq 0$ for all $t \in I$. An *admissible curve* is a curve $\gamma : [a, b] \rightarrow M$ such that there exists a partition $a = a_0 < a_1 < \dots < a_k = b$ such that $\gamma|_{[a_i, a_{i+1}]}$ is regular for $i = 0, \dots, k-1$.

The *length* of a smooth curve $\gamma : I \rightarrow M$ is defined to be

$$L_g(\gamma) = \int_a^b \|\gamma'(t)\|_g dt \quad (3.1)$$

Proposition 3.2 (Properties of the length). *Let (M, g) be a Riemannian manifold and $\gamma : [a, b] \rightarrow M$ be an admissible curve.*

- **Additivity:** *If $a < c < b$, then $L_g(\gamma) = L_g(\gamma|_{[a, c]}) + L_g(\gamma|_{[c, b]})$.*
- **Parameter Independence:** *If $\tilde{\gamma}$ is a reparameterization of γ , that is $\gamma \circ \varphi : I' \rightarrow M$ where $\varphi : I' \rightarrow [a, b]$ is a diffeomorphism, then $L_g(\gamma) = L_g(\tilde{\gamma})$.*
- **Isometry Invariance :** *If $(N, \tilde{g})$ is a Riemannian manifold and $\psi : M \rightarrow N$ is a local isometry, then $L_g(\gamma) = L_{\tilde{g}}(\psi \circ \gamma)$.*

The *arc-length* of γ is the function $s : [a, b] \rightarrow \mathbb{R} : s(t) = L_g(\gamma|_{[a, t]}) = \int_a^t \|\gamma'(u)\|_g du$. The arc-length s is continuous everywhere, and from the fundamental theorem of calculus, it is smooth whenever $s'(t) = \|\gamma'(t)\|_g$. From this we say that a smooth curve $\gamma : I \rightarrow M$

is a *unit-speed curve* if $\|\gamma'(t)\|_g = 1$ for all t and a *constant-speed curve* if $\|\gamma'(t)\|_g = c$ for some constant $c \in \mathbb{R}$.

If $\gamma : [a, b] \rightarrow M$ is a unit-speed admissible curve, then its arc length function has the form $s(t) = t - a$. If the interval is of the form $[0, b]$, with $b > 0$, then $s(t) = t$. So we will say a curve is *parametrized by arc length* if its parameter interval is of the form $[0, b]$.

3.2. Variation of a Curve

We say that an admissible curve γ in M is said to be *minimizing* if $L_g(\gamma) \leq L_g(\tilde{\gamma})$ for every admissible curve $\tilde{\gamma}$ with the same endpoints. When M is connected, γ is minimizing if and only if $L_g(\gamma)$ is equal to the distance between its endpoints.

Definition 3.3 (One-parameter variation). Let $[a, b], (-\epsilon, \epsilon) \subseteq \mathbb{R}$ be intervals, with $\epsilon > 0$ and $a, b \in \mathbb{R}$ arbitrary. Given an admissible curve $\gamma : [a, b] \rightarrow M$, a continuous map $F : (-\epsilon, \epsilon) \times [a, b] \rightarrow M$ is called (an admissible) *one-parameter variation of γ* if

- (i) there is a partition $(a = a_0 < a_1 < \dots < a_k = b)$ of $[a, b]$ such that F is smooth on each rectangle $(-\epsilon, \epsilon) \times [a_{i-1}, a_i]$.
- (ii) $F_s(\cdot) := F(s, \cdot)$ is an admissible curve for each $s \in (-\epsilon, \epsilon)$,
- (iii) $(-\epsilon, \epsilon)$ is an open interval containing 0 and $F_0(t) = \gamma(t)$.

F defines two collections of curves in M :

- **Main Curves:** $F_s(t) = F(s, t)$ defined for all $t \in [a, b]$ and by keeping s constant and
- **Transverse Curves:** $F^t(s) = F(s, t)$ defined for all $s \in (-\epsilon, \epsilon)$ and by keeping t constant.

It is called a *proper variation* if all the main curves have the same starting and ending points ($F_s(a) = \gamma(a)$ and $F_s(b) = \gamma(b)$ for all $s \in (-\epsilon, \epsilon)$).

Remark 3.2.1. A equivalent definition will produce an n -parameter family of curves, by replacing $(-\epsilon, \epsilon)$ with $(-\epsilon, \epsilon)^n \subseteq \mathbb{R}^n$, as well as n -parameter variation.

If F is smooth, then the velocity vectors of the main and transverse curves are $\partial_t F(s, t) = F'_s(t) \in T_{F(s, t)}M$ and $\partial_s F(s, t) = (F^t)'(s) \in T_{F(s, t)}M$. Following Definition A.13, these are examples of vector fields along F .

Remark 3.2.2. For all intents and purposes, we will assume that variations are smooth.

Definition 3.4. If F is a variation of γ , the *variation field of F* is the smooth vector field along γ defined by $V_t = \partial_s F(0, t)$. Following the definition of a proper variation of γ , we say that a vector field V along γ is proper if $V_a = 0 = V_b$.

Lemma 3.5. *If γ is an admissible curve and V is a vector field along γ , then V is the variation field of some variation of γ . If V is proper, the variation is proper.*

Proof. Suppose γ and V satisfy the hypotheses and set

$$F(s, t) = \exp_{\gamma(t)}(sV_t) \quad (3.2)$$

If F is indeed a variation, then Equation 3.2 gives the variation field:

$$\partial_s F(0, t) = \partial_s \left(\exp_{\gamma(t)}(sV_t) \right) = \exp_{\gamma(t)}(sV_t) V_t|_{0,t} = V_t.$$

By compactness of $[a, b]$, there is some $\epsilon > 0$ such that F is defined on $(-\epsilon, \epsilon) \times [a, b]$. By composition, F is smooth on each $(-\epsilon, \epsilon) \times [a_{i-1}, a_i]$ where V is smooth, thus F is a variation of γ . Furthermore, if V is proper, $V_a = 0 = V_b$, then

$$\begin{aligned} F(s, a) &= \exp_{\gamma(a)}(sV_0) = \exp_{\gamma(a)}(0) = \gamma(a) \\ F(s, b) &= \exp_{\gamma(b)}(sV_0) = \exp_{\gamma(b)}(0) = \gamma(b). \end{aligned}$$

□

3.3. Critical points and First Variation Formula

Theorem 3.6 (First Variation Formula). *Let (M, g) be a Riemannian manifold. Consider a unit-speed admissible curve $\gamma : [a, b] \rightarrow M$ with variation $F : (-\epsilon, \epsilon) \times [a, b] \rightarrow M$ and variation field V . Then $L_g(\gamma)$ is a smooth function of s , and*

$$\begin{aligned} \frac{d}{ds} \Big|_{s=0} L_g(F_s) &= - \int_a^b \left\langle V, \frac{D}{dt} \gamma' \right\rangle_g dt - \sum_{i=1}^{k-1} \langle V_{a_i}, \Delta_i \gamma' \rangle_g \\ &\quad + \langle V_b, \gamma'(b) \rangle_g - \langle V_a, \gamma'(a) \rangle_g \end{aligned} \quad (3.3)$$

Where $(a = a_0 < a_1 < \dots < a_k = b)$ is a partition of V and for each $i = 1, \dots, k-1$

$$\Delta_i \gamma' = \gamma'(a_i^+) - \gamma'(a_i^-)$$

is the jump, or discontinuity, in the velocity vector field of γ . If F is a proper variation, then

$$\frac{d}{ds} \Big|_{s=0} L_g(F_s) = - \int_a^b \left\langle V, \frac{D}{dt} \gamma' \right\rangle_g dt - \sum_{i=1}^{k-1} \langle V_{a_i}, \Delta_i \gamma' \rangle_g \quad (3.4)$$

Proof. For brevity, we will denote $T(s, t) = \partial_t F(s, t)$ and $S(s, t) = \partial_s F(s, t)$.

- **Length is smooth:** On every rectangle $(-\epsilon, \epsilon) \times [a_{i-1}, a_i]$ where F is smooth and the domain of integration (see Equation 3.1) is compact, we can differentiate under the integral sign infinitely many times. Since L_g is the finite sum of such integrals, it is a smooth function of s .
- **Formulas:** We will work on the rectangles $(-\epsilon, \epsilon) \times [a_{i-1}, a_i]$ where F is smooth and the domain of integration (see Equation 3.1) is compact, such that the following

equalities hold :

$$\begin{aligned}
\frac{d}{ds} L_g (F_s|_{[a_{i-1}, a_i]}) &= \frac{d}{ds} \int_{a_{i-1}}^{a_i} \langle T, T \rangle_g^{\frac{1}{2}} dt \\
&= \int_{a_{i-1}}^{a_i} \frac{\partial}{\partial s} \langle T, T \rangle_g^{\frac{1}{2}} dt \\
&= \int_{a_{i-1}}^{a_i} \frac{1}{2} \langle T, T \rangle_g^{-\frac{1}{2}} \frac{\partial}{\partial s} \langle T, T \rangle_g dt \\
&= \int_{a_{i-1}}^{a_i} \frac{1}{2} \langle T, T \rangle_g^{-\frac{1}{2}} 2 \left\langle \frac{D}{\partial s} T, T \right\rangle_g dt \\
&= \int_{a_{i-1}}^{a_i} \frac{1}{\|T\|_g} \left\langle \frac{D}{\partial t} S, T \right\rangle_g dt
\end{aligned}$$

where the second-to-last equality is due to the properties Equation A.12 and Equation A.11 of the Levi-Civita connection, while the last is due to Lemma A.14.

When $s = 0$, note that $F_0(t) = F(0, t) = \gamma$ so $T(0, t) = (F_0)'(t) = \gamma'(t)$ and it has norm 1, as γ is unit-speed and by definition of a variation vector field $S(0, t) = V_t$, thus

$$\frac{d}{ds} L_g (F_0|_{[a_{i-1}, a_i]}) = \int_{a_{i-1}}^{a_i} \left\langle \frac{D}{\partial t} V, \gamma' \right\rangle_g dt \quad (3.5)$$

$$= \int_{a_{i-1}}^{a_i} \left(\frac{d}{dt} \langle V, \gamma' \rangle_g - \left\langle V, \frac{D}{dt} \gamma' \right\rangle_g \right) dt \quad (3.6)$$

$$= \langle V_{a_i}, \gamma'(a_i) \rangle_g - \langle V_{a_{i-1}}, \gamma'(a_{i-1}) \rangle_g - \int_{a_{i-1}}^{a_i} \left\langle V, \frac{D}{dt} \gamma' \right\rangle_g dt \quad (3.7)$$

□

Remark 3.3.1. The terms $-\sum_{i=1}^{k-1} \langle V_{a_i}, \Delta_i \gamma' \rangle_g + \langle V_b, \gamma'(b) \rangle_g - \langle V_a, \gamma'(a) \rangle_g$ show that varying a curve γ in the direction of its ‘jumps’ $\Delta_i \gamma'$, decreases its length. While the term $-\int_a^b \left\langle V, \frac{D}{dt} \gamma' \right\rangle_g dt$ shows that varying a curve γ in the direction of its acceleration vector field decreases its length.

Recall ¹, that we call an admissible curve γ a geodesic if it satisfies the following

$$\frac{D}{dt} \gamma' \equiv 0. \quad (3.8)$$

Theorem 3.7. *In a Riemannian manifold, every minimizing curve is a geodesic when it is given a unit speed parametrization.*

Proof. Suppose $\gamma : [a, b] \rightarrow M$ is a minimizing unit-speed curve and $a = a_0 < a_1 < \dots < a_k = b$ is a partition for γ . If F is any proper variation of γ , then $L_g(F_s)$ is a smooth function of s that achieves its minimum at $s = 0$, hence $\frac{d}{ds} L_g(F_s) \Big|_{s=0} = 0$. Since every proper vector field along γ is a variation field, the right-hand side of Equation 3.4 must vanish for each V .

¹See Definition A.18 in Appendix A.

- **Broken Geodesic:**

We wish to show that $\frac{D}{dt}\gamma' = 0$ on each $[a_{i-1}, a_i]$, so γ is a broken geodesic. Let $\varphi \in C^\infty(\mathbb{R})$ be a bump function such that $\varphi > 0$ on (a_{i-1}, a_i) and $\varphi = 0$ otherwise. Then from Equation 3.4 with $V = \varphi \frac{D}{dt}\gamma'$, we have

$$0 = \langle V_{a_i}, \gamma'(a_i) \rangle_g - \langle V_{a_{i-1}}, \gamma'(a_{i-1}) \rangle_g - \int_{a_{i-1}}^{a_i} \left\langle \varphi \frac{D}{dt}\gamma', \frac{D}{dt}\gamma' \right\rangle_g dt = \int_{a_{i-1}}^{a_i} \varphi \left\| \frac{D}{dt}\gamma' \right\|_g^2 dt$$

Since the integrand is non-negative, this shows that $\frac{D}{dt}\gamma' = 0$ on each sub-interval.

- **Non-corners:**

We want to show that $\Delta_i \gamma' = 0$ for each $i \in \{0, \dots, k\}$, i.e no corners. Similarly to the broken geodesic part, we will use a bump function in a coordinate chart to construct a piecewise smooth vector field V along γ such that $V_{a_i} = \Delta_i \gamma'$ and $V_{a_j} = 0$ for $j \neq i$. Then Equation 3.4 with this vector field is

$$0 = \langle V_{a_i}, \Delta_i \gamma' \rangle_g - \int_{a_{i-1}}^{a_i} \left\langle V, \frac{D}{dt}\gamma' \right\rangle_g dt = \langle \Delta_i \gamma', \Delta_i \gamma' \rangle_g \quad (3.9)$$

thus $\Delta_i \gamma' = 0$ for each i .

Since the two one-sided velocity vectors of γ ($\gamma'(a_i^+)$ and $\gamma'(a_i^-)$) match up for each a_i , it follows from uniqueness of geodesics that $\gamma|_{[a_i, a_{i+1}]}$ is the continuation of $\gamma|_{a_{i-1}, a_i}$ and thus γ is smooth. $\square$

From this we can say that an admissible curve γ is a *critical point of L_g* if for every proper variation F of γ

$$\frac{d}{ds} L_g(F_s) \Big|_{s=0} = 0.$$

Corollary 3.7.1. *A unit-speed admissible curve γ is a critical point for L_g if and only if it is a geodesic.*

Proof. From the previous theorem, we know that if γ is a critical point, then it is a geodesic. On the other hand, if γ is a geodesic, the first term on the right-hand side of Equation 3.4 vanishes by the geodesic equation, and the second term vanishes because γ' has no jumps. $\square$

4. Morse theory and Geodesics: The Path Space

This chapter references *Morse Theory*[22], *Introduction to Riemannian manifolds* [15] and *Riemannian Geometry and Geometric Analysis* [14].

In the previous chapter, we established that geodesics are the critical points of the length functional. As demonstrated in chapter 2, critical points of a function f play a crucial role in constructing a CW-decomposition of its domain. In this chapter, we will explore how a similar construction can be applied to geodesics on an arbitrary Riemannian manifold, using the energy functional instead of the length functional. This will allow us to construct a CW-decomposition of the domain of the energy functional. The CW-decomposition from chapter 2 requires the knowledge of the Morse index of the critical points. To calculate the Morse index, we will analyze the Hessian of the energy functional. This analysis will yield the Jacobi operator, which is crucial for the later results of this thesis.

4.1. Path Space and the Energy Functional

Let M be a smooth manifold and let p and q be two – not necessarily distinct – points in M . The set of all admissible curves from p to q in M will be denoted by $\mathcal{PM}(p, q)$, or simply $\mathcal{PM}$, and will be called the *Path space*.

In the following chapter, we will show that $\mathcal{PM}$ is equivalent to the *Sobolev space* $H^1(I, M)$. We consider $H^1(I, M)$ because it is the completion of $C^\infty(I, M)$, the space of smooth paths in M . This allows us to regard any type of Riemannian manifold, not necessarily compact, as it ensures the existence of minima. This space works well with the *energy functional*, which is a more manageable functional when generalizing the study of geodesics to higher dimensions.

Definition 4.1. Given a Riemannian manifold (M, g) . The *energy functional* is the functional $\mathcal{E}_g : \mathcal{PM} \rightarrow \mathbb{R}$ defined by

$$\mathcal{E}_g(\omega) = \frac{1}{2} \int_0^1 \left\| \frac{d\omega}{dt} \right\|_g^2 dt \quad \forall \omega \in \mathcal{PM}. \quad (4.1)$$

Remark 4.1.1. For $\omega \in \mathcal{PM}$, the *energy of* ω from a to b (with $0 \leq a < b \leq 1$) is

$$\mathcal{E}_g(\omega|_{[a,b]}) = \frac{1}{2} \int_a^b \left\| \frac{d\omega}{dt} \right\|_g^2 dt.$$

By Schwartz's inequality, we have

$$\begin{aligned} (L_g(\omega|_{[a,b]}))^2 &= \left(\int_a^b \left\| \frac{d\omega}{dt} \right\|_g dt \right)^2 \\ &\leq \left(\int_a^b \|1\|_g^2 dt \right) \left(\int_a^b \left\| \frac{d\omega}{dt} \right\|_g^2 dt \right) = 2(b-a)\mathcal{E}_g(\omega|_{[a,b]}), \end{aligned} \quad (4.2)$$

with equality if and only if $\left\| \frac{d\omega}{dt} \right\|_g^2$ is constant. Furthermore, if there exists a minimal geodesic γ from $p = \omega(0)$ to $q = \omega(1)$, then

$$E_g(\gamma) = (L_g(\gamma))^2 \leq (L_g(\omega))^2 \leq E_g(\omega). \quad (4.3)$$

From the above remark, we have the energy version of Theorem 3.7.

Theorem 4.2. *Let (M, g) be a complete¹ Riemannian manifold and let $p, q \in M$ have distance d . Then the energy functional $\mathcal{E}_g : \mathcal{PM} \rightarrow \mathbb{R}$ takes on its minimum d^2 precisely on the set of minimal geodesics from p to q .*

Similarly, one has a *First Variation Formula* for the energy functional, whose proof is exactly the one of Theorem 3.6. For this, we define the *tangent space of $\mathcal{PM}$* at a path ω to be the vector space of all piecewise smooth proper vector fields V along ω :

$$T_\omega \mathcal{PM} := \left\{ V : [0, 1] \rightarrow TM : V_t \in T_{\omega(t)}M \text{ and } V_0 = 0 = V_1 \right\}. \quad (4.4)$$

Theorem 4.3 (First Variation Formula for $\mathcal{E}$). *Let (M, g) be a Riemannian manifold. Consider $\omega \in \mathcal{PM}$ with a one-parameter proper variation $F : (-\epsilon, \epsilon) \times [0, 1] \rightarrow M$ and variation field V . Then*

$$\left. \frac{d\mathcal{E}_g(F_s)}{ds} \right|_{s=0} = - \int_0^1 \left\langle V, \frac{D}{dt} \frac{d\omega}{dt} \right\rangle_g dt - \sum_{i=1}^{k-1} \left\langle V_{a_i}, \Delta_i \frac{d\omega}{dt} \right\rangle_g \quad (4.5)$$

Where $(a = a_0 < a_1 < \dots < a_k = b)$ is a partition of V and for each $i = 1, \dots, k-1$

$$\Delta_i \frac{d\omega}{dt} = \frac{d\omega}{dt}(a_i^+) - \frac{d\omega}{dt}(a_i^-)$$

is the jump, or discontinuity, in the velocity vector field of ω . Of course, $\Delta_i \frac{d\omega}{dt} = 0$ for all but a finite number of values.

Similarly to the length functional, geodesics are the critical points of $\mathcal{E}$.

Corollary 4.3.1. *A path $\omega \in \mathcal{PM}$ is a critical point of $\mathcal{E}_g$ if and only if ω is a geodesic.*

¹See Appendix B for more details on the relation between geodesically completeness and topological completeness

Following this corollary and Remark 4.1.1, we have that geodesics are both the critical points of the energy functional and the length functional.

Proposition 4.4. *A curve $\gamma : [a, b] \rightarrow M$ in $\mathcal{PM}(p, q)$ minimizes the energy functional $\mathcal{E}_g$ if and only if it has constant speed and minimizes the length functional L_g*

Proof. Suppose γ minimize $\mathcal{E}_g$ but there exists $\omega \in \mathcal{PM}(p, q)$ such that $L_g(\omega) < L_g(\gamma)$, then for the “constant speed re-parametrization” $\tilde{\omega}$ we have

$$\mathcal{E}(\tilde{\omega}) = \frac{1}{2(b-a)} L(\tilde{\omega})^2 = \frac{1}{2(b-a)} L(\omega)^2 < \frac{1}{2(b-a)} L(\gamma)^2 \leq \mathcal{E}(\omega),$$

which is a contradiction.

Conversely, if γ has constant speed and minimize L_g , but there exists $\omega \in \mathcal{PM}(p, q)$ $\mathcal{E}_g(\omega) < \mathcal{E}_g(\gamma)$, then

$$L_g(\omega) \leq \sqrt{2(b-a)\mathcal{E}_g(\omega)} < \sqrt{2(b-a)\mathcal{E}_g(\gamma)} = L_g(\gamma),$$

a contradiction. □

4.2. Hessian of the Energy Functional: The Second Variation Formula

To give the Morse Theory for the path space $\mathcal{PM}$ via the energy functional, we need to define the Hessian of $\mathcal{E}_g$ and the Morse index of the geodesics.

Fix a path $\omega \in \mathcal{PM}$. Consider two vectors $W, V \in T_\omega \mathcal{PM}$ and a *two-parameter* proper variation $F : (-\epsilon, \epsilon)^2 \times [0, 1] \rightarrow M$, such that $F(0, 0, t) = \omega(t)$ and

$$W = \left. \frac{\partial F_{s_1, s_2}}{\partial s_1} \right|_{(s_1, s_2) = (0, 0)}, \quad V = \left. \frac{\partial F_{s_1, s_2}}{\partial s_2} \right|_{(s_1, s_2) = (0, 0)}.$$

Definition 4.5. The *Hessian* of the energy functional $\mathcal{E}_g : \mathcal{PM} \rightarrow \mathbb{R}$ at a geodesic ω , $\mathcal{E}_{**} : T_\omega \mathcal{PM} \times T_\omega \mathcal{PM} \rightarrow \mathbb{R}$, is defined by the second variation of $\mathcal{E}_g$:

$$(W, V) \mapsto \left. \frac{\partial^2 \mathcal{E}_g(F_{s_1, s_2}(W, V))}{\partial s_1 \partial s_2} \right|_{(s_1, s_2) = (0, 0)}. \quad (4.6)$$

Theorem 4.6 (Second Variation Formula). *Let $F : (-\epsilon, \epsilon)^2 \times [0, 1] \rightarrow M$ be a two-parameter proper variation of the geodesic γ in the Riemannian manifold (M, g) , with variation vector fields*

$$W = \left. \frac{\partial F_{s_1, s_2}}{\partial s_1} \right|_{(s_1, s_2) = (0, 0)}, \quad V = \left. \frac{\partial F_{s_1, s_2}}{\partial s_2} \right|_{(s_1, s_2) = (0, 0)}.$$

Then the second variation of the energy functional has the form

$$\left. \frac{\partial^2 \mathcal{E}_g(F_{s_1, s_2})}{\partial s_1 \partial s_2} \right|_{(s_1, s_2)=0} = - \sum_i \left\langle V_t, \Delta_i \frac{D}{dt} W \right\rangle_g - \int_0^1 \left\langle V, \frac{D^2}{dt^2} W + R \left(W, \frac{d\gamma}{dt}, \frac{d\gamma}{dt} \right) \right\rangle_g dt \quad (4.7)$$

Where $(a = a_0 < a_1 < \dots < a_k = b)$ is a partition of W and for each $i = 1, \dots, k-1$

$$\Delta_i \frac{DW}{dt} = \frac{DW}{dt}(a_i^+) - \frac{DW}{dt}(a_i^-)$$

is the jump, or discontinuity, in $\frac{DW}{dt}$ at one of its finitely many points of discontinuity in the open unit interval.

Proof. By the First Variation formula for $\mathcal{E}_g$, we have

$$\frac{\partial \mathcal{E}_g(F)}{\partial s_2} = - \sum_{i=1}^{k-1} \left\langle \frac{\partial F}{\partial s_2} \Big|_{a_i}, \Delta_i \frac{\partial F}{\partial t} \right\rangle_g - \int_0^1 \left\langle \frac{\partial F}{\partial s_2}, \frac{D}{dt} \frac{\partial F}{\partial t} \right\rangle_g dt \quad (4.8)$$

Therefore

$$\begin{aligned} \frac{\partial^2 \mathcal{E}_g(F)}{\partial s_1 \partial s_2} &= - \sum_{i=1}^{k-1} \left\langle \frac{D}{\partial s_1} \frac{\partial F}{\partial s_2} \Big|_{a_i}, \Delta_i \frac{\partial F}{\partial t} \right\rangle_g - \sum_{i=1}^{k-1} \left\langle \frac{\partial F}{\partial s_2} \Big|_{a_i}, \frac{D}{\partial s_1} \Delta_i \frac{\partial F}{\partial t} \right\rangle_g \\ &\quad - \int_0^1 \left\langle \frac{D}{\partial s_1} \frac{\partial F}{\partial s_2}, \frac{D}{dt} \frac{\partial F}{\partial t} \right\rangle_g dt - \int_0^1 \left\langle \frac{D}{\partial s_1} \frac{\partial F}{\partial s_2}, \frac{D}{\partial s_1} \frac{D}{dt} \frac{\partial F}{\partial t} \right\rangle_g dt \end{aligned}$$

By evaluating it at $(s_1, s_2) = (0, 0)$, substituting for W and V , and recalling that $\gamma(t) = F(0, 0, t)$ is a geodesic, we have

$$\Delta_i \frac{\partial F}{\partial t} = 0 \text{ and } \frac{D}{dt} \frac{\partial F}{\partial t} = 0.$$

Thus

$$\begin{aligned} \left. \frac{\partial^2 \mathcal{E}_g(F)}{\partial s_1 \partial s_2} \right|_{s_1=s_2=0} &= - \sum_{i=1}^{k-1} \left\langle V_{a_i}, \frac{D}{\partial s_1} \Delta_i \frac{d\gamma}{dt} \right\rangle_g - \int_0^1 \left\langle \frac{D}{\partial s_1} V, \frac{D}{\partial s_1} \frac{D}{dt} \frac{d\gamma}{dt} \right\rangle_g dt \\ &= - \sum_{i=1}^{k-1} \left\langle V_{a_i}, \Delta_i \frac{D}{\partial s_1} \frac{d\gamma}{dt} \right\rangle_g - \int_0^1 \left\langle \frac{D}{\partial s_1} V, \frac{D}{\partial s_1} \frac{D}{dt} \frac{d\gamma}{dt} \right\rangle_g dt \\ &= - \sum_{i=1}^{k-1} \left\langle V_{a_i}, \Delta_i \frac{D}{\partial t_1} \frac{\partial F}{\partial s_1} \right\rangle_g - \int_0^1 \left\langle \frac{D}{\partial s_1} V, \frac{D}{\partial s_1} \frac{D}{dt} \frac{d\gamma}{dt} \right\rangle_g dt, \end{aligned}$$

where the last two equalities are due to Lemma A.14. By substituting for W and applying the relation between curvature and velocity vectors from Lemma A.16, we have

$$\begin{aligned}
& \left. \frac{\partial^2 \mathcal{E}_g(F)}{\partial s_1 \partial s_2} \right|_{s_1=s_2=0} \\
&= - \sum_{i=1}^{k-1} \left\langle V_{a_i}, \Delta_i \frac{DW}{\partial t_1} \right\rangle_g - \int_0^1 \left\langle \frac{D}{\partial s_1} V, R \left(W, \frac{d\gamma}{dt}, \frac{d\gamma}{dt} \right) + \frac{D}{\partial t} \frac{D}{\partial s_1} \frac{d\gamma}{dt} \right\rangle_g dt \\
&= - \sum_{i=1}^{k-1} \left\langle V_{a_i}, \Delta_i \frac{DW}{\partial t_1} \right\rangle_g - \int_0^1 \left\langle \frac{D}{\partial s_1} V, R \left(W, \frac{d\gamma}{dt}, \frac{d\gamma}{dt} \right) + \frac{D}{dt} \frac{D}{dt} W \right\rangle_g dt \\
&= - \sum_{i=1}^{k-1} \left\langle V_{a_i}, \Delta_i \frac{DW}{\partial t} \right\rangle_g - \int_0^1 \left\langle \frac{D}{\partial s_1} V, R \left(W, \frac{d\gamma}{dt}, \frac{d\gamma}{dt} \right) + \frac{D^2}{dt^2} W \right\rangle_g dt,
\end{aligned}$$

where the second last equality follows from Lemma A.14. $\square$

Corollary 4.6.1. *The Hessian of the energy functional $\mathcal{E}_{**}$ is a well-defined symmetric and bilinear function of W and V .*

Proof. • **Well-defined:** The second variation formula shows that $\left. \frac{\partial^2 \mathcal{E}_g(F_{s_1, s_2})}{\partial s_1 \partial s_2} \right|_{(s_1=s_2=0)}$ depends only on the variation vector fields W and V , so $\mathcal{E}_{**}(W, V)$ is well-defined.

- **Symmetry:** The symmetry property $\mathcal{E}_{**}(W, V) = \mathcal{E}_{**}(V, W)$ follows from the symmetry property $\frac{\partial^2 \mathcal{E}_g}{\partial s_1 \partial s_2} = \frac{\partial^2 \mathcal{E}_g}{\partial s_2 \partial s_1}$.
- **Bilinearity:** To show bilinearity, we will use the following lemma.

Lemma 4.7. *Consider the smooth function $s : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that interchanges the variable $(x, y) \mapsto (y, x)$ and an interval $I \subset \mathbb{R}$. Define $U = I^2$ and consider any function $f \in C^\infty(U, \mathbb{R})$. Then*

$$\frac{\partial}{\partial x} \frac{\partial}{\partial y} (f \circ s) = \left(\frac{\partial}{\partial y} \frac{\partial}{\partial x} f \right) \circ s$$

and

$$\frac{\partial}{\partial y} \frac{\partial}{\partial x} (f \circ s) = \left(\frac{\partial}{\partial x} \frac{\partial}{\partial y} f \right) \circ s$$

To ensure clarity, we will consider a two-parameter variation $T : (-\epsilon, \epsilon) \times [0, 1] \rightarrow M$ as the map $\tilde{T} : (-\epsilon, \epsilon)^2 \rightarrow \mathcal{PM}$.

Now consider vector fields $W, V \in T_\gamma \mathcal{PM}$ and a 2-parameter proper variation $\tilde{F} : (-\epsilon, \epsilon)^2 \rightarrow \mathcal{PM}$ of γ such that W, V are the variation fields:

$$W = \left. \frac{\partial \tilde{F}}{\partial s_1} \right|_{(s_1=s_2=0)} \quad \text{and} \quad V = \left. \frac{\partial \tilde{F}}{\partial s_2} \right|_{(s_1=s_2=0)}$$

Define a two-parameter variation $\tilde{B} : (-\epsilon, \epsilon)^2 \rightarrow \mathcal{PM}$ such that $\tilde{B} = \tilde{F} \circ s$, where s is the switching function from the lemma above, then

$$V = \left. \frac{\partial \tilde{B}}{\partial s_1} \right|_{(s_1=s_2=0)} \quad \text{and} \quad W = \left. \frac{\partial \tilde{B}}{\partial s_2} \right|_{(s_1=s_2=0)}$$

Since $\frac{\partial^2 \mathcal{E}_g(F_{s_1, s_2})}{\partial s_1 \partial s_2} \Big|_{s_1=s_2=0}$ does not depend on the chosen variation, we may use $\tilde{B}$:

$$\begin{aligned} \mathcal{E}_{**}(W, V) &= \frac{\partial}{\partial s_1} \frac{\partial}{\partial s_2} (\mathcal{E}_g \circ \tilde{B}) (0, 0) = \frac{\partial}{\partial s_1} \frac{\partial}{\partial s_2} (\mathcal{E}_g \circ \tilde{F} \circ s) (0, 0) \\ &= \frac{\partial}{\partial s_2} \frac{\partial}{\partial s_1} (\mathcal{E}_g \circ \tilde{F}) (s(0, 0)) = \frac{\partial}{\partial s_2} \frac{\partial}{\partial s_1} (\mathcal{E}_g \circ \tilde{F}) (0, 0) \\ &= \frac{\partial}{\partial s_1} \frac{\partial}{\partial s_2} (\mathcal{E}_g \circ \tilde{F}) (0, 0) = \mathcal{E}_{**}(V, W). \end{aligned}$$

□

Remark 4.2.1. The diagonal terms $\mathcal{E}_{**}(W, W)$ of the Bilinear pairing $\mathcal{E}_{**}$ can be defined in terms of a 1-parameter proper variation of γ .

From a proper variation $F : (-\epsilon, \epsilon) \times [0, 1] \rightarrow M$ of γ with variation field W , we can define a two-parameter proper variation $B : (-\epsilon, \epsilon)^2 \times [0, 1] \rightarrow M$ by

$$B_{s_1, s_2} = F_{s_1 + s_2}.$$

Then, the variation fields of B coincide with the variation field of F :

$$\frac{\partial B}{\partial s_1} = \frac{\partial B}{\partial s_2} = \frac{dF}{ds},$$

so the second variation becomes

$$\frac{\partial^2 (\mathcal{E} \circ B)}{\partial s_1 \partial s_2} = \frac{d^2 (\mathcal{E} \circ F)}{ds^2}. \quad (4.9)$$

4.3. Morse Index of the Energy Functional.

As in chapter 2, we say that a geodesic $\gamma \in \mathcal{PM}$ is *non-degenerate* if the *null-space* of the Hessian $\mathcal{E}_{**}$,

$$\text{Nul}(\mathcal{E}_{**}) := \{V \in T_\gamma \mathcal{PM} : \mathcal{E}_{**}(V, W) = 0 \quad \forall W \in T_\gamma \mathcal{PM}\}, \quad (4.10)$$

is trivial, equivalently if

$$\nu = \dim \text{Nul} \mathcal{E}_{**} = 0.$$

Likewise, we defined the Morse index of $\mathcal{E}_{**}$ at a non-degenerate geodesic γ as the dimensions of the maximal subspace of $T_\gamma \mathcal{PM}$ such that $\mathcal{E}_{**}$ is negative definite. From Remark 4.2.1, the Morse index of a minimal geodesic is straightforward.

Lemma 4.8. *If $\gamma \in \mathcal{PM}$ is a minimal geodesic, then the Hessian $\mathcal{E}_{**}$ is positive definite. Hence, the Morse index of $\mathcal{E}_{**}$ is $\mu_M(\gamma) = 0$.*

Proof. Let γ be a minimal geodesic, then it is a minimum of the energy functional $\mathcal{E}$. The inequality $\mathcal{E}_g(\tilde{F}(s)) \geq \mathcal{E}_g(\gamma) = E_g(\tilde{F}(0))$ implies that

$$\frac{d^2 (\mathcal{E} \circ \tilde{F})}{ds^2} \Big|_{s=0} \geq 0.$$

Thus $\mathcal{E}_{**}(W, W) \geq 0$ for all $W \in T_\gamma \mathcal{PM}$, hence the index of $\mathcal{E}_{**}$ is zero. □

In this section, we will introduce an equivalent method for determining the nullity and the Morse index of a geodesic by identifying specific points along it. This will be done by looking at the *Jacobi operator*, which arises from Equation 4.7, $J_\gamma : T_\gamma \mathcal{P}M \rightarrow T_\gamma \mathcal{P}M$ defined by

$$W \mapsto \frac{D^2}{dt^2}W + R\left(W, \frac{d\gamma}{dt}, \frac{d\gamma}{dt}\right). \quad (4.11)$$

What is of interest in our calculations is the kernel of the Jacobi operator, so we give a name to its elements.

Definition 4.9. A vector field J along a geodesic γ is called a *Jacobi field* if it is in the kernel of the *Jacobi Operator*: i.e if

$$\frac{D^2}{dt^2}J + R\left(J, \frac{d\gamma}{dt}, \frac{d\gamma}{dt}\right) = 0. \quad (4.12)$$

Proposition 4.10 (Existence and Uniqueness of Jacobi fields). *Let (M, g) be a Riemannian manifold. Let $\gamma : I \rightarrow M$ be a geodesic and consider $a \in I$ with $\gamma(a) = p$. For every pair of vectors $v, w \in T_p M$ there is a unique Jacobi field J along γ satisfying the initial condition: $\left(J_a, \frac{D}{dt}J_a\right) = (v, w)$.*

Proof. Choose a parallel orthonormal frame $\{E_i\}$ along γ , and write $v = \sum_i v^i E_i(a)$, $w = \sum_i w^i E_i(a)$ and $\frac{d\gamma}{dt} = \sum_i y^i E_i$, where $v^i, w^i, y^i : I \rightarrow \mathbb{R}$ are all smooth functions. Consider a vector field $J = \sum_i J^i E_i$ along γ . In local coordinates the Jacobi equation for J becomes a system of linear second-order ODEs for the functions $J^i : I \rightarrow \mathbb{R}$:

$$\frac{d^2 J^i}{dt^2} + R_{jkl}^i(\gamma(t)) J^j y^k y^l = 0.$$

By placing $W^i = \frac{dJ^i}{dt}$, then the Jacobi equation becomes a first-order linear system in $2n$ unknowns:

$$\frac{dJ^i}{dt} = W^i \quad (4.13)$$

$$\frac{dW^i}{dt} = -R_{jkl}^i(\gamma(t)) J^j y^k y^l. \quad (4.14)$$

Thus, we establish the existence and uniqueness of a smooth solution with arbitrary initial conditions $J^i(a) = v^i$ and $W^i(a) = w^i$. Since $\frac{DJ}{dt}(a) = \sum \frac{dJ^i}{dt}(a) E_i(a) = \sum_i W^i(a) E_i(a) = w$. It follows that J satisfies the Jacobi equation. $\square$

Corollary 4.10.1. *The space of Jacobi fields along a geodesic γ of M^n , $\ker J_\gamma$, is a $2n$ -dimensional linear subspace of the space of vector fields along γ .*

Proof. Linearity follows from the Jacobi equation, while the dimension is from the isomorphism $\ker J(\gamma) \xrightarrow{\cong} T_p M \times T_p M$ defined by $J \mapsto \left(J_a, \frac{DJ}{dt}(a)\right)$. $\square$

Jacobi fields fit well in the study of geodesics, not only because they come from the formula of the Hessian of $\mathcal{E}_g$, but because they are the variation field of *variations through geodesics*.

Definition 4.11. We say that a one-parameter proper variation $F : (-\epsilon, \epsilon) \times I \rightarrow M$ of a geodesic $\gamma \in \mathcal{PM}$ is a *variation through geodesics* if the main curves $F_s : I \rightarrow M$ are geodesics.

Given a variation through geodesics, it is straightforward to see that the variation field is a Jacobi field.

Theorem 4.12 (Variation fields are Jacobi). *Let (M, g) be a Riemannian manifold, γ a geodesic in M , and J a vector field along $\gamma : I \rightarrow M$. If J is the variation field of a variation through geodesics, then J is a Jacobi field.*

Proof. Let $F : (-\epsilon, \epsilon) \times I \rightarrow M$ be the variation to which J is the variation field. Write $T(s, t) = \partial_t F(s, t)$ and $S(s, t) = \partial_s F(s, t)$. The geodesic equation (Equation A.25) states that $\frac{D}{\partial t} T = 0$ for all (s, t) , hence $\frac{D}{\partial s} \frac{D}{\partial t} T = 0$. By Lemma A.16 and Lemma A.14

$$0 = \frac{D}{\partial s} \frac{D}{\partial t} T = \frac{D}{\partial t} \frac{D}{\partial s} T + R(S, T, T) = \frac{D}{\partial t} \frac{D}{\partial t} S + R(S, T, T).$$

Thus at $s = 0$, where $S(0, t) = J_t$ and $T(0, t) = \frac{d\gamma}{dt}$: $0 = \frac{D^2}{dt^2} J + R\left(J, \frac{d\gamma}{dt}, \frac{d\gamma}{dt}\right)$. $\square$

The converse, however, is not guaranteed, and it is called *integrability*.

Definition 4.13. A Jacobi field J along a geodesic γ is called *integrable* if there exists a proper variation $F : (-\epsilon, \epsilon) \times I \rightarrow M$ of γ through geodesics such that J is its variation field: $J = \frac{\partial F_s}{\partial s} \Big|_{s=0}$.

As the following proposition shows, under certain conditions, all Jacobi fields are integrable.

Proposition 4.14 (Integrability of Jacobi fields). *Let (M, g) be a Riemannian manifold, $\gamma : I \rightarrow M$ a geodesic in M . If M is complete², or I is a compact interval, then every Jacobi field along γ is the variation field of a variation of γ through geodesics.*

Proof. Let J be a Jacobi field of γ . Without loss of generality, we may assume that $0 \in I$ and consider $\gamma(0) = p$ and $\frac{d\gamma}{dt}(0) = v$. Remark that under these assumptions, $\gamma(t) = \exp_p(tv)$ for all $t \in I$.

Choose a smooth curve $\sigma : (-\epsilon, \epsilon) \rightarrow M$ and a smooth vector field V along σ such that

$$\begin{aligned} \sigma(0) &= p & V(0) &= v \\ \frac{d\sigma}{ds}(0) &= J_0 & \frac{DV}{ds}(0) &= \frac{DJ}{dt}(0). \end{aligned}$$

We will show that

$$F(s, t) = \exp_{\sigma(s)}(tV(s)) \tag{4.15}$$

is a variation of γ . If M is complete, then F is defined for all $(-\epsilon, \epsilon) \times I$. If I is compact, since $\mathcal{E}_p$ is an open set of $T_p M$ that contains the compact set $\{(p, tv) \in TM : t \in I\}$

²See Appendix B for more details on the relation between geodesically completeness and topological completeness

guarantees that there is some $\delta > 0$ such that F is defined on $(-\delta, \delta) \times I$. By construction, F is continuous and

$$F(0, t) = \exp_{\sigma(0)}(tV(0)) = \exp_p(tv) = \gamma(t)$$

and

$$F(s, 0) = \exp_{\sigma(s)}(0V(s)) = \sigma(s).$$

So F is a variation of γ .

The properties of the exponential map guarantee that F is a variation through geodesics, and therefore, its variation field $W(t) = \partial_s F(0, t)$ is a Jacobi field along γ . Remark that

$$W(0) = \partial_s F(0, t)|_{s=0} = \frac{d\sigma}{dt}(0) = J_0,$$

thus if $\frac{DW}{dt}(0) = \frac{DJ}{dt}(0)$, by uniqueness of Jacobi fields, $W = J$.

The main curves $F_s(t)$ are geodesics with initial vector fields $V(s)$, so $\partial_t F(s, 0) = V(s)$. By Lemma A.14, $\frac{D}{dt}\partial_s F(s, t) = \frac{D}{ds}\partial_t F(s, t)$ and by substituting for W and V we have

$$\frac{DW}{dt}(0) = \frac{D}{dt}\partial_s F(0, 0) = \frac{D}{ds}\partial_t F(0, 0) = \frac{DV}{ds}(0) = \frac{DJ}{dt}(0).$$

□

Corollary 4.14.1. *Jacobi fields are determined by their values at the endpoints of the geodesic.*

As it will become clear in Theorem 4.18 and Theorem 4.20, we will look at the cases when a Jacobi field vanishes, specifically at two distinct points. The next corollary gives us a formula for a variation through geodesics when a Jacobi field vanishes at a point, which we can then use to obtain an explicit formula of the fields.

Corollary 4.14.2. *Let (M, g) be a Riemannian manifold, I an interval of $\mathbb{R}$ containing 0, and $\gamma : I \rightarrow M$ a geodesic. Suppose J is a Jacobi field such that $J_0 = 0$. If M is complete, or I is compact, then J is the variation field of the variation through geodesics*

$$F(s, t) = \exp_p(t(v + sw)) \tag{4.16}$$

where $\gamma(0) = p$, $v = \frac{d\gamma}{dt}$ and $w = \frac{DJ}{dt}(0)$.

Proof. Take the variation $F(s, t) = \exp_{\sigma(s)}(tV(s))$ from the proof above and take $\sigma(s) \equiv p$ and $V(s) = v + sw$. □

Proposition 4.15 (Jacobi fields vanishing at a point). *Let (M, g) be a Riemannian manifold and $p \in M$. Consider a geodesic, $\gamma : I \rightarrow M$ with $0 \in I$ and $\gamma(0) = p$. For every $w \in T_p M$, the Jacobi field J along γ defined by the pair $(J_0 = 0, \frac{DJ}{dt}(0))$ is given by*

$$J_t = d(\exp_p)_{tv} tw \tag{4.17}$$

where $v = \frac{d\gamma}{dt}(0)$ and $tw \in T_{tv}(T_p M) \cong T_p M$.

If $\{x^i\}$ are normal coordinates on a normal neighborhood of p containing the image of γ then J is given by the formula

$$J_t = \sum_i t w^i \partial_i|_{\gamma(t)}, \quad (4.18)$$

where $\sum_i w^i \partial_i|_p$ is the normal coordinate representation of w .

Proof. By restricting J to any compact interval containing 0, then it is a variation field of variation through geodesics of the form $F(s, t) = \exp_p(t(v + sw))$.

$$\begin{aligned} \partial_s F(s, t) &= \partial_s \exp_p(t(v + sw)) = d(\exp_p(t(v + sw))) tw \\ J_t &= \partial_s F(0, t) = d(\exp_p)_{tv} tw \end{aligned}$$

Because every t in the domain of γ is contained in some such compact interval, the formula holds for all such t .

In normal coordinates, the exponential map is the identity, so F can be written explicitly as

$$\left((t(v^1 + sw^1)), \dots, (t(v^n + sw^n)) \right)$$

and differentiating with respect to s at $s = 0$, we see that the variation field J is of the desired form. $\square$

What is of importance to us is looking at when a Jacobi field along a geodesic γ vanishes at two points on γ . These points, called *conjugate points*, will be shown in the upcoming theorem to be the critical points of the exponential map defined by γ . Their existence plays a crucial role in determining the nullity and the Morse index of the Hessian

Definition 4.16. Two points $p = \gamma(a)$ and $q = \gamma(b)$ in M are *conjugate along γ* , for some geodesic $\gamma : I \rightarrow M$, if there exists a non-zero Jacobi field along γ which vanishes for $t = a$ and $t = b$. If either a or b is in the interior of I , then we say that p and q are *internal conjugate points*. The *multiplicity* of p and q as conjugate points is equal to the dimension of the vector space consisting of all such Jacobi fields.

Remark 4.3.1. From the definition, we know that conjugate points occur along a geodesic $\gamma(t)$ where non-zero Jacobi fields vanish. This means that one can find a family of geodesic segments that start at p and end around q .

Following from this definition and the previous proposition, it is no surprise that conjugate points are the critical points of the exponential map.

Theorem 4.17. Let (M, g) be a Riemannian manifold, $p \in M$ and $v \in \mathcal{E}_p \subseteq T_p M$. The point $\exp_p(v)$ is conjugate to p along the geodesic $\gamma_v : [0, 1] \rightarrow M$ from p to $\exp_p(v)$ if and only if the map $\exp_v$ is critical at v .

Proof. ‘ $\Leftarrow$ ’ Suppose $\exp_p$ is critical at $v \in T_p M$. Then $d(\exp_p)_v(w) = 0$ for some non-zero $w \in T_v(T_p M)$. Let F be the variation field through geodesics defined as in Equation 4.16 and the variation field $J_t = \partial_s F(0, t)$. Then

$$J_1 = \partial_s F(0, 1) = \frac{\partial}{\partial s} \exp_p(v + sw)|_{s=0} = d(\exp_p)_v(w) = 0.$$

Since $w = \frac{DJ}{dt}(0)$, J is non-trivial and vanishes at $t = 0$ and $t = 1$. Hence p and $\exp_p(v)$ are conjugate along γ_v .

‘ $\Rightarrow$ ’ If p and $\exp_p(v)$ are conjugate along a geodesic γ_v , then there is a non-trivial Jacobi field J such that $J_0 = 0$ and $J_1 = 0$, by Corollary 4.14.2 J is the variation field of a variation

$$F(s, t) = \exp_p(t(v + sw))$$

with $w = \frac{DJ}{dt}(0) \in T_p M$. By the previous paragraph we have that $d(\exp_p)_v(w) = J_1 = 0$, so $\exp_p$ is critical at v . □

Due to the close relation between geodesics and the exponential map, we have the following corollary.

Corollary 4.17.1. *Let $p \in M$. Then for almost all $q \in M$, p is not conjugate to q along any geodesics.*

Proof. Follows from Sard’s theorem. □

Finally, the following theorem shows the essential role that Jacobi fields and the conjugate points along a geodesic play in determining the nullity of $\mathcal{E}_{**}$.

Theorem 4.18 (Nullity is the multiplicity of endpoints). *A vector field $V \in T_\gamma \mathcal{P}M$ belongs to the null-space of $\mathcal{E}_{**}$ if and only if it is a Jacobi field. Hence $\mathcal{E}_{**}$ is degenerate if and only if the endpoints p and q are conjugate along γ . The nullity of $\mathcal{E}_{**}$ is equal to the multiplicity of p and q as conjugate points.*

The following proof follows the same outline as the proof of Theorem 3.7.

Proof. If J is a Jacobi field which vanishes at p and q , then $J \in T_\gamma \mathcal{P}M$. The second variation formula states that for any $W \in T_\gamma \mathcal{P}M$

$$\mathcal{E}_{**}(J, W) = - \sum_i \langle W_t, 0 \rangle_g - \int_0^1 \langle W, 0 \rangle_g dt = 0. \quad (4.19)$$

Hence $J \in \text{Nul}(\mathcal{E}_{**})$.

Conversely, suppose $V \in \text{Nul}(\mathcal{E}_{**})$ and choose a partition $0 = t_0 < t_1 < \dots < t_k = 1$ of $[0, 1]$ so that $V|_{[t_{i-1}, t_i]}$ is smooth for each i . Let $f : [0, 1] \rightarrow [0, 1]$ be a smooth function which vanishes from the parameter values $\{t_i\}$ and is positive otherwise, and consider

$$W(t) = f(t) \left(\frac{D^2}{dt^2} V + R \left(V, \frac{d\gamma}{dt}, \frac{d\gamma}{dt} \right) \right).$$

Then

$$\mathcal{E}_{**}(V, W) = \sum 0 + \int_0^1 f(t) \left\| \frac{D^2}{dt^2} V + R \left(V, \frac{d\gamma}{dt}, \frac{d\gamma}{dt} \right) \right\|_g^2 dt = 0$$

and each $V|_{[t_{j-1}, t_j]}$ is a Jacobian for each i .

Now, consider a vector field $\widetilde{W}^2 \in T_\gamma \mathcal{P}M$ such that for $i = 1, \dots, k-1$

$$\widetilde{W}_{t_i} = \Delta_i \frac{DV}{dt}.$$

Then

$$\mathcal{E}_{**}(V, \widetilde{W}) = \sum_{i=1}^{k-1} \left\| \Delta_i \frac{DV}{dt} \right\|_g^2 + \int_0^1 0 dt = 0.$$

Hence $\frac{DV}{dt}$ has no jump. Since a solution J of the Jacobi equation is completely determined by the pair of vectors $(J_{t_i}, \frac{DJ}{dt}(t_i))$. Thus it follows that the k Jacobi fields $V|_{[t_{i-1}, t_i]}$ for $i = 1, \dots, k$ fit together to give a smooth Jacobi field. $\square$

Remark 4.3.2. As $\dim \ker J_\gamma = 2n$, one can see that the nullity $\nu(\mathcal{E}_{**})$ is finite.

Proposition 4.19. *The nullity satisfies $0 \leq \nu(\mathcal{E}_{**}) < n$, where $\dim M = n$.*

Proof. Remark that the space of Jacobi fields that vanish at $t = 0$, $\ker J_{\gamma,0}$, has dimension n :

$$\ker J_{\gamma,0} \xrightarrow{\sim} \{0\} \times T_{\gamma(0)}M \cong T_{\gamma(0)}M. \quad (4.20)$$

Thus, we have that $\nu(\mathcal{E}_{**}) \leq n$. To demonstrate that strict inequality holds, we will construct a Jacobi field that vanishes at $t = 0$ and $t = 1$. Consider $J_t = t \frac{d\gamma}{dt}$, since γ is a geodesic we have

$$\frac{DJ}{dt} = \frac{d\gamma}{dt} + t \frac{D}{dt} \frac{d\gamma}{dt} = \frac{d\gamma}{dt} \quad (4.21)$$

and

$$\frac{D^2 J}{dt^2} = 0.$$

Furthermore, since the curvature tensor is skew-symmetric in the first two variables

$$R\left(J, \frac{d\gamma}{dt}, \frac{d\gamma}{dt}\right) = t R\left(\frac{d\gamma}{dt}, \frac{d\gamma}{dt}, \frac{d\gamma}{dt}\right) = 0.$$

Thus J is a Jacobi field but $J_1 = \frac{d\gamma}{dt} \neq 0$. $\square$

4.4. Index Theorem and the Topology of the Path Space

From Theorem 4.18, we have that the endpoints of a non-degenerate geodesic γ are not conjugate along γ . In this section, it will be shown that the multiplicity of *interior conjugate points* defines the Morse index.

Theorem 4.20 (Morse Index is the multiplicity of interior points). *The Morse index of $\mathcal{E}_{**}$ at a (non-degenerate) geodesic $\gamma \in \mathcal{PM}$ is equal to the number of points $\gamma(t)$ with $0 < t < 1$, such that $\gamma(t)$ is conjugate to $\gamma(0)$ along γ ; each such conjugate point being counted with its multiplicity. This Morse index $\mu_M(\gamma)$ is always finite.*

To prove the above theorem, we will estimate $\mu_M(\gamma)$ by splitting the vector space $T_\gamma \mathcal{PM}$ into two orthogonal subspaces.

To this end, we consider geodesically convex open sets.

Definition 4.21. An open set $U \subseteq M$ is said to be geodesically convex if any two points of U are joined by a unique minimal geodesic.³

Consider a partition $0 = t_0 < t_1 < \dots < t_k = 1$ such that each segment $\gamma|_{[t_{i-1}, t_i]}$ lies within a geodesically convex open set U , and each $\gamma|_{[t_{i-1}, t_i]}$ is minimal. Define the finite-dimensional vector subspace consisting of *broken Jacobi fields along γ* as

$$T_\gamma \mathcal{P}M(t_0, t_1, \dots, t_k) := \left\{ W \in T_\gamma \mathcal{P}M : W|_{[t_{i-1}, t_i]} \in \ker J_{\gamma|_{[t_{i-1}, t_i]}} \text{ for each } i \right\}, \quad (4.22)$$

and a second space

$$T' : \{ W \in T_\gamma \mathcal{P}M : W_{t_i} = 0 \quad i = 0, 1, \dots, k \}. \quad (4.23)$$

Lemma 4.22. *The tangent space of $\mathcal{P}M$ at a geodesic γ is the direct sum of the space of broken Jacobi fields along γ and the vector space T' :*

$$T_\gamma \mathcal{P}M = T_\gamma \mathcal{P}M(t_0, t_1, \dots, t_k) \oplus T'. \quad (4.24)$$

*These two subspaces are mutually perpendicular with respect to the inner product on $\mathcal{E}_{**}$. Furthermore, $\mathcal{E}_{**}$ restricted to T' is positive definite.*

Proof. • **Direct Sum:**

Given any vector field $W \in T_\gamma \mathcal{P}M$, let V denote the unique *broken Jacobi field* in $T_\gamma \mathcal{P}M(t_0, t_1, \dots, t_k)$ such that $V_{t_i} = W_{t_i}$ for $i = 0, \dots, k$. It follows from Corollary 4.14.1 that V exists and is unique. Clearly, $W - V \in T'$, thus the two subspaces generate $T_\gamma \mathcal{P}M$, and have only the zero vector in common.

• **Orthogonality:**

If $W \in T_\gamma \mathcal{P}M(t_0, t_1, \dots, t_k)$ and $V \in T'$, then the second variation formula takes the form

$$\mathcal{E}_{**}(W, V) = - \sum_i \left\langle V_t, \Delta_i \frac{D}{dt} V \right\rangle_g - \int_0^1 \langle V, 0 \rangle_g dt = 0. \quad (4.25)$$

Thus, the two subspaces are mutually perpendicular with respect to $\mathcal{E}_{**}$.

• **Positive-definite:**

For any $W \in T_\gamma \mathcal{P}M$, the hessian $\mathcal{E}_{**}$ can be defined by a 1-parameter proper variation of γ , $F : (-\epsilon, \epsilon) \times [0, 1] \rightarrow M$ - see Remark 4.2.1. If $V \in T'$, then we may assume that F is chosen such that for $i = 0, \dots, k$ $F_s(t_i) = \gamma(t_i)$.

★ **Claim:** $\mathcal{E}_{**}(V, V) \geq 0$ for $V \in T'$

Each $F_s \in \mathcal{P}M$ is a piecewise smooth path from $\gamma(t_{i-1})$ to $\gamma(t_i)$ for $i = 0, 1, \dots, k$. But each $\gamma|_{[t_{i-1}, t_i]}$ is a minimal geodesic, and therefore has smaller energy than any other path between the endpoints. Thus $\mathcal{E}(F_s) \geq \mathcal{E}(\gamma) = \mathcal{E}(F_0)$, and

$$\left. \frac{d^2(\mathcal{E}(F_s))}{ds^2} \right|_{s=0} \geq 0.$$

³See Appendix B for more details on the relation between geodesically completeness and topological completeness

★ **Claim:** $\mathcal{E}_{**}(V, V) > 0$ for $V \in T' \setminus \{0\}$

Suppose that $\mathcal{E}_{**}(V, V) = 0$. We know that for any $W \in T_\gamma \mathcal{P}M(t_0, t_1, \dots, t_k)$ $\mathcal{E}_{**}(V,) = 0$, and for any $\tilde{V} \in T'$ the inequality

$$0 \leq \mathcal{E}_{**}(V + c\tilde{V}, V + c\tilde{V}) = 2c\mathcal{E}_{**}(\tilde{V}, V) + c^2\mathcal{E}_{**}(\tilde{V}, \tilde{V})$$

for any c implies $\mathcal{E}_{**}(\tilde{V}, V) = 0$. Therefore, $V \in \text{Nul}(\mathcal{E}_{**}) = \ker J_\gamma$. However $T' \cap \ker J_\gamma = \{0\}$, so $V = 0$.

Finally, we have that $\mathcal{E}_{**}$ is positive definite on T' . □

Corollary 4.22.1. *The Morse index (the nullity) of $\mathcal{E}_{**}$ is equal to the Morse index (nullity) of $\mathcal{E}_{**}$ restricted to the space $T_\gamma \mathcal{P}M(t_0, t_1, \dots, t_k)$ of broken Jacobi fields. In particular, the index $\mu_M(\gamma)$ is always finite.*

The proof of Theorem 4.20 follows the following steps, for further details we refer to *Morse Theory*, Chapter 15: The Index Theorem [22]. We begin by considering γ_τ , the restriction of a geodesic $\gamma : [0, 1] \rightarrow M$ to $[0, \tau] \subset [0, 1]$, with $\mathcal{E}_g^\tau = \mathcal{E}_g(\gamma_\tau)$ and $\mu_M(\gamma_\tau)$. By the splitting of $T_\gamma \mathcal{P}M$ and the above corollary, we solely look at the space of the broken geodesics. We calculate the ‘broken’ Morse indices $\mu_M(\gamma_\tau)$ to then obtain $\mu_M(\gamma_1) = \mu_M(\gamma)$.

This is done by first showing some properties of $\mu_M(\gamma_\tau)$, i.e Claim 1 to 3, and then giving the relation between the Morse index and the nullity.

- **Claim 1:** $\mu_M(\gamma_\tau)$ is a monotone function of τ .
- **Claim 2 :** $\mu_M(\gamma_\tau) = 0$ for some small value of τ .
- **Claim 3 :** For all sufficiently small $\epsilon > 0$: $\mu_M(\gamma_{(\tau-\epsilon)}) = \mu_M(\gamma_\tau)$.
- **Claim 4 :** Let ν^τ be the nullity of $\mathcal{E}_{**}^\tau$. For sufficiently small $\epsilon > 0$, $(\mu_M(\gamma_{(\tau+\epsilon)})) = \mu_M(\gamma_\tau) + \nu^\tau$.

Corollary 4.22.2 (of Theorem 4.20). *A geodesic segment $\gamma : [0, 1] \rightarrow M$ can contain only finitely many points which are conjugate to $\gamma(0)$ along γ .*

4.5. Topology of the Path Space Via Morse Theory

Finally, we can define the topology of the Path space $\mathcal{P}M = \mathcal{P}M(p, q)$. Throughout this section, we refer to the proofs of *Morse Theory*, Chapter 16.1 [22].

Firstly, we define the metric that will induce the desired topology on $\mathcal{P}M$. For $\omega, \omega' \in \mathcal{P}M$, the distance $d(\omega, \omega')$ is defined to be

$$\max_{0 \leq t \leq 1} \rho(\omega(t), \omega'(t)) + \sqrt{\int_0^1 \left\| \frac{ds}{dt} - \frac{ds'}{dt} \right\|_g^2 dt} \quad (4.26)$$

where $s(t)$ and $s'(t)$ are the arc-lengths of ω and ω' , respectively, and ρ is the metric on (M, g) .

Secondly, we follow the example of chapter 2. That is, given $c > 0$, we consider the level set $\mathcal{P}M^c := \mathcal{E}_g^{-1}([0, c]) \subseteq \mathcal{P}M$ with interior $(\mathcal{P}M^c)^\circ := \mathcal{E}_g^{-1}([0, c))$. The topology of $\mathcal{P}M^c$ is constructed via a finite-dimensional approximation via ‘broken geodesics’.

Consider a partition $0 = t_0 < t_1 < \dots < t_k = 1$ of $[0, 1]$. Let $\mathcal{P}M(t_0, \dots, t_k)$ be the space of ‘broken geodesic’:

$$\mathcal{P}M(t_0, \dots, t_k) = \left\{ \omega \in \mathcal{P}M : \omega|_{[t_{i-1}, t_i]} \text{ is a geodesic for each } i \right\}, \quad (4.27)$$

with level sets $\mathcal{P}M(t_0, \dots, t_k)^c = \mathcal{P}M^c \cap \mathcal{P}M(t_0, \dots, t_k)$ and interiors $(\mathcal{P}M(t_0, \dots, t_k)^c)^\circ = (\mathcal{P}M^c)^\circ \cap \mathcal{P}M(t_0, \dots, t_k)$.

Lemma 4.23. *Let M be a complete Riemannian manifold, let c be a fixed positive number such that $\mathcal{P}M^c \neq \emptyset$. Then for all sufficiently fine partitions $(t_0, \dots, t_k)$ of $[0, 1]$ the set $(\mathcal{P}M(t_0, \dots, t_k)^c)^\circ$ can be given the structure of a smooth finite-dimensional manifold in a natural way.*

Let $\mathcal{E}'_g : (\mathcal{P}M(t_0, \dots, t_k)^c)^\circ \rightarrow \mathbb{R}$ be the restriction of $\mathcal{E}_g$ to $(\mathcal{P}M(t_0, \dots, t_k)^c)^\circ$.

Theorem 4.24. *The function $\mathcal{E}'_g : (\mathcal{P}M(t_0, \dots, t_k)^c)^\circ \rightarrow \mathbb{R}$ is smooth. Furthermore, for each $a < c$, the set*

$$((\mathcal{P}M(t_0, \dots, t_k)^c)^\circ)^a = (\mathcal{E}'_g)^{-1}[0, a]$$

*is compact and is a deformation retract of the corresponding set $\mathcal{P}M^a$. The critical points of $\mathcal{E}'_g$ are the same as the critical points of $\mathcal{E}_g$ in $(\mathcal{P}M^a)^\circ$: namely, the unbroken geodesics from p to q of length less than $\sqrt{c}$. The index (or the nullity) of $\mathcal{E}'_{**}$ at each such critical point γ is equal to the index (or nullity) of $\mathcal{E}_{**}$ at γ .*

Thus the finite dimensional manifold $(\mathcal{P}M(t_0, \dots, t_k)^c)^\circ$ provides a faithful model for the infinite dimensional Path space $(\mathcal{P}M^c)^\circ$.

Theorem 4.25. *Let M be a complete Riemannian manifold and let $p, q \in M$ be two points which are not conjugate along any geodesic of length $L_g(\gamma) \leq \sqrt{a}$. Then $\mathcal{P}M^a$ has the homotopy type of a finite CW-complex, with one-cell of dimension $\mu_M(\gamma)$ for each geodesic $\gamma \in \mathcal{P}M^a$.*

Proof. Follows from Theorem 2.10 and the previous theorem. □

Remark 4.5.1. In particular, $\mathcal{P}M^a$ contains only finitely many geodesics

To complete the study of the topology of $\mathcal{P}M$, we relate it to the Path space in homotopy theory.

Theorem 4.26. *Let $\mathcal{P}M^\star$ be the space of all continuous paths $\omega : [0, 1] \rightarrow M$ from p to q in the compact open topology, defined by the metric $d^\star(\omega, \omega') = \max_t \rho(\omega, \omega')$. The natural inclusion $i : \mathcal{P}M \hookrightarrow \mathcal{P}M^\star$.*

We refer to the proof in *Morse Theory*, Theorem 17.1. Since $\mathcal{P}M^\star$ has the homotopy type of a CW-complex, so does our $\mathcal{P}M$.

Theorem 4.27 (Fundamental Theorem of Morse Theory). *Let (M, g) be a complete Riemannian manifold, $p, q \in M$ be two points which are not conjugate along any geodesic. Then $\mathcal{PM}(p, q)$ has the homotopy type of a countable CW-complex which contains one cell of dimension $\mu_M(\gamma)$ for each geodesic γ between p and q .*

Proof. The proof is analogous to Theorem 2.10. □

4.6. Applications

Spheres

Proposition 4.28. *Any antipodal points on the unit sphere S^n are conjugate with multiplicity $n - 1$. Hence, great circles are not minimal geodesics.*

Proof. Suppose p and q are antipodal points on S^n , and let γ be a great circle arc from p to q . Since every Jacobi field can be obtained by a proper variation of γ through geodesics. By rotating γ and keeping the endpoints fixed, one obtains a proper variation of γ , which will give a Jacobi field. Thus, rotating γ in $n - 1$ different directions gives $n - 1$ distinct Jacobi fields. Thus p and q are conjugate along γ with multiplicity $n - 1$. □

Proposition 4.29. *The Path space $\mathcal{PS}^n(p, q)$ (for non-conjugate points) has the homotopy type of a CW-complex with one cell in the dimension $0, n - 1, 2(n - 1), 3(n - 1), \dots$*

Proof. Since p and q are non-conjugate, it means that they are not antipodal $p \neq q'$ and $q \neq p'$ (for p' the antipodal point of p and q' the antipodal point of q). Then there are infinitely many geodesics from p to q . Let γ_0 denote the short great circle arc from p to q and γ_1 the long great circle arc $pq'p'q$, γ_2 the long great circle arc $pqp'q'pq$, and so on. The index $\lambda(\gamma_k) = \mu_1 + \dots + \mu_k = k(n - 1)$. □

Corollary 4.29.1. *Let M have the homotopy type of S^n , for $n > 2$. Then any two non-conjugate points of M are joined by infinitely many geodesics.*

Remark 4.6.1. If M is any complete manifold which is not contractible, then any two non-conjugate points of M are joined by infinitely many geodesics.

Curvature and Topology

Proposition 4.30. *On flat manifolds, every geodesic is minimal.*

Proof. Let M be a flat manifold, that is $R \equiv 0$. Then the Jacobi equation simplifies to $\frac{D^2 J}{dt^2} = 0$. For a parallel frame $\{P^i\}$ and f^i smooth functions, a Jacobi vector field has the form $J_t = \sum_i f^i(t)P_t^i$. By the Jacobi equation $0 = \frac{D^2 J}{dt^2} = \frac{\partial^2 f}{\partial t^2}$, so every Jacobi field is trivial, $\ker J_\gamma = \{0\}$. Thus, there are no conjugate points and $\mathcal{E}_{**}$ is non-degenerate. □

Proposition 4.30, can be generalized for a manifold with non-positive curvature.

Lemma 4.31. *Suppose $\langle R(A, B, A), B \rangle_g \leq 0$ for every pair of vectors (A, B) in $T_p M$ and for all $p \in M$. Then no two points of M are conjugate along any geodesic.*

Proof. Let γ be a geodesic with velocity vector field V and let J , be a Jacobi field along γ . Then $\frac{D^2 J}{dt^2} = -\langle R(V, J), V \rangle_g \geq 0$ and

$$\frac{d}{dt} \left\langle \frac{DJ}{dt}, J \right\rangle_g = \left\langle \frac{D^2 J}{dt^2}, J \right\rangle_g + \left\| \frac{DJ}{dt} \right\|_g^2 \geq 0. \quad (4.28)$$

Thus, the function $\left\langle \frac{DJ}{dt}, J \right\rangle_g$ is monotonically increasing and strictly so if $\frac{DJ}{dt} \neq 0$. If J vanishes both at 0 and $t_0 > 0$, then the function $\left\langle \frac{DJ}{dt}, J \right\rangle_g$ also vanishes at 0 and $t_0 > 0$, and hence must vanish on $[0, t_0]$. This implies that $J_0 = \frac{DJ}{dt}(0) = 0$ so J is identically zero. $\square$

Definition 4.32. If A and B are orthogonal unit vectors at p , then we call $\langle R(A, B, A), B \rangle_g$ the *sectional curvature* determined by A and B .

Theorem 4.33 (Cartan). *Suppose M is a simply connected complete Riemannian manifold, and the sectional curvature $\langle R(A, B, A), B \rangle_g \leq 0$ everywhere. Then any two points of M are joined by a unique geodesic. Furthermore, M is diffeomorphic to $\mathbb{R}^n$.*

Proof. Since there are no conjugate points, it follows from the index theorem that every geodesic from p to q has index $\mu_M = 0$. Thus $\mathcal{PM}(p, q)$ has the homotopy type of a 0-dimension CW-complex with one vertex for each geodesic. The hypothesis that M is simply connected implies that $\mathcal{PM}(p, q)$ is connected. Since a connected 0-dimensional CW-complex must consist of a single point, it follows that there is precisely one geodesic from p to q . Therefore the exponential map $\exp_p : T_p M \rightarrow M$ is one-to-one and surjective. Furthermore, since $\exp_p$ is non-critical everywhere, it must be a local diffeomorphism. Thus, it is a global diffeomorphism. $\square$

Corollary 4.33.1. *If M is complete with non-positive sectional curvature, then the homotopy groups $\pi_i(M) = 0$ for $i > 1$ and $\pi_1(M)$ contains no elements of finite order other than the identity.*

5. Understanding the Path Space: Sobolev Spaces

This chapter references *Sobolev Spaces* [2], *The differential topology of loop spaces*[28], and *Functional Analysis, Sobolev Spaces and Partial Differential Equations* [5].

In the previous chapter, we mentioned that the Path Space was an infinite-dimensional Hilbert manifold. In this chapter, we will show that this is the case.

Firstly, we shall explain what an infinite-dimensional manifold is. Unlike finite-dimensional manifolds, which are modeled on $\mathbb{R}^n$, infinite-dimensional manifolds are modeled on *Banach spaces*: i.e., complete normed spaces $(X, \|\cdot\| : X \rightarrow \mathbb{R})$. Some examples of Banach spaces are the Euclidean space $\mathbb{R}^n$ with the Euclidean norm $\|x\| = \sqrt{\sum_{i=1}^n (x^i)^2}$ and the function space $L^p(S)$ of all Lebesgue p -integrable functions on a measurable set S with the p -norm

$$\|f\|_{L^p} = \begin{cases} (\int_S |f|^p)^{\frac{1}{p}} & p \in [1, \infty) \\ \inf\{C \in \mathbb{R}_{>0} : |f(s)| < C \text{ a.e } s\} & p = \infty \end{cases}.$$

The definition of a Banach manifold is equivalent to the well-known definition for finite-dimensional manifolds.

Definition 5.1. A *Banach Manifold* M is Hausdorff topological space endowed with a maximal atlas $\{U_i, \varphi_i\}$, with $\varphi_i : U_i \xrightarrow{\sim} V_i$ where

1. $M = \cup_{i \in I} U_i$.
2. Each V_i is a Banach space.
3. For each i and j , $\varphi_i \circ \varphi_j^{-1} : \varphi_j(U_i \cap U_j) \xrightarrow{o} V_j \rightarrow \varphi_i(U_i \cap U_j) \xrightarrow{o} V_i$ is smooth map between Banach Spaces

Remark 5.0.1. In general, the manifolds need not be modeled on a *single* locally convex space. However, since the transition functions are diffeomorphisms, the spaces are isomorphic on connected components.

Remark 5.0.2. Every smooth n -manifold is a Banach manifold: the Banach spaces V_i are all $\mathbb{R}^n$.

A subtype of Banach spaces that have been conceptualized as a generalization of $\mathbb{R}^n$ are the *Hilbert spaces*. Hilbert spaces are complete inner product spaces $(H, \langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{R})$ such that they are complete metric spaces with respect to the distance function induced by the inner product. An example that is $L^2(S)$ with the inner product $\langle f, g \rangle = \int_S fg$.

We shall see that the Path Space is modeled on *Sobolev Spaces*. Sobolev spaces are spaces of Lebesgue p -integrable maps between finite-dimensional manifolds endowed with a norm consisting of the sum of the L^p norms of *weak derivatives*. It will be in later chapters, when considering minimal sub-manifolds and harmonic maps, that Sobolev Spaces will appear again and will be the higher-dimensional variant of the Path Space.

5.1. Weak Derivatives and Sobolev Spaces

A *weak derivative* is a generalization of *derivative* for integrable functions – not necessarily differentiable functions. It arises from *integration by parts*: i.e, for u, ϕ smooth functions

$$\int_a^b u(x)\phi'(x)dx = [u(x)\phi(x)]_a^b - \int_a^b u'(x)\phi(x)dx.$$

A *weak derivative* of $u \in L^1[a, b]$, is a function $v \in L^1[a, b]$ such that

$$\int_a^b u(x)\phi'(x)dx = - \int_a^b v(x)\phi(x)dx. \quad (5.1)$$

for all infinitely differentiable functions ϕ such that $\phi(a) = \phi(b) = 0$. For n -dimensions, consider locally integrable functions $u, v \in L^1_{\text{loc}}(U)$, on some open $U \subseteq \mathbb{R}^n$, and a multi-index α . We say that $D^\alpha u$ is the α th - *weak derivative* of u if for all compactly supported $\phi \in C_c^\infty(U)$,

$$\int_U u D^\alpha \phi = (-1)^{|\alpha|} \int_U D^\alpha u \phi \quad (5.2)$$

and

$$D^\alpha u = \frac{\partial^{|\alpha|} u}{\partial^{\alpha_1} x_1 \dots \partial^{\alpha_n} x_n} \quad (5.3)$$

There are some nice properties of functions and their weak derivatives that will aid in the construction of Sobolev spaces.

Lemma 5.2. 1. If ν and v are weak derivatives of a function u , then $\nu = v$ almost everywhere.

2. If f' is the weak derivative of the zero function, then $f = c$ for some constant $c \in \mathbb{R}$.

Lemma 5.3. Let $-\infty < a < x_0 < b < \infty$, and $f, g \in L^1_{\text{loc}}((a, b))$. Then $f' = g$ is a weak derivative of f if and only if there exists $c \in \mathbb{R}$ such that

$$f(x) = c + \int_{x_0}^x g(y)dy \quad (\text{a.e. } x \in (a, b)). \quad (5.4)$$

Proof. ‘ $\Rightarrow$ ’ Let f' be the weak derivative of g , and consider the function $h(x) = f(x) - \int_{x_0}^x g(y)dy$. Then $h' = f' - g = 0$, thus by item 2 of Lemma 5.2 $h(x) = c$ for some $c \in \mathbb{R}$.

‘ $\Leftarrow$ ’ Consider $\varphi \in C_c^\infty(a, b)$ and $f(x) = c + \int_{x_0}^x g(y)dy$. Then

$$\begin{aligned} \int_a^b \varphi'(x)f(x)dx &= \int_a^b \varphi'(x) \left(c + \int_{x_0}^x g(y)dy \right) \\ &= c \int_a^b \varphi'(x) + \iint_{x_0 < y < x < b} \varphi'(x)g(y)dy \\ &= \iint_{x_0 < y < x < b} \varphi'(x)g(y)dy = - \int_a^b \varphi(y)g(y)dy, \end{aligned}$$

and f' is a weak derivative of g . □

Remark 5.1.1. Observe that the right-hand-side of Equation 5.4 is continuous as a function of x .

Let $k \in \mathbb{N}$, $1 \leq p \leq \infty$ and $U \stackrel{o}{\subseteq} \mathbb{R}^n$. The *Sobolev Space* $W^{k,p}(U)$ is defined as the set of all Lebesgue p -integrable functions such that the α th-weak derivatives, for $|\alpha| \leq k$, are also p -integrable:

$$W^{k,p}(U) = \{u \in L^p(U) : \forall |\alpha| \leq k, D^\alpha u \in L^p(U)\}. \quad (5.5)$$

The Sobolev norm depends on the L^p norm and on the value of p .

- For $p \in [1, \infty)$ the $W^{k,p}$ -norm of a class $u \in W^{k,p}(U)$ is defined as

$$\|u\|_{k,p} = \left(\int_U \sum_{|\alpha| \leq k} |D^\alpha u|^p d\mathbf{x} \right)^{\frac{1}{p}} = \left(\sum_{|\alpha| \leq k} \int_U \|D^\alpha u\|^p d\mathbf{x} \right)^{\frac{1}{p}}, \quad (5.6)$$

which is equivalent to

$$\|u\|_{W^{k,p}} = \|D^k u\|_{L^p} + \|u\|_{L^p}. \quad (5.7)$$

- For $p = \infty$ the $W^{k,\infty}$ -norm of a class $u \in W^{k,\infty}(U)$ is defined as

$$\|u\|_{W^{k,\infty}} := \max_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty}. \quad (5.8)$$

Remark 5.1.2. Throughout this paper, we will omit the notation $[u]$ for a function of class $W^{k,p}$ and only use it when making it explicit that representatives are being used.

Proposition 5.4 (Properties of weak derivatives in Sobolev spaces). *Let $u, v \in W^{k,p}(U)$, $p \in [1, \infty]$ and $|\alpha| \leq k$*

- (i) $D^\alpha u \in W^{k-|\alpha|,p}(U)$ and for all multi-indices β such that $|\alpha| + |\beta| \leq k$

$$D^\beta (D^\alpha u) = D^\alpha (D^\beta u) = D^{\alpha+\beta} u.$$

- (ii) For $\lambda, \mu \in \mathbb{R}$ $\lambda u + \mu v \in W^{k,p}(U)$ and $D^\alpha (\lambda u + \mu v) = \lambda D^\alpha u + \mu D^\alpha v$.

- (iii) For an open subset, $\tilde{U} \stackrel{o}{\subseteq} U$, $W^{k,p}(\tilde{U}) \subseteq W^{k,p}(U)$.

(iv) If $\phi \in C_c^\infty(U)$ then $\phi u \in W^{k,p}(U)$ and $D^\alpha(\phi u) = \sum_{\beta: s.t. |\beta| \leq |\alpha|} \binom{\alpha}{\beta} D^\beta \phi D^{\alpha-\beta} u$.

It is not hard to see that for $p \in [1, \infty]$, $W^{k,p}(U)$ are Banach spaces. Furthermore, since L^2 is a Hilbert space, for any k , $W^{k,2}(U)$ are Hilbert spaces with inner product $\langle u, v \rangle_{W^{k,2}} = \langle u, v \rangle_{L^2} + \langle D^k u, D^k v \rangle_{L^2}$. These Hilbert spaces will be denoted by the symbol $H^k(U)$.

Via the canonical inclusion of

$$\phi \mapsto [\phi] : C^\infty(U) \hookrightarrow W^{k,p}(U), \quad (5.9)$$

$C^\infty(U)$ is embedded in the Sobolev space $W^{k,p}(U)$. In fact, $W^{k,p}(U)$ is the completion of $C^\infty(U)$ with norm $\|\cdot\|_{W^{k,p}}$, as formalized as a theorem below and proven in *Sobolev Spaces*, Theorem 3.17 (H = W) [2]. This foundational result allows us to simply consider smooth maps between manifolds throughout this thesis.

Theorem 5.5. For $1 \leq p < \infty$ and $k \in \mathbb{N}$, $\overline{C^\infty(U)}^{\|\cdot\|_{W^{k,p}}} = W^{k,p}(U)$

Another crucial theorem for this thesis is the *Sobolev Embedding Theorem*, which will be stated but not proved. The *Sobolev Embedding Theorem* states when a Sobolev space can be contained in other Sobolev spaces and other Banach Spaces, such as L^p and Hölder spaces.

Definition 5.6. A *Hölder Space* is the space of functions that satisfy the *Hölder condition*: that is, real (or complex) valued functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ such that there exist real constants $C, \alpha \in \mathbb{R}_{\geq 0}$ for which the inequality $|f(x) - f(y)| \leq C \|x - y\|^\alpha$ holds.

In symbols, it is,

$$C^{k,\alpha}(U) := \{f \in C^k(U) : f^{(i)} \text{ satisfy the Hölder condition } \forall i \leq k\}$$

for $U \subseteq \mathbb{R}^n$, $k \in \mathbb{Z}_{\geq 0}$ and $\alpha \in (0, 1]$. A Hölder Space $C^{k,\alpha}(U)$ has $C^{k,\alpha}$ - norm:

$$\|f\|_{C^{k,\alpha}} = \|f\|_{C^k} + \sum_{i \leq k} \|f^{(i)}\|_{C^{0,\alpha}},$$

where

$$\|f\|_{C^{0,\alpha}} = \sup_{x,y \in U, x \neq y} \frac{|f(x) - f(y)|}{\|x - y\|^\alpha}. \quad (5.10)$$

Remark 5.1.3. If a function satisfies the Hölder condition for $\alpha = 0$, then the function is bounded. For $\alpha > 0$, the condition implies that the function is uniformly continuous, while for $\alpha = 1$, the Hölder condition is the *Lipschitz condition*.

Theorem 5.7 (Sobolev Embedding Theorem). Consider the Sobolev space $W^{k,p}(U)$ with $U \subseteq \mathbb{R}^n$, $k \in \mathbb{Z}_{\geq 0}$ and $1 \leq p < \infty$.

1. If $k > l$, $p < n$ and $1 \leq p < q < \infty$ such that $\frac{1}{p} - \frac{k}{n} = \frac{1}{q} - \frac{l}{n}$, then $W^{k,p}(U) \subseteq W^{l,q}(U)$ and the embedding is continuous.

2. If $n < pk$ and $\frac{1}{p} - \frac{k}{n} = -\frac{r+\alpha}{n}$, or equivalently $r + \alpha = k - \frac{n}{p}$ with $\alpha \in (0, 1)$ then one has the embedding $W^{k,p}(U) \subseteq C^{r,\alpha}(U)$.

Remark 5.1.4. The second part of the Sobolev embedding theorem states that if $kp > n$, then the embedding criterion will hold for $r = 0$ and some $\alpha > 0$, which gives us that $W^{1,2}(\mathbb{R}) \subseteq C^0(\mathbb{R})$; however, for higher dimensions, some complications arise, such as

$$W^{2,2}(\mathbb{R}^2) \subseteq C^0(\mathbb{R}^2) \quad \text{but} \quad W^{1,2}(\mathbb{R}^2) \not\subseteq C^0(\mathbb{R}^2).$$

Item 2 of *Sobolev Embedding Theorem* for the case of the Sobolev Space $W^{1,2}([a, b])$ with real numbers $a < b$ can be shown via the known properties of the weak derivatives.

Theorem 5.8. $W^{1,2}([a, b])$ is compactly embedded in $C^0([a, b])$:

$$(W^{1,2}([a, b]), \|\cdot\|_{H^1}) \hookrightarrow (C^0([a, b]), \|\cdot\|_{\infty}) \quad (5.11)$$

Proof. • **Well-defined:**

Recall that we defined the Sobolev space as

$$W^{1,2}([a, b]) = \{u \in L^2([a, b]) : u' \in L^2([a, b])\}. \quad (5.12)$$

By Lemma 5.3, we can see that there exists a continuous representative for the class of u , and thus the mapping $W^{1,2}([a, b]) \hookrightarrow C^0([a, b])$ with $[u] \mapsto u$ is well-defined.

• **Continuity of embedding:**

Consider a class $u \in W^{1,2}([a, b])$ and choose a continuous representative. Fix $s, c \in [a, b]$ such that $\|u\|_{\infty} = |u(s)|$ and $\int_a^b u(x)dx = u(c)(b - a)$.

If $c < s$, then

$$\begin{aligned} \|u\|_{\infty} &\leq |u(s) - u(c)| + |u(c)| = |u(c)| + \int_a^b |u'(t)|dt \\ &\leq (b - a)^{-1} \int_a^b |u(t)|dt + \int_a^b |u'(t)|dt \\ &\leq (b - a)^{-\frac{1}{2}} \|u\|_{L^2} + (b - a)^{\frac{1}{2}} \|u'\|_{L^2} \\ &\leq \left((b - a)^{-\frac{1}{2}} + (b - a)^{\frac{1}{2}} \right) \|u\|_{1,2} \end{aligned}$$

• **Compactness of embedding:**

Recall that a linear operator $T : X \rightarrow Y$ between normed spaces is compact if for all bounded sequences $\{x_i\}$ of X , then $\{T(x_i)\}$ has a convergent subsequence.

To show this, we will use the Arzela-Ascoli Theorem, which states that if a sequence in C^0 is bounded and equicontinuous, then it has a uniformly convergent subsequence.

Let $\{u_i\}$ be a bounded sequence in $W^{1,2}([a, b])$, that is $\|u_i\|_{1,2} \leq M$ for some $M \in \mathbb{N}$. Then for every $\epsilon > 0$, there exists a $\delta = \frac{\epsilon^2}{M^2}$ such that $|y - x| < \delta$

$$|v_i(x) - v_i(y)| = \left| \int_x^y v_i'(t)dt \right| \leq \sqrt{|y - x|} \|v_i\|_{1,2} < \sqrt{\delta} M = \epsilon.$$

Thus, we have that $\{u_i\}$ is bounded and equicontinuous in $C^0([a, b])$, and thus has a convergent subsequence. □

There exists another embedding theorem of Sobolev spaces; this time, it states when a Sobolev space is compactly embedded in an L^p space.

Theorem 5.9 (Rellich–Kondrachov Embedding Theorem). *Let $U \subseteq \mathbb{R}^n$ be an open, bounded Lipschitz domain, and $1 \leq p < n$. Consider $p^* := \frac{np}{n-p}$. The Sobolev space $W^{1,p}(U)$ is continuously embedded in the $L^{p^*}(U)$ and is compactly embedded in $L^q(U)$ for every $1 \leq q < p^*$.*

Lastly, we can consider the Sobolev Spaces themselves as Banach Manifolds.

Definition 5.10 (Sobolev spaces as Banach Manifolds). For $U \stackrel{o}{\subseteq} \mathbb{R}^n$ and $k \in \mathbb{R}_{>0}$, $(W^{k,p}(U), \|\cdot\|_{W^{k,p}})$ is a Banach Manifold with maximal atlas given by the identity operator $\text{Id} : W^{k,p}(U) \rightarrow W^{k,p}(U)$.

In particular, when $p = 2$, Sobolev spaces $(W^{k,2}(U), \|\cdot\|_{W^{k,2}}) = (H^k(U), \|\cdot\|_{H^1})$ are Hilbert manifolds with inner product

$$\langle u, v \rangle_{H^k} = \langle u, v \rangle_{L^2} + \langle D^k u, D^k v \rangle_{L^2}. \quad (5.13)$$

Remark 5.1.5. Throughout this section, Sobolev spaces have been defined as the space of functions $f : U \rightarrow \mathbb{R}$; an equivalent definition can be made for maps $u : U \rightarrow \mathbb{R}^d$, where the $W^{k,p}$ -norm is defined similarly, but instead of an absolute value, one has the Euclidean norm. The same can be done for any other space of functions - Hölder space, C^j . Via this, all results that were introduced will still hold.

5.2. Sobolev Spaces on Manifolds

Finally, we can demonstrate that the Path Space $\mathcal{PM}$ of a Riemannian manifold is indeed a Hilbert manifold. From the description of the Sobolev spaces and what is known about the Path Space, it is not hard to see that $\mathcal{PM}$ can be identified with $H^1([0, 1], M)$.

In the following pages, we will show that that $H^1(I, M)$ is a Hilbert Manifold modeled after $H^1(I, \mathbb{R}^n)$ in two ways: one as the *completion* of the Banach Manifold $C^\infty(I, M)$ and the other by considering it the a *sub-manifold* of $C^0(I, M)$.

Remark 5.2.1. Given Riemannian manifolds M and N , we will call the manifolds $W^{k,p}(M, N)$ *Sobolev Spaces on M and N* . For the case of $W^{k,p}(I, M)$, we will simply call it *Sobolev space on M* to stay consistent with *the Path Space of M* .

Easy does it: Sub-manifold

One straightforward way to see a geometric structure for $W^{k,p}(I, M)$ for any k and Riemannian manifold M is by considering the *Whitney Embedding Theorem*. Whitney Embedding supplies us with an embedding $M \hookrightarrow \mathbb{R}^d$ for some $d > 0$ large enough. Hence, we can define any (nice) Sobolev space over a manifold $M \stackrel{o}{\subseteq} \mathbb{R}^d$ as a sub-manifold of $W^{k,p}(I, \mathbb{R}^d)$

Definition 5.11. Let M and $\mathbb{R}^d$ be as above, then a (k, p) -Sobolev space on M is defined as

$$W^{k,p}(I, M) := \left\{ [u] \in W^{k,p}(I, \mathbb{R}^d) : u(t) \in M \stackrel{o}{\subseteq} \mathbb{R}^d \quad \forall t \right\}. \quad (5.14)$$

For ‘nice’ k and p , the *Sobolev Embedding Theorem* guarantees that $W^{k,p}(I, \mathbb{R}^d) \subseteq C^0(I, \mathbb{R}^d)$, thus we can always choose a continuous representative for a class in $W^{k,p}(I, M)$. Hence, $W^{k,p}(I, M)$ is a Banach Manifold modeled on $W^{k,p}(I, \mathbb{R}^d)$ and $H^k(I, M)$ is Hilbert manifold modeled on $H^k(I, \mathbb{R}^d)$.

Let’s get our hands dirty: Completion

Now we will just consider the space $H^1(I, M)$; an equivalent computation will produce the same results for $H^k(I, M)$.

What the previous method lacked was a description of the atlas for $H^1(I, M)$, which we provide using the *Completion Method*. To do so, we will provide a different definition of $H^1(I, M)$.

Definition 5.12. If M is an m -dimensional manifold, a continuous curve $\gamma : I \rightarrow M$ is a *class* H^1 if $\varphi_i \circ \gamma : I \rightarrow \mathbb{R}^m$ is a class H^1 curve for any chart $\varphi_i : U_i \rightarrow \mathbb{R}^m$. The set of all class H^1 curves $I \mapsto M$ is denoted by $H^1(I, M)$.

The crux of this method is to

1. construct explicit charts of $C^\infty(I, M)$ and see it as a Banach Manifold modeled on $C^\infty(I, \mathbb{R}^m)$,
2. apply Theorem 5.5 on $C^\infty(I, \mathbb{R}^m)$, to see that $\overline{C^\infty(I, \mathbb{R}^m)}^{\|\cdot\|_{H^1}} = H^1(I, \mathbb{R}^m)$ and using the above definition one sees that
 - a) $\overline{C^\infty(I, M)}^{\|\cdot\|_{H^1}} = H^1(I, M)$
 - b) and that $H^1(I, M) \hookrightarrow C^0(I, M)$.

Thus the ‘completion of the charts’ of $C^\infty(I, M)$ will be the charts for $H^1(I, M)$.

Firstly, let us look at M and the exponential map. Recall that the exponential map $\exp : \mathcal{E} \subseteq TM \rightarrow M$ defined by $(p, v) \mapsto \exp_p(v)$, at any $p \in P$ becomes a diffeomorphism from a neighborhood of the zero vector in $T_p M$ onto its image in $M - \exp_p : V \xrightarrow{\sim} U$. Thus, for any $p \in M$ by considering an open ball centered at zero of radius $\epsilon > 0$, $B_\epsilon(0)$ in $T_p M \cong \mathbb{R}^m$, and its image, a geodesic ball $B_\epsilon(p)$, centered at the p , we obtain a chart via the exponential map.

$$\begin{array}{ccc}
\mathbb{R}^m \cong T_p M & & M \\
\cup & & \cup \\
B_\epsilon(0) & \longrightarrow & B_\epsilon(p) \\
\psi & & \psi \\
v & \longrightarrow & \exp_p v
\end{array} \tag{5.15}$$

This defines a smooth atlas for the manifold M .

This is the method that we will use to construct an atlas for $C^\infty(I, M)$. First, we describe the tangent bundle of the space of smooth maps between manifolds.

Proposition 5.13. *Let M and N be smooth manifolds. The tangent space of $C^\infty(M, N)$ at a map f , $T_f C^\infty(M, N)$, is isomorphic to the space of smooth sections of the pullback of TN via f , $\Gamma(f^*TN)$:*

$$T_f C^\infty(M, N) \cong \Gamma(f^*TN)$$

Proof. Let $f \in C^\infty(M, N)$ and $v \in T_f C^\infty(M, N)$. Via integral curves, we know that there exists a $\sigma : (-\epsilon, \epsilon) \rightarrow C^\infty(M, N)$ such that $\sigma(0) = f$ and $\sigma'(0) = v$. From this, we can construct

$$\begin{aligned} \Gamma : (-\epsilon, \epsilon) \times M &\rightarrow N \\ (s, p) &\mapsto \sigma(s)(p), \end{aligned}$$

from which we see that $v(p) = \sigma'(0)(p) = \partial_s \Gamma(0, p) \in T_{f(p)} N$.

For the converse, we remark that any smooth section of $\Gamma(f^*TN)$ generates a 1-parameter family of smooth maps Γ such that $\Gamma(0) = f$. $\square$

Remark 5.2.2. Given a smooth map $f : M \rightarrow N$, the pullback bundle of TN via f is

$$f^*(TN) := \{(p, v) \in M \times TN \mid f(p) = \pi(v)\}$$

for $\pi : TN \rightarrow N$ the natural projection. Thus $f^*(TN) \subseteq M \times TN$.

For a smooth path $\alpha \in C^\infty(I, M)$, the tangent space of $C^\infty(I, M)$ at α is

$$T_\alpha C^\infty(I, M) \cong \Gamma(\alpha^*TM).$$

Via an appropriate trivialization by isomorphisms $\Phi(t) : \mathbb{R}^m \rightarrow T_{\alpha(t)} M$, we have

$$\Gamma(\alpha^*(TM)) \subseteq C^\infty(I, I \times TM) \cong C^\infty(I, I \times \mathbb{R}^m) \cong C^\infty(I, \mathbb{R}^m). \quad (5.16)$$

Similarly, to the case above, we define a map from an open ball in $C^\infty(I, \mathbb{R}^m)$ to an open ball centered at α in $C^\infty(I, M)$:

$$\begin{aligned} \exp_\alpha : B_{\epsilon, \alpha}(0) &\rightarrow B_\epsilon(\alpha) \\ \xi &\mapsto \exp_\alpha(\xi) \end{aligned}$$

such that $\exp_\alpha(\xi)(t) = \exp_{\alpha(t)} \xi(t)$. The following diagram illustrates this map.

$$\begin{array}{ccc} C^\infty(I, \mathbb{R}^m) & & \\ \cup & & \\ T_\alpha C^\infty(I, M) & \Longleftrightarrow & \Gamma(\alpha^*TM) \quad C^\infty(I, M) \\ \cup & & \cup \\ B_{\epsilon, \alpha}(0) & \longrightarrow & B_\epsilon(\alpha) \\ \Psi & & \Psi \\ \xi & \longrightarrow & \exp_\alpha \xi \end{array}$$

Theorem 5.14. *The diffeomorphisms described above define a maximal atlas for $C^\infty(I, M)$ modeled on $C^\infty(I, \mathbb{R}^m)$:*

$$\{B_\epsilon(\alpha), \exp_\alpha^{-1} : B_\epsilon(\alpha) \rightarrow B_{\epsilon, \alpha}(0)\}_{\alpha \in C^\infty(I, M)}$$

Proof. Consider two arbitrary paths α_1, α_2 with respective exponential maps $\exp_{\alpha_1} : B_{\epsilon_1, \alpha_1}(0) \rightarrow B_{\epsilon_1}(\alpha_1)$, $\exp_{\alpha_2} : B_{\epsilon_2, \alpha_2}(0) \rightarrow B_{\epsilon_2}(\alpha_2)$.

• **Claim 1:**

– The set

$$W_{12} : \{(t, v) \in \alpha_1^*(TM) \mid \alpha_1(t) = \alpha_2(t) = \pi(v) \text{ and } \exp_{\alpha_1(t)}(v) \in B_{\epsilon_2}(\alpha_2)\}$$

is open in α_1^*TM and $\exp_{\alpha_1}^{-1}(B_{\epsilon_1}(\alpha_1) \cap B_{\epsilon_2}(\alpha_2)) = \Gamma(W_{12})$

– The set

$$W_{21} : \{(t, v) \in \alpha_2^*(TM) \mid \alpha_1(t) = \alpha_2(t) = \pi(v) \text{ and } \exp_{\alpha_2(t)}(v) \in B_{\epsilon_1}(\alpha_1)\}$$

is open in α_2^*TM and $\exp_{\alpha_2}^{-1}(B_{\epsilon_1}(\alpha_1) \cap B_{\epsilon_2}(\alpha_2)) = \Gamma(W_{21})$

Proof. Only the first bullet point will be shown, as the proof of the others is equivalent.

Firstly, W_{12} is indeed open in α_1^*TM as it is the preimage of an open condition. Secondly, to show that $\exp_{\alpha_1}^{-1}(B_{\epsilon_1}(\alpha_1) \cap B_{\epsilon_2}(\alpha_2)) = \Gamma(W_{12})$, consider $\gamma \in \Gamma(\alpha_1^*TM)$ and let $\tilde{\gamma}$ be the image in TM : i.e $\gamma(t) = (t, \tilde{\gamma}(t))$. Observe that γ takes values in W_{12} if and only if $\alpha_2(t) = \alpha_1(t) = \pi(\tilde{\gamma}(t))$ and $\exp_{\alpha_1(t)}\tilde{\gamma}(t) \in B_{\epsilon_2}(\alpha_2)$, in other words if $\exp_{\alpha_1}(\gamma) \in B_{\epsilon_1}(\alpha_1) \cap B_{\epsilon_2}(\alpha_2)$. Thus, γ takes values in W_{12} if and only if $\exp_{\alpha_1}(\gamma) \in B_{\epsilon_1}(\alpha_1) \cap B_{\epsilon_2}(\alpha_2)$. $\square$

• **Claim 2:** The transition function

$$\exp_{\alpha_1}^{-1} \circ \exp_{\alpha_2} : \exp_{\alpha_2}^{-1}((B_{\epsilon_1}(\alpha_1) \cap B_{\epsilon_2}(\alpha_2))) \rightarrow \exp_{\alpha_1}^{-1}((B_{\epsilon_1}(\alpha_1) \cap B_{\epsilon_2}(\alpha_2)))$$

is a diffeomorphism.

Proof. Since each $\exp_i$ is a diffeomorphism from their respective neighborhoods of 0, we have that their composition is again a diffeomorphism. $\square$

The above claims show that $\{B_\epsilon(\alpha), \exp_\alpha^{-1}\}_{\alpha \in C^\infty(I, M)}$ defines an atlas for $C^\infty(I, M)$ $\square$

We have now defined the geometric structure for $C^\infty(I, M)$. To obtain the one for $H^1(I, M)$, we will use Definition 5.12 and Theorem 5.5 in tandem to obtain the desired inclusions.

Proposition 5.15. *It remains true that all the things that we saw for Sobolev spaces $H^k(U)$ will also hold for $H^1(I, M)$:*

$$C^\infty(I, M) \subseteq H^1(I, M) \subseteq C^0(I, M) \tag{5.17}$$

$$\overline{C^\infty(I, M)}^{\|\cdot\|_{H^1}} = H^1(I, M). \tag{5.18}$$

As the space of smooth sections of the pull-back bundle can also be ‘completed’ with the H^1 -norm, we have for any $\alpha \in C^\infty(I, M)$ the following bijection

$$T_\alpha H^1(I, M) \cong \overline{\Gamma(\alpha^* TM)}^{\|\cdot\|_{H^1}}. \quad (5.19)$$

Finally, we can describe the atlas of $H^1(I, M)$.

Theorem 5.16. *For any M^m and the Sobolev space $H^1(I, M)$ defined as in Definition 5.12, the exponential map*

$$\exp_\alpha : B_{\epsilon, \alpha}(0) \subseteq \Gamma(\alpha^* TM) \rightarrow B_\epsilon(\alpha) \subseteq C^\infty(I, M)$$

extends to a unique well-defined map on the completions :

$$\widetilde{\exp}_\alpha : \widetilde{B_{\epsilon, \alpha}(0)} \subseteq \overline{\Gamma(\alpha^* TM)}^{\|\cdot\|_{H^1}} \rightarrow \widetilde{B_\epsilon(\alpha)} \subseteq H^1(I, M).$$

The $\widetilde{\exp}_\alpha$ define charts for $H^1(I, M)$ modeled on $H^1(I, \mathbb{R}^m)$ with maximal atlas:

$$\left\{ \widetilde{B_\epsilon(\alpha)}, \widetilde{\exp}_\alpha : B_{\epsilon, \alpha}(0) \rightarrow B_\epsilon(\alpha) \right\}_{\alpha \in H^1(I, M)}.$$

Proof. Proof follows immediately from Proposition 5.15 and Theorem 5.14.

$$\begin{array}{ccc} H^1(I, \mathbb{R}^m) & \Longleftrightarrow & \overline{C^\infty(I, \mathbb{R}^m)}^{\|\cdot\|_{H^1}} \\ \cup & & \cup \\ T_\alpha H^1(I, M) & \Longleftrightarrow & \overline{\Gamma(\alpha^* TM)}^{\|\cdot\|_{H^1}} \quad \quad \quad \overline{C^\infty(I, M)}^{\|\cdot\|_{H^1}} \cong H^1(I, M) \\ \cup & & \cup \\ \widetilde{B_{\epsilon, \alpha}(0)} & \longrightarrow & \widetilde{B_\epsilon(\alpha)} \\ \Psi & & \Psi \\ \xi & \longmapsto & \widetilde{\exp}_\alpha \xi. \end{array}$$

□

Remark 5.2.3. If instead, we define the exponential map $\exp$ on the space $\mathfrak{g}$ the space of continuous curves in M , $C^0(I, M)$, we can define an atlas similarly as in Theorem 5.14.

$$\begin{array}{ccc} C^0(I, \mathbb{R}^m) & \supseteq & C^0(I, \alpha^* TM) \quad \quad \quad C^0(I, M) \\ \cup & & \cup \\ B_{\epsilon, \alpha}(0) & \longrightarrow & B_\epsilon(\alpha) \\ \Psi & & \Psi \\ \xi & \longmapsto & \exp_\alpha \xi. \end{array}$$

From Proposition 5.15, by restricting this atlas to the Sobolev space $H^1(I, M)$ we obtain the same atlas as in the above theorem.

Both methods to construct geometric structure for the Sobolev spaces on M , $W^{k,p}(I, M)$ and $H^k(I, M)$, can be used to construct geometric structures for Sobolev spaces $W^{k,p}(M, N)$ and $H^k(M, N)$, for M and N smooth manifolds.

For the *sub-manifold methodology*, we will use the *Sobolev Embedding Theorem* twice: for $M \hookrightarrow \mathbb{R}^{d_1}$ with some d_1 large enough, and for $N \hookrightarrow \mathbb{R}^{d_2}$ with some d_2 . This gives the first definition of $W^{k,p}(M, N)$.

Definition 5.17. Let M , N , $\mathbb{R}^{d_1}$, $\mathbb{R}^{d_2}$ be as above. Consider then a (k, p) -Sobolev space on M and N is defined as

$$W^{k,p}(M, N) := \left\{ [f] \in W^{k,p}(\mathbb{R}^{d_1}, \mathbb{R}^{d_2}) : f(x) \in N \stackrel{o}{\subseteq} \mathbb{R}^{d_2} \quad \forall x \in M \stackrel{o}{\subseteq} \mathbb{R}^{d_1} \right\}. \quad (5.20)$$

For ‘nice’ k and p , the *Sobolev Embedding Theorem* guarantees that $W^{k,p}(\mathbb{R}^{d_1}, \mathbb{R}^{d_2}) \subseteq C^0(\mathbb{R}^{d_1}, \mathbb{R}^{d_2})$, thus we can always choose a continuous representative for a class.

While for the *completion method*, we use the following definition for $H^k(M, N)$.

Definition 5.18. Consider smooth manifolds M^m and N^n , a smooth map $f : M \rightarrow N$ is of class H^k for $k > \frac{n}{2}$ if $\varphi_i \circ f \circ \phi_j^{-1} : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a class H^k for any chart $\varphi_i : U_i \rightarrow \mathbb{R}^n$ of M and for any chart $\phi_j : V_j \rightarrow \mathbb{R}^m$ of N . The set of all class H^k maps $M \mapsto N$ is then denoted by $H^k(M, N)$.

Following the same methodology, one then obtains the desired charts for $H^k(M, N)$.

Theorem 5.19. For smooth manifolds M^m , N^n , $2k > m$ and the Sobolev space $H^k(M, N)$ defined above, the exponential map

$$\exp_f : B_{\epsilon, f}(0) \subseteq \Gamma(f^*TN) \rightarrow B_\epsilon(f) \subseteq C^\infty(M, N)$$

extends to a unique well-defined map on the completions :

$$\widetilde{\exp}_f : \widetilde{B_{\epsilon, f}(0)} \subseteq \overline{\Gamma(f^*TN)}^{\|\cdot\|_{H^1}} \rightarrow \widetilde{B_\epsilon(f)} \subseteq H^k(M, N).$$

The $\widetilde{\exp}_f$ define charts for $H^k(M, N)$ modeled on $H^k(\mathbb{R}^m, T\mathbb{R}^n)$ with maximal atlas:

$$\left\{ \widetilde{B_\epsilon(f)}, \widetilde{\exp}_f : B_{\epsilon, f}(0) \rightarrow B_\epsilon(f) \right\}_{f \in H^k(M, N)}.$$

Part II.

Generalizing Geodesics to Higher Dimensions: Minimal Sub-manifolds and Harmonic Maps

6. Minimal Sub-manifolds

This chapter references *Introduction to Riemannian manifolds*[15], *Geometric Analysis*[19], *Introduction to Minimal Surfaces: Lecture Notes for MATH286, Stanford, Winter 2025*[8], *SURFACES MINIMALES: THEORIE VARIATIONNELLE ET APPLICATIONS*[20], “La formule de la variation seconde de l’énergie au voisinage d’une application harmonique” [21], and *The Matrix Cookbook*[25].

We have now arrived at the portion of the thesis where we generalize the theory to minimal sub-manifolds. We begin by discussing some geometric properties of sub-manifolds. Specifically, it will be shown that given an embedding Riemannian $(M^m, g) \hookrightarrow (N^n, h)$, with Levi-Civita connections ∇^M and ∇^N , respectively, one has a decomposition of ∇ into ∇^M and a ‘normal component’. Following the scheme of the previous part, we tackle the definition of *minimal sub-manifolds*. We define a variation of a sub-manifold as was done for a path, from which we construct the first variation of the volume functional and define its critical points as minimal sub-manifolds. In the following chapter, we will explain why it is not necessary to define the second variation of the volume functional; however, a formula is still shown in section 6.3.

6.1. The Gauss Formula and Equation

We first remark that it is sufficient to consider embedded sub-manifolds (M^m, g) of an ambient manifold (N^n, h) . Indeed, for general smooth manifolds M and N , with $F : M \rightarrow N$ an isometric immersion, every point $p \in M$ has a neighborhood $U \subseteq M$ such that $F|_U$ is an embedding, so the results that follow can be applied to $F(U) \subseteq N$. Furthermore, the metric of M we will consider is the induced metric $\iota^*h = g$ where $\iota : M \hookrightarrow N$. Since we are interested in restrictions to M and $\langle \cdot, \cdot \rangle_h|_M = \langle \cdot, \cdot \rangle_g$, we will just use $\langle \cdot, \cdot \rangle$.

Note that one has an orthogonal decomposition for the tangent bundle of the ambient manifold N into the tangent and normal bundle of the sub-manifold M :

$$TN|_M = TM \oplus NM. \quad (6.1)$$

From this decomposition we have the *tangent* projection, $\pi^\top : TN|_M \rightarrow TM$, and the *normal* projection, $\pi^\perp : TN|_M \rightarrow NM$, such that for an adapted orthonormal frame $(e_1, \dots, e_n, \dots, e_m) :$

$$\pi^\top(e_1, \dots, e_n, \dots, e_m) = \text{Span}(e_1, \dots, e_n)$$

and

$$\pi^\perp(e_1, \dots, e_n, \dots, e_m) = \text{Span}(e_{n+1}, \dots, e_m).$$

With the orthonormal decomposition and its projections, we can split the Levi-Civita connection of N at M as well: for $\tilde{X}, \tilde{Y} \in \mathfrak{X}(N)$

$$\nabla_{\tilde{X}}^N \tilde{Y}|_M = \pi^\top (\nabla_{\tilde{X}}^N \tilde{Y}|_M) + \pi^\perp (\nabla_{\tilde{X}}^N \tilde{Y}|_M) = (\nabla_{\tilde{X}}^N \tilde{Y}|_M)^\top + (\nabla_{\tilde{X}}^N \tilde{Y}|_M)^\perp \quad (6.2)$$

The first term is called the *tangential connection*, and it is no surprise that it is the Levi-Civita connection on M .

Theorem 6.1 (The Tangential connection on a sub-manifold). *Suppose (M, g) is an embedded Riemannian sub-manifold of a Riemannian manifold (N, h) . Let ∇^N be the Levi-Civita connection on N . Then, for $X, Y \in \mathfrak{X}M$ and arbitrary extensions $\tilde{X}, \tilde{Y} \in \mathfrak{X}(N)$, the tangential connection is the Levi-Civita connection on M , ∇^M :*

$$\nabla_X^M Y = \pi^\top (\nabla_{\tilde{X}}^N \tilde{Y}|_M) = (\nabla_{\tilde{X}}^N \tilde{Y}|_M)^\top \quad (6.3)$$

Proof. • **Well-defined:** Since the value of $\nabla_{\tilde{X}}^N \tilde{Y}$ at $p \in M$ depended on $\tilde{X}_p = X_p$, which is independent of the choice of extension, $\nabla^\top$ is well-defined.

• **Smoothness and linearity** over $C^\infty(M)$ in X and over $\mathbb{R}$ in Y are clear from construction.

• **Leibniz identity:**

Let $f \in C^\infty$ and $\tilde{f}$ be the extension, then $\tilde{f}\tilde{Y}$ is a smooth extension of fY . Then

$$\begin{aligned} \nabla_X^\top (fY) &= \pi^\top (\nabla_{\tilde{X}}^N (\tilde{f}\tilde{Y})|_M) \\ &= \pi^\top (\tilde{X}(\tilde{f})\tilde{Y}|_M) + \pi^\top (\tilde{f}\nabla_{\tilde{X}}^N (\tilde{Y})|_M) = X(f)Y + \nabla_X^\top Y. \end{aligned}$$

Thus $\nabla^\top$ is a connection.

• **Compatible with metric:**

Suppose $X, Y, Z \in \mathfrak{X}(M)$ and let $\tilde{X}, \tilde{Y}, \tilde{Z}$ be the smooth extensions. At points of M we have

$$\begin{aligned} \nabla_X^\top \langle Y, Z \rangle &= X \langle Y, Z \rangle = (\tilde{X} \langle \tilde{Y}, \tilde{Z} \rangle)|_M = \nabla_{\tilde{X}}^N \langle \tilde{Y}, \tilde{Z} \rangle \\ &= \langle \nabla_{\tilde{X}}^N \tilde{Y}|_M, \tilde{Z}|_M \rangle + \langle \tilde{Y}|_M, \nabla_{\tilde{X}}^N \tilde{Z}|_M \rangle \\ &= \langle \pi^\top (\nabla_{\tilde{X}}^N \tilde{Y}|_M), \tilde{Z}|_M \rangle + \langle \tilde{Y}|_M, \pi^\top (\nabla_{\tilde{X}}^N \tilde{Z}|_M) \rangle \\ &= \langle \nabla_X^\top Y, Z \rangle + \langle Y, \nabla_X^\top Z \rangle. \end{aligned}$$

Thus, we have that $\nabla^\top$ is compatible with the induced metric of M .

• **Symmetry:**

Suppose $X, Y \in \mathfrak{X}(M)$ and let $\tilde{X}, \tilde{Y}$ be the smooth extensions. With respect to the

inclusion $\iota : M \rightarrow N$ X and Y are ι -related to $\tilde{X}$ and $\tilde{Y}$, respectively, so $[X, Y]$ is ι -related to $[\tilde{X}, \tilde{Y}]$. Furthermore, $\pi^\top \left([\tilde{X}, \tilde{Y}] \Big|_M \right) = [\tilde{X}, \tilde{Y}] \Big|_M$, and

$$\begin{aligned} \nabla_X^\top Y - \nabla_Y^\top X &= \pi^\top \left(\nabla_{\tilde{X}}^N \tilde{Y} \Big|_M - \nabla_{\tilde{Y}}^N \tilde{X} \Big|_M \right) \\ &= \pi^\top \left([\tilde{X}, \tilde{Y}] \Big|_M \right) = [\tilde{X}, \tilde{Y}] \Big|_M = [X, Y]. \end{aligned}$$

By the uniqueness of the Levi-Civita connection, $\nabla^M = \nabla^\top$. $\square$

The second term of Equation 6.2 is called the *second fundamental form*.

Definition 6.2. The *Second Fundamental Form* of M is the map $\mathbb{I} : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \Gamma(NM)$ defined by

$$\mathbb{I}(X, Y) = \left(\nabla_{\tilde{X}}^N \tilde{Y} \Big|_M \right)^\perp \quad (6.4)$$

for $X, Y \in \mathfrak{X}(M)$ and arbitrary extensions $\tilde{X}, \tilde{Y} \in \mathfrak{X}(N)$.

One of the properties of the fundamental form is that it is smooth— as the normal projection is smooth. The Following proposition introduces some more properties of $\mathbb{I}$.

Proposition 6.3 (Properties of the Second Fundamental Form). *Suppose (M, g) is an embedded Riemannian sub-manifold of a Riemannian manifold (N, h) . Let ∇^N be the Levi-Civita connection on N and let $X, Y \in \mathfrak{X}(M)$.*

1. $\mathbb{I}(X, Y)$ is independent of the extension of X and Y on an open subset of N .
2. $\mathbb{I}(X, Y)$ is bilinear over $C^\infty(M)$ in X and Y .
3. $\mathbb{I}(X, Y)$ is symmetric in X and Y .
4. The value of $\mathbb{I}(X, Y)$ at a point $p \in M$ depends only on X_p and Y_p .

Proof. Consider an arbitrary extension $\tilde{X}, \tilde{Y} \in \mathfrak{X}(N)$.

- **Symmetry:** The symmetry of the connection ∇^N implies

$$\mathbb{I}(X, Y) - \mathbb{I}(Y, X) = \left(\nabla_{\tilde{X}}^N \tilde{Y} \Big|_M - \nabla_{\tilde{Y}}^N \tilde{X} \Big|_M \right)^\perp = \left([\tilde{X}, \tilde{Y}] \Big|_M \right)^\perp = 0.$$

The last equality is because both $\tilde{X}$ and $\tilde{Y}$ are tangent to M .

- **Independence of extension and Value at a point:** Since $\nabla_{\tilde{X}}^N \tilde{Y} \Big|_p$ is dependent only on X_p , it follows that $\mathbb{I}(X, Y)|_p$ depended only on X_p . Similarly, as $\nabla_{\tilde{X}}^N \tilde{Y}$ is independent on extension of X , so is $\mathbb{I}(X, Y)$. By symmetry of the second fundamental form, the same applies to Y .
- **Bi-linearity:** Since $\nabla_{\tilde{X}}^N \tilde{Y} \Big|_M$ is linear over $C^\infty(M)$ in N and every $f \in C^\infty(M)$ can be extended to a smooth function on a neighborhood of M in N , $\mathbb{I}(X, Y)$ is linear over $C^\infty(M)$ in X . By symmetry of the second fundamental form, the same applies to Y .

□

The decomposition Equation 6.2 can be restated in terms of the second fundamental form and the Levi-Civita connection of M , and it is called the *Gauss Formula*.

Theorem 6.4 (Gauss Formula). *Suppose (M, g, ∇^M) is an embedded Riemannian sub-manifold of a Riemannian manifold (N, h, ∇^N) . If $X, Y \in \mathfrak{X}(M)$ are extended arbitrarily to smooth vector fields on a neighborhood of M in N , the following formula holds along M :*

$$\nabla_X^N Y = \nabla_X^M Y + \mathbb{I}(X, Y). \quad (6.5)$$

From the second fundamental form, we can define a curvature of M as a sub-manifold called the *mean curvature* of M in N .

Definition 6.5. For an adapted orthonormal frame $(e_1, \dots, e_n)$ the *mean curvature* of M is defined as

$$H_M := \sum_i \mathbb{I}(e_i, e_i).$$

In terms of local coordinates $(x^1, \dots, x^m)$ with local orthonormal frame $(\partial_1, \dots, \partial_m)$ of TM , we have $H_M = \sum_{i,j} g_{ij} \mathbb{I}(\partial_i, \partial_j)$.

6.1.1. Divergence on a Sub-manifold

An operator that is crucial for the formula of the first variation of the volume functional is the *divergence*¹

Definition 6.6. Let $(M^m, g) \subseteq (N^n, h)$ and $(e_1, \dots, e_m)$ be an adapted orthonormal frame for M . The *tangential divergence* $\text{div}_M : \Gamma(TN|_M) \rightarrow C^\infty(M)$ defined by

$$\text{div}_M(X) = \sum_i^m \langle \nabla_{e_i}^N X, e_i \rangle,$$

and if $(x_1, \dots, x_m)$ are local coordinates for M with $(\partial_1, \dots, \partial_m)$ a local orthonormal frame,

$$\text{div}_M(X) = g^{ij} \langle \partial_i X, \partial_j \rangle.$$

The following proposition provides some properties of the tangential divergence, one of which is an immediate consequence of the Gauss Formula.

Proposition 6.7 (Properties of div_M). *Suppose (M, g) is an embedded Riemannian sub-manifold of a Riemannian manifold (N, h) . Consider $X \in \Gamma(TN|_M)$ with decomposition $X = X^\top + X^\perp$ (as in Equation 6.1) and $f \in C^\infty(M)$*

1. $\text{div}_M(fX) = X^\top(f) + f \text{div}_M(X)$
2. $\text{div}_M(X) = \text{div}_g(X^\top)$, where div_g is as in Equation A.34
3. $\text{div}_M(X) = \text{div}_M(X^\top) - \langle H_M, X \rangle$.

¹See subsection A.4.2 for further background on this topic.

Proof. 1. By the Leibniz rule of the connection, we have

$$\begin{aligned}\operatorname{div}_M(fX) &= \sum_{i,j}^m g_{ij} \langle \partial_i(fX), \partial_j \rangle = \sum_{i,j}^m g_{ij} \langle (\partial_i f)X + f \nabla_{\partial_i}^N X, \partial_j \rangle \\ &= \sum_{i,j}^m g_{ij} (\partial_i f) \langle X, \partial_j \rangle + \sum_{i,j}^m f g_{ij} \langle \nabla_{\partial_i}^N X, \partial_j \rangle = X^\top(f) + f \operatorname{div}_M(X).\end{aligned}$$

2. Note that $\langle \nabla_{\partial_i}^N X, \partial_j \rangle = \langle \nabla_{\partial_i} X, \partial_j \rangle$ as $\{\partial_i\}_{i=1}^m$ are from the local coordinate on M . Then by applying g_{ij} and the Gauss Formula, we have the desired result.

3. By the decomposition of X , we have

$$\operatorname{div}_M(X) = \sum_{i,j}^n g_{ij} \langle \partial_i(X^\top + X^\perp), \partial_j \rangle = \operatorname{div}_M(X^\top) + \operatorname{div}_M(X^\perp).$$

Now, we focus on the normal component:

$$\begin{aligned}\operatorname{div}_M(X^\perp) &= \sum_{ij} g_{ij} \langle \partial_i X^\perp, \partial_j \rangle = \sum_{ij} -g_{ij} \langle X^\perp, \nabla_{\partial_i}^N \partial_j \rangle \\ &= \sum_{ij} -g_{ij} \langle X^\perp, \mathbb{I}(\partial_i, \partial_j) \rangle = -\langle X^\perp, H_M \rangle = -\langle X, H_M \rangle\end{aligned}$$

Hence, we have the desired result. □

Remark 6.1.1. The Laplacian $\Delta_M f$ for a $f : M \rightarrow \mathbb{R}$, $\Delta_M(f) = \operatorname{div}_{\iota^*g}(\nabla f)$, can be computed as $\Delta_M f = \operatorname{div}_M(\nabla f)$.

6.2. Minimal Sub-manifolds and the First Variation Formula

Throughout this section, we will consider the following pair of Riemannian manifolds: $(M^m, \iota^*g, \nabla^M) \subseteq (N^n, g, \nabla^N)$. Furthermore, the Einstein summation notation will be used.

Finally, to define the first variation of the volume, we need to define a 1-parameter variation of M .

Definition 6.8. A *one-parameter variation* of M is a smooth map $F : (-\epsilon, \epsilon) \times M \rightarrow N$ such that $F(0, x) = x$ for $x \in M$.

- (a) A variation is *proper*, if $F(s, x) = x$ for all $(s, x) \in (-\epsilon, \epsilon) \times \partial M$.
- (b) A variation is *compactly supported*, if there exists a compact subset in the interior of M , $K \subseteq \overset{\circ}{M}$, such that $F(s, x) = x$ for all $(s, x) \in (-\epsilon, \epsilon) \times (M \setminus K)$.
- (c) For each fixed $s \in (-\epsilon, \epsilon)$, the *s-stage of the variation* is the sub-manifold $F(s, M) = M_s \subseteq N$. Specifically, $M_0 = M$.

(d) $V \in \Gamma(TN|_M)$ is a *variation field of F* if $V_x = \frac{\partial}{\partial s}f(x) \in T_{F(s,x)}N$. If F is proper, then so is V . In general, one can define vector fields along each s -stage

$$V_{s,x} = \frac{\partial}{\partial s}F(s, x) \in T_{F(s,x)}N$$

Remark 6.2.1. In the case of curves, section 3.2, whose images are just one-dimensional sub-manifolds, we see that the definition of a one-parameter variation is consistent with the definition just displayed.

Furthermore, A similar definition produces an n -parameter variation of M by considering $(-\epsilon, \epsilon)^n$ instead of $(-\epsilon, \epsilon)$.

Observe that each s -stage M_s has its own induced metric

$$(F_s)_* \langle \partial_i, \partial_j \rangle_g = \langle \partial_i F_s, \partial_j F_s \rangle_g \quad (6.6)$$

and thus its own induced volume form

$$d\mu_s = \sqrt{\det \left(\langle \partial_i F_s, \partial_j F_s \rangle_g \right)} dx^1 \wedge \cdots \wedge x^m, \quad (6.7)$$

where $\{x^i\}_{i=1}^m$ is a local coordinate for M .

We now have all the ingredients needed to formulate the First Variation of the Volume.

Theorem 6.9 (First Variation of the Volume). *Consider a compactly supported 1-parameter variation of M , $F : (-\epsilon, \epsilon) \times M \rightarrow N$ with s -stages $\{M_s\}$, and variation field $V = \partial_s F_0$. Then the following equality holds*

$$\left. \frac{d}{ds} \text{Vol } M_s \right|_{s=0} = \int_M \text{div}_M(V) d\mu, \quad (6.8)$$

as $M_0 = M$ and $d\mu_0 = d\mu$.

Proof. Without loss of generality, we can consider for each point $p \in M \subseteq N$ the normal coordinates centered at p , such that by Theorem A.26 $g_{ij} = \delta_{ij}$. Since the local frame $\{\partial_1, \dots, \partial_m\}$ are orthonormal, $\{\partial_1 F_0, \dots, \partial_m F_0\}$ is also an orthonormal frame of $T_p M$.

Thus, the derivative of the volume form at $s = 0$ is as follows:

$$\begin{aligned}
& \left. \frac{d}{ds} \sqrt{\det(\langle \partial_i F_s, \partial_j F_s \rangle_g)} \right|_{s=0} \\
&= \frac{1}{2} \left(\det(\langle \partial_i F_0, \partial_j F_0 \rangle_g) \right)^{-1/2} \left(\left. \frac{d}{ds} \det(\langle \partial_i F_s, \partial_j F_s \rangle_g) \right|_{s=0} \right) \\
&= \frac{1}{2} \left(\det(\langle \partial_i F_0, \partial_j F_0 \rangle_g) \right)^{-1/2} \det(\langle \partial_i F_0, \partial_j F_0 \rangle_g) \operatorname{Tr} \left(\langle \partial_i F_0, \partial_j F_0 \rangle_g^{-1} \frac{d}{ds} (\langle \partial_i F_0, \partial_j F_0 \rangle_g) \right) \\
&= \frac{1}{2} \left(\det(\langle \partial_i F_0, \partial_j F_0 \rangle_g) \right)^{1/2} \operatorname{Tr} \left(\langle \partial_i F_0, \partial_j F_0 \rangle_g^{-1} \frac{d}{ds} (\langle \partial_i F_0, \partial_j F_0 \rangle_g) \right) \\
&= \frac{1}{2} \left(\det(\langle \partial_i F_0, \partial_j F_0 \rangle_g) \right)^{1/2} \sum_{i=1}^m \left(\langle \partial_i F_0, \partial_i F_0 \rangle_g^{-1} \frac{d}{ds} (\langle \partial_i F_0, \partial_i F_0 \rangle_g) \right) \\
&= \frac{1}{2} \left(\det(\langle \partial_i F_0, \partial_j F_0 \rangle_g) \right)^{1/2} \sum_{i=1}^m 2 \left(\langle \partial_i F_0, \partial_i F_0 \rangle_g^{-1} (\langle \partial_s \partial_i F_0, \partial_i F_0 \rangle_g) \right) \\
&= \sum_{i=1}^m \left\langle \nabla_{\partial_i F_0}^N \partial_s F_0, \partial_i F_0 \right\rangle_g \\
&= \operatorname{div}_M(\partial_s F_0) = \operatorname{div}_M(V). \tag{6.9}
\end{aligned}$$

Where the second equality is due to the following proposition ², while in the penultimate line, Equation 6.9, we use the fact that the local frame is orthonormal, making the determinant is 1, and Lemma A.14.

Proposition 6.10. *Let $(e_1, \dots, e_n)$ be the standard orthonormal basis of $\mathbb{R}^n$. For any $n \times n$ matrix $A(t) = (A_1(t), \dots, A_n(t)) = (A(t)e_1, \dots, A(t)e_n)$, where $A_i(t)$ is the i th-column of $A(t)$. If $\det A(t) \neq 0$, then*

$$\frac{d}{dt} \det(A(t)) = \det(A(t)) \operatorname{Tr} \left(A^{-1}(t) \frac{d}{dt} A(t) \right). \tag{6.10}$$

Thus

$$\begin{aligned}
\left. \frac{d}{ds} \right|_{s=0} \operatorname{Vol} M_s &= \int_{M_s} \left. \frac{d}{ds} \right|_{s=0} \sqrt{\det(\langle \partial_i F_s, \partial_j F_s \rangle_g)} dx^1 \wedge \dots \wedge x^m \Big|_{s=0} \\
&= \int_M \operatorname{div}_M(V) dx^1 \wedge \dots \wedge x^m = \int_M \operatorname{div}_M(V) d\mu.
\end{aligned}$$

□

Using our knowledge of the Divergence Theorem, the first variation can be further simplified, when the sub-manifold M has non-empty boundary.

Theorem 6.11 (First Variation of Volume with non-empty boundary). *Let M and $\{M_s\}$ as in theorem Theorem 6.9. Suppose ∂M is non-empty, then the first variation simplifies to*

$$\left. \frac{d}{ds} \operatorname{Vol} M_s \right|_{s=0} = - \int_M \langle H_M, V \rangle_g d\mu. \tag{6.11}$$

²For the proof, we refer to the identities in *The Matrix Cookbook*.

Proof. From First Variation of the Volume, the third item of Properties of div_M , and Divergence Theorem, we have the desired result :

$$\begin{aligned} \left. \frac{d}{ds} \text{Vol } M_s \right|_{s=0} &= \int_M \text{div}_M(V) d\mu = \int_M (\text{div}_M(V) - \langle H_M, V \rangle_g) d\mu \\ &= \int_{\partial M} \langle V, \eta \rangle_{\hat{g}} d\mu_{\hat{g}} - \int_M \langle H_M, V \rangle_g d\mu. \end{aligned}$$

Where $\eta \in \Gamma(TN|_M)$ be the outward-pointing normal unit vector on ∂M and $\hat{g}$ is the induced metric on the boundary ∂M . Since the variation is compactly supported, the variation field V vanishes on the boundary and we have the desired result. $\square$

Corollary 6.11.1. *Let $(M^m, \iota^*g, \nabla^M) \subseteq (N^n, g, \nabla^N)$ be a sub-manifold. If $\left. \frac{d}{ds} \text{Vol } M_s \right|_{s=0} = 0$ among all compactly supported variations, then $H_M = 0$.*

From this, we have the definition of a minimal sub-manifold.

Definition 6.12. A sub-manifold is *minimal* if its mean curvature is zero.

Some applications

Proposition 6.13. *No closed sub-manifold of $\mathbb{R}^n$ is minimal.*

Proof. Supposed $M \subseteq \mathbb{R}^n$ is a closed minimal m -dimensional sub-manifold and consider the compactly-supported variation $\{(1+s)M\}_{s \in (-\epsilon, \epsilon)}$. Then

$$0 = \left. \frac{d}{ds} \text{Vol } (1+s)M \right|_{s=0} = \left. \frac{d}{ds} (1+s)^m \text{Vol } M \right|_{s=0} = m \text{Vol } M > 0.$$

$\square$

The following proposition gives a little taste of what will be shown in the following chapter regarding the relation between minimal sub-manifolds and harmonic maps.

Proposition 6.14. *The compact sub-manifold $(M^m, \iota^*g, \nabla^M) \subseteq (\mathbb{R}^n, g, \nabla)$ is minimal if and only if the coordinate functions on $\mathbb{R}^n$ restrict to harmonic functions.*

Proof. Let E_r be a coordinate vector field on $\mathbb{R}^n$ for a coordinate function $x_r : \mathbb{R}^n \rightarrow \mathbb{R}$ and $X = \varphi E_r$ for some smooth function on M . Since E_r is parallel, we have for an adapted frame $\{e_1, \dots, e_n\}$:

$$\begin{aligned} \text{div}_M X &= \sum_{i=1}^m \langle \nabla_{e_i} X, e_i \rangle_g = \sum_{i=1}^m \text{grad}_M \varphi \langle E_r, e_i \rangle_g = \langle \text{grad}_M \varphi, E_r \rangle \\ &= \langle \text{grad}_M \varphi, \text{grad}_M x_r|_M \rangle_g \end{aligned}$$

Thus if $\varphi \in C_c^\infty(M \setminus \partial M)$, by the first Green identity Equation (A.48) on page 116 :

$$\begin{aligned} \int_M \varphi \Delta_M x_r|_M d\mu &= - \int_M \langle \text{grad}_M \varphi, \text{grad}_M x_r|_M \rangle_g d\mu \\ &= \int_M \text{div}_M(X) d\mu = \int_M \varphi \langle H_M, e_r \rangle d\mu. \end{aligned}$$

Since φ was chosen arbitrary, we have that $\Delta_M x|_M = H_M$, and the assertion is proven. $\square$

6.3. What Actually Minimizes the Volume?

We end this chapter by giving the formula of the second variation of the volume functional with respect to a one-parameter compactly supported variation.

Theorem 6.15 (Second Variation Formula (Volume Version)). *Let $(M, \iota^*g, \nabla^M) \subseteq (N, g, \nabla)$ be an immersed Riemannian sub-manifold and R^M and R the curvature tensors of M and N , respectively. Consider a compactly supported 1-parameter variation of M , $F : (-\epsilon, \epsilon) \times M \rightarrow N$ with s -stages $\{M_s\}$, and variation field $V = \partial_s F_0$. Then the following equality holds*

$$\begin{aligned} \left. \frac{d^2}{ds} \right|_{s=0} \text{Vol}(M_s) = \int_M \left(\sum_{i=1}^k \langle R(V, \partial_i F_0, V), \partial_i F_0 \rangle_g + \text{div}_M (\nabla_V V) \right. \\ \left. + \sum_{i=1}^k \|\nabla_{\partial_i F_0} V\|_g^2 - \sum_{i,j=1}^k \langle \nabla_{\partial_i F_0} V, \partial_j F_0 \rangle_g \langle \nabla_{\partial_j F_0} V, \partial_i F_0 \rangle_g \right) d\mu. \end{aligned} \quad (6.12)$$

Proof. To compute the second derivative of $\text{Vol } M_s$, we will use the fact that for a function $f \in C^2(M)$ such that $f(0) = 1$ we have

$$\left. \frac{d^2}{ds} \right|_{s=0} \sqrt{f(s)} = -\frac{1}{4} \left(\left. \frac{d}{ds} \right|_{s=0} f(s) \right)^2 + \frac{1}{2} \left. \frac{d^2}{ds^2} \right|_{s=0} f(s),$$

and the following proposition on the second derivative of the determinant of an invertible matrix ³.

Proposition 6.16. *For any $n \times n$ matrix $A(t) = (A_1(t), \dots, A_n(t))$, where $A_i(t)$ is the i th-column of $A(t)$. If $\det A(t) \neq 0$, then*

$$\begin{aligned} \frac{d^2}{dt^2} \det A(t) = \text{Tr} \left(-A^{-1}(t) \left(\frac{d}{dt} A(t) \right)^2 + A^{-1}(t) \left(\frac{d^2}{dt^2} A(t) \right) \right) \det A(t) \\ + \text{Tr} \left(A^{-1}(t) \frac{d}{dt} A(t) \right)^2 \det A(t) \end{aligned}$$

If $A(0) = \text{Id}_{n \times n}$, then at $t = 0$ we have

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \det A(t) = \text{Tr} \left(\frac{d}{dt} A(0) \right)^2 + \text{Tr} \left(\frac{d^2}{dt^2} A(0) \right) - \text{Tr} \left(\left(\frac{d}{dt} A(0) \right)^2 \right).$$

Furthermore, if $A(t)$ is symmetric then

$$\text{Tr} \left(\left(\frac{d}{dt} A(0) \right)^2 \right) = \sum_{i=1}^k \left(\frac{d}{dt} A(0) \right)_{ii}^2 = \sum_{i,j=1}^k \left(\frac{d}{dt} A^2(0) \right)_{ij} = \left\| \frac{d}{dt} A(0) \right\|^2$$

then

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \det A(t) = \text{Tr} \left(\frac{d^2}{dt^2} A(0) \right) - \left\| \left(\frac{d}{dt} A(0) \right) \right\|^2 + \text{Tr} \left(\frac{d}{dt} A(0) \right)^2$$

³For the proof, we refer to the identities in *The Matrix Cookbook* [25].

Applying the above formulas, we have

$$\begin{aligned}
\frac{d^2}{d^2s} \langle \partial_i F_0, \partial_j F_0 \rangle_g &= \frac{d}{ds} \left(\langle \nabla_{\partial_s F_0} \partial_i F_0, \partial_j F_0 \rangle_g + \langle \partial_i F_0, \nabla_{\partial_s F_0} \partial_j F_0 \rangle_g \right) \\
&= \langle \nabla_{\partial_s F_0} \nabla_{\partial_s F_0} \partial_i F_0, \partial_j F_0 \rangle_g + 2 \langle \nabla_{\partial_s F_0} \partial_i F_0, \nabla_{\partial_s F_0} \partial_j F_0 \rangle_g \\
&\quad + \langle \partial_i F_0, \nabla_{\partial_s F_0} \nabla_{\partial_s F_0} \partial_j F_0 \rangle_g \\
&= \langle \nabla_V \nabla_V \partial_i F_0, \partial_j F_0 \rangle_g + 2 \langle \nabla_{\partial_i F_0} V, \nabla_{\partial_j F_0} V \rangle_g \\
&\quad + \langle \partial_i F_0, \nabla_V \nabla_V \partial_j F_0 \rangle_g,
\end{aligned}$$

wherein the last equality, we use $V = \partial_s F_0$ and Lemma A.14. Since

$$\begin{aligned}
\langle \nabla_V \nabla_V \partial_i F_0, \partial_j F_0 \rangle_g &= \langle (\nabla_V \nabla_{\partial_i F_0} - \nabla_{\partial_i F_0} \nabla_V) V, \partial_j F_0 \rangle_g + \langle \nabla_V \nabla_{\partial_i F_0} V, \partial_j F_0 \rangle_g \\
&= \langle R(V, \partial_i F_0, V), \partial_j F_0 \rangle_g + \langle \nabla_V \nabla_{\partial_i F_0} V, \partial_j F_0 \rangle_g,
\end{aligned}$$

and

$$\langle \partial_i F_0, \nabla_V \nabla_V \partial_j F_0 \rangle_g = \langle \partial_i F_0, R(V, \partial_j F_0, V) \rangle_g + \langle \partial_i F_0, \nabla_V \nabla_{\partial_j F_0} V \rangle_g,$$

we simplify the second derivative to

$$\begin{aligned}
\frac{d^2}{d^2s} \langle \partial_i F_0, \partial_j F_0 \rangle_g &= \langle R(V, \partial_i F_0, V), \partial_j F_0 \rangle_g + \langle \nabla_V \nabla_{\partial_i F_0} V, \partial_j F_0 \rangle_g \\
&\quad + 2 \langle \nabla_{\partial_i F_0} V, \nabla_{\partial_j F_0} V \rangle_g + \langle \partial_i F_0, R(V, \partial_j F_0, V) \rangle_g + \langle \partial_i F_0, \nabla_V \nabla_{\partial_j F_0} V \rangle_g.
\end{aligned}$$

Observe that by the symmetry of the inner product and Lemma A.17, we have the equality

$$\langle \partial_i F_0, R(V, \partial_j F_0, V) \rangle_g = \langle R(V, \partial_i F_0, V), \partial_j F_0 \rangle_g,$$

and by the symmetry of the inner product and Lemma A.14, we have

$$\begin{aligned}
\langle \nabla_V \nabla_{\partial_i F_0} V, \partial_j F_0 \rangle_g &= \langle \nabla_{\partial_i F_0} (\nabla_V V), \partial_j F_0 \rangle_g \\
\langle \partial_i F_0, \nabla_V \nabla_{\partial_j F_0} V \rangle_g &= \langle \partial_i F_0, \nabla_{\partial_j F_0} \nabla_V V \rangle_g = \langle \nabla_{\partial_j F_0} \nabla_V V, \partial_i F_0 \rangle_g.
\end{aligned}$$

Now we have all the ingredients needed for the formula of $\frac{d^2}{d^2s} \langle \partial_i F_0, \partial_j F_0 \rangle_g$. By the proposition on matrices, we have

$$\begin{aligned}
\text{Tr} \left(\frac{d^2}{d^2s} \langle \partial_i F_0, \partial_j F_0 \rangle_g \right) &= \sum_{i=1}^k 2 \langle R(V, \partial_i F_0, V), \partial_i F_0 \rangle_g + \sum_{i=1}^k 2 \langle \nabla_{\partial_i F_0} (\nabla_V V), \partial_i F_0 \rangle_g \\
&\quad + \sum_{i=1}^k 2 \langle \nabla_{\partial_i F_0} V, \nabla_{\partial_i F_0} V \rangle_g \\
&= \sum_{i=1}^k 2 \langle R(V, \partial_i F_0, V), \partial_i F_0 \rangle_g + 2 \text{div}_M (\nabla_V V) + \sum_{i=1}^k 2 \|\nabla_{\partial_i F_0} V\|_g^2.
\end{aligned}$$

Finally, from Theorem 6.9, $\text{Tr} \left(\frac{d}{ds} \langle \partial_i F_0, \partial_j F_0 \rangle_g \right)^2 = (2 \text{div}_M(V))^2 = 4 \text{div}_M^2(V)$, and

$$\begin{aligned}
\left\| \frac{d}{ds} \langle \partial_i F_0, \partial_j F_0 \rangle_g \right\|_g^2 &= \left\| \langle \nabla_V \partial_i F_0, \partial_j F_0 \rangle_g + \langle \partial_i F_0, \nabla_V \partial_j F_0 \rangle_g \right\|_g^2 \\
&= \sum_{i,j=1}^k \left(\langle \nabla_V \partial_i F_0, \partial_j F_0 \rangle_g + \langle \partial_i F_0, \nabla_V \partial_j F_0 \rangle_g \right)^2 \\
&= \sum_{i=1}^k 2 \langle \nabla_V \partial_i F_0, \partial_i F_0 \rangle_g^2 + \sum_{i,j=1}^k 2 \langle \nabla_V \partial_i F_0, \partial_j F_0 \rangle_g \langle \nabla_V \partial_j F_0, \partial_i F_0 \rangle_g \\
&= \sum_{i=1}^k 2 \langle \nabla_{\partial_i F_0} V, \partial_i F_0 \rangle_g^2 + \sum_{i,j=1}^k 2 \langle \nabla_{\partial_i F_0} V, \partial_j F_0 \rangle_g \langle \nabla_{\partial_j F_0} V, \partial_i F_0 \rangle_g \\
&= 2 \text{div}_M^2(V) + \sum_{i,j=1}^k 2 \langle \nabla_{\partial_i F_0} V, \partial_j F_0 \rangle_g \langle \nabla_{\partial_j F_0} V, \partial_i F_0 \rangle_g.
\end{aligned}$$

Placing everything together, we have

$$\begin{aligned}
\frac{d^2}{d^2 s} \Big|_{s=0} \sqrt{\det \langle \partial_i F_s, \partial_j F_s \rangle_g} &= -\frac{1}{4} \left(\frac{d}{ds} \Big|_{s=0} \det \langle \partial_i F_s, \partial_j F_s \rangle_g \right)^2 + \frac{1}{2} \frac{d^2}{d^2 s} \Big|_{s=0} \det \langle \partial_i F_s, \partial_j F_s \rangle_g \\
&= -\frac{1}{4} (2 \text{div}_M(V))^2 + \frac{1}{2} \text{Tr} \left(\frac{d^2}{d^2 s} \langle \partial_i F_0, \partial_j F_0 \rangle_g \right) \\
&\quad - \frac{1}{2} \left\| \frac{d}{ds} \langle \partial_i F_0, \partial_j F_0 \rangle_g \right\|_g^2 + \frac{1}{2} \text{Tr} \left(\frac{d}{ds} \langle \partial_i F_0, \partial_j F_0 \rangle_g \right)^2 \\
&= -\text{div}_M^2(V) + \sum_{i=1}^k \langle R(V, \partial_i F_0, V), \partial_i F_0 \rangle_g \\
&\quad + \text{div}_M(\nabla_V V) + \sum_{i=1}^k \|\nabla_{\partial_i F_0} V\|_g^2 - \text{div}_M^2(V) \\
&\quad - \sum_{i,j=1}^k \langle \nabla_{\partial_i F_0} V, \partial_j F_0 \rangle_g \langle \nabla_{\partial_j F_0} V, \partial_i F_0 \rangle_g + 2 \text{div}_M^2(V) \\
&= \sum_{i=1}^k \langle R(V, \partial_i F_0, V), \partial_i F_0 \rangle_g + \text{div}_M(\nabla_V V) + \sum_{i=1}^k \|\nabla_{\partial_i F_0} V\|_g^2 \\
&\quad - \sum_{i,j=1}^k \langle \nabla_{\partial_i F_0} V, \partial_j F_0 \rangle_g \langle \nabla_{\partial_j F_0} V, \partial_i F_0 \rangle_g.
\end{aligned}$$

Finally, we have the desired formula

$$\begin{aligned}
\frac{d^2}{d^2s} \Big|_{s=0} \text{Vol } M_s &= \int_{M_s} \frac{d^2}{d^2s} \Big|_{s=0} \sqrt{\det \left(\langle \partial_i F_s, \partial_j F_s \rangle_g \right)} dx^1 \wedge \cdots \wedge x^k \\
&= \int_M \sum_{i=1}^k \langle R(V, \partial_i F_0, V), \partial_i F_0 \rangle_g + \text{div}_M (\nabla_V V) dx^1 \wedge \cdots \wedge x^k \\
&\quad + \int_M \sum_{i=1}^k \|\nabla_{\partial_i F_0} V\|_g^2 dx^1 \wedge \cdots \wedge x^k \\
&\quad - \int_M \sum_{i,j=1}^k \langle \nabla_{\partial_i F_0} V, \partial_j F_0 \rangle_g \langle \nabla_{\partial_j F_0} V, \partial_i F_0 \rangle_g dx^1 \wedge \cdots \wedge x^k.
\end{aligned}$$

□

7. Energy Functional

This chapter references *Riemannian Geometry and Geometric Analysis*[14], “Harmonic Mappings of Riemannian Manifolds”[10]

Determining whether the critical points of the volume functional are minima can be very challenging, as seen from section 6.3. Thus, as we did for geodesics, we want to work on the energy functional.

We begin the chapter by establishing the geometric objects that are essential in the definition of the energy functional as well as in deriving its first and second variation formulas. Once the critical points of the energy functionals are established, we show the relation between volume-minimizing sub-manifolds and harmonic maps with respect to specific inclusions. With this knowledge, we unreservedly look at the second variation formula for the energy functional. As was done for geodesics and the Path Space, we turn to the Jacobi operator for non-degeneracy and the Morse index. This is when problems emerge. It will be explained that a generalization for conjugate points has not been found, but most importantly, that a Handle-Body decomposition for the energy functional cannot be defined.

Remark 7.0.1. For clarity, the Einstein summation notation will be used throughout this chapter.

7.1. Setting the Stage

Consider a smooth map between Riemannian manifolds $f : (M^m, (g_{\alpha\beta})) \rightarrow (N^n, (h_{ij}))$, with ∇^M and ∇^N are the respective Levi-Civita connections. In this chapter, we will think of df as a vector bundle morphism covering f , that is

$$df \in \text{Hom}_f(TM, TN) \cong \text{Hom}(TM, f^*TN) \cong T^*M \otimes f^*TN,$$

where f^*TN is the pull-back vector bundle over M .

$$\begin{array}{ccc} TM & \xrightarrow{df} & TN \\ \downarrow \pi_M & & \downarrow \pi_N \\ M & \xrightarrow{f} & N \end{array}$$

As orthonormal basis for $\text{Hom}_f(TM, TN)_p$ is $(dx^\alpha)_p \otimes \left(\frac{\partial}{\partial y^i}\right)_{f(p)}$, the differential df at a point p in coordinates is

$$df_p = \sum_{\alpha=1}^m \sum_{i=1}^n \frac{\partial f^i(p)}{\partial x^\alpha} (dx^\alpha)_p \otimes \left(\frac{\partial}{\partial y^i}\right)_{f(p)}. \quad (7.1)$$

For simplicity of notation, we will use the shorthand $df_p = \partial x^\alpha f^i(p) dx^\alpha \otimes \partial y^i$. From the induced inner metrics on T^*M and f^*TN , we have a metric on $T^*M \otimes f^*TN$:

$$\left\langle (dx^\alpha)_p \otimes \left(\frac{\partial}{\partial y^i} \right)_{f(p)}, (dx^\beta)_p \otimes \left(\frac{\partial}{\partial y^j} \right)_{f(p)} \right\rangle_{T^*M \otimes f^*TN} = g^{\alpha\beta}(p) h_{ij}(f(p)); \quad (7.2)$$

hence

$$\begin{aligned} \|df\|_{T^*M \otimes f^*TN}^2 &= \langle df, df \rangle_{T^*M \otimes f^*TN} \\ &= g^{\alpha\beta} \langle \partial x^\alpha f, \partial x^\beta f \rangle_{f^*TN} = g^{\alpha\beta}(p) h_{ij}(f(p)) (\partial x^\alpha f^i) (\partial x^\beta f^j). \end{aligned} \quad (7.3)$$

Definition 7.1. The *energy* of a smooth map $f : (M, g) \rightarrow (N, h)$ is

$$\mathcal{E}_g(f) = \frac{1}{2} \int_M \|df\|_{\text{Hom}_f(TM, TN)}^2 d\mu_g \quad (7.4)$$

where $d\mu_g$ is the volume form of M . Where $\mathcal{E}_g : H^k(M, N) \rightarrow \mathbb{R}$ is the *energy function* on the Sobolev space $H^k(M, N)$ with $2k > \dim M$.

Remark that for a tensor product bundle $E \otimes E'$ with ∇ and ∇' connections of E and E' , respectively, the canonical connection is $\nabla \otimes \nabla'$ which is defined by

$$(\nabla \otimes \nabla')_X (s \otimes s') = \nabla_X s \otimes s' + s \otimes \nabla'_X s' \quad (s \in \Gamma(E), s' \in \Gamma(E')).$$

Thus, the connection on $T^*M \otimes f^*TN$ comes from the pullback connection on f^*TN ,

$$\widetilde{\nabla}_{\partial x^\alpha} \partial y^i = \nabla_{\frac{\partial f^j}{\partial x^\alpha} \partial y^j}^N \partial y^i = \frac{\partial f^j}{\partial x^\alpha} \nabla_{\partial y^j}^N \partial y^i = \frac{\partial f^j}{\partial x^\alpha} \Gamma_{ij}^k \partial y^k \quad (7.5)$$

where Γ_{ij}^k are the Christoffel symbols for N , and the induced connection for T^*M

$$\nabla_{\partial x^\alpha}^* dx^\beta = -\Gamma_{\alpha\beta}^\eta dx^\eta, \quad (7.6)$$

where $\Gamma_{\alpha\beta}^\eta$ are the Christoffel symbols for M . Explicitly, the tensor connection of $T^*M \otimes f^*TN$ in local coordinates is

$$\begin{aligned} \nabla_{\partial x^\alpha} (dx^\beta \otimes \partial y^i) &= \nabla_{\partial x^\alpha}^* (dx^\beta) \otimes \partial y^i + dx^\beta \otimes' \nabla_{\partial x^\alpha} (\partial y^i) \\ &= -\Gamma_{\alpha\beta}^\eta dx^\eta \otimes \partial y^i + \Gamma_{ij}^k \frac{\partial f^j}{\partial x^\alpha} \partial y^k. \end{aligned} \quad (7.7)$$

Since we are working with the Levi-Civita connections on M and N , the tensor connection is also torsion-free and metric. From this we can describe what the tensor connection on $df \in \text{Hom}_f(TM, TN)$:

$$\begin{aligned} \nabla_{\partial x^\alpha} (df) &= \nabla_{\partial x^\alpha} \left(\frac{\partial f^i}{\partial x^\beta} dx^\beta \otimes \partial y^i \right) \\ &= \partial x^\alpha \left(\frac{\partial f^i}{\partial x^\beta} \right) dx^\beta \otimes \partial y^i + \frac{\partial f^i}{\partial x^\beta} \nabla_{\partial x^\alpha} (dx^\beta \otimes \partial y^i) \\ &= \frac{\partial^2 f^i}{\partial x^\alpha \partial x^\beta} dx^\beta \otimes \partial y^i - \frac{\partial f^i}{\partial x^\eta} \Gamma_{\alpha\beta}^\eta dx^\eta \otimes \partial y^i + \Gamma_{ij}^k \frac{\partial f^i}{\partial x^\beta} \frac{\partial f^j}{\partial x^\alpha} \partial y^k. \end{aligned}$$

Similarly to how one defines the Laplacian of a smooth function $f : M \rightarrow \mathbb{R}$ via the connection on M (see Corollary A.32.1 in section A.4), we define the Laplacian of a smooth map $f : (M, g) \rightarrow (N, h)$ via the tensor connection.

Definition 7.2. Let $f : (M, g) \rightarrow (N, h)$ be a smooth map between Riemannian manifolds, then

$$\Delta_M f = g^{\alpha\beta} \left(\frac{\partial^2 f^i}{\partial x^\alpha \partial x^\beta} - \Gamma_{\alpha\beta}^\eta \frac{\partial f^i}{\partial x^\eta} \right). \quad (7.8)$$

7.2. Harmonic Maps and the First Variation

Variations aren't restricted to inclusions and paths; they can be defined for any smooth mapping.

Definition 7.3. An n -parameter variation F of a smooth mapping $f : (M, g) \rightarrow (N, h)$ is a smooth map $F : (-\epsilon, \epsilon) \times M \rightarrow N$ such that $F(0, x) = f(x)$ for $x \in M$. With equivalent definitions as (a) – (e) in Definition 6.8.

Lemma 7.4. If V is a vector field along $f : (M, g) \rightarrow (N, h)$, $V \in \Gamma(f^*TN)$, then V is the variation field of some variation of f .

Proof. Equivalent to the proof of Lemma 3.5. $\square$

Now that we have the definition of a variation of a smooth map, we can define the first variation of the energy functional.

Theorem 7.5. Let $f : (M, g, \nabla^M) \rightarrow (N, h, \nabla^N)$ be a smooth map between Riemannian manifolds. Consider a compactly supported one-parameter variation $F : (-\epsilon, \epsilon) \times M \rightarrow N$ with variation field

$$V = \left. \frac{\partial F_s}{\partial s} \right|_{s=0}.$$

Then the following equality holds

$$\left. \frac{d}{ds} \mathcal{E}_g(F_s) \right|_{s=0} = \int_M - \langle \text{Tr}_g \nabla(df), V \rangle_{f^*TN} d\mu_g, \quad (7.9)$$

where $d\mu_g$ is the volume form of M .

Proof. From direct computation, we have

$$\begin{aligned} \left. \frac{d}{ds} \mathcal{E}_g(F_s) \right|_{s=0} &= \frac{1}{2} \int_M \left. \frac{d}{ds} \right|_{s=0} \langle df, df \rangle_{T^*M \otimes f^*TN} d\mu_g \\ &= \frac{1}{2} \int_M \left. \frac{d}{ds} \right|_{s=0} \left(g^{\alpha\beta} h_{ij} \frac{\partial F_s^i}{\partial x^\alpha} \frac{\partial F_s^j}{\partial x^\beta} \right) d\mu_g \\ &= \int_M g^{\alpha\beta} h_{ij} \nabla_{\partial s} \left(\frac{\partial F_0^i}{\partial x^\alpha} \right) \frac{\partial F_0^j}{\partial x^\beta} d\mu_g \\ &= \int_M g^{\alpha\beta} h_{ij} \widetilde{\nabla}_{\partial x^\alpha} \left(\frac{\partial F_0^i}{\partial s} \right) \frac{\partial F_0^j}{\partial x^\beta} d\mu_g, \end{aligned}$$

where the second-to-last and last equalities are due to the tensor connection being metric and torsion-free. By substituting f for F_0 , V^i for $\frac{\partial F_0^i}{\partial s}$, and

$$\nabla V = \widetilde{\nabla}_{\partial x^\alpha} (V^i \partial y^i) \otimes dx^\alpha = \frac{\partial V^i}{\partial x^\alpha} \partial y^i \otimes dx^\alpha + V^i \frac{\partial f^j}{\partial x^\alpha} \Gamma_{ij}^k \partial y^k \otimes dx^\alpha,$$

we have

$$\begin{aligned} \left. \frac{d}{ds} \mathcal{E}_g(F_s) \right|_{s=0} &= \int_M \langle \nabla V, df \rangle_{T^*M \otimes f^*TN} d\mu_g \\ &= \int_M - \langle \nabla_{\partial x^\alpha} (df), V \otimes dx^\alpha \rangle_{T^*M \otimes f^*TN} d\mu_g \\ &= \int_M - \langle \text{Tr}_g \nabla(df), V \rangle_{f^*TN} d\mu_g, \end{aligned}$$

where the second-to-last equality is due to the tensor connection being metric. $\square$

The component $\text{Tr}_g \nabla(df)$ is called the *tension tensor* of f , and it will be denoted by $\tau(f)$. It follows that the critical points of the energy functional are the maps f that satisfy $\tau(f) = 0$. It can be seen that the components of the tension tensors, $\tau(f) = \sum_{k=1}^n \tau(f)^k \partial y^k$ are

$$\begin{aligned} \tau(f)^k &= g^{\alpha\beta} \left(\frac{\partial^2 f^k}{\partial x^\alpha \partial x^\beta} - \Gamma_{\alpha\beta}^\eta \frac{\partial f^k}{\partial x^\eta} + \Gamma_{ij}^k \frac{\partial f^i}{\partial x^\alpha} \frac{\partial f^j}{\partial x^\beta} \right) \\ &= \Delta_M(f) + g^{\alpha\beta} \Gamma_{ij}^k \frac{\partial f^i}{\partial x^\alpha} \frac{\partial f^j}{\partial x^\beta}, \quad (7.10) \end{aligned}$$

Δ_M is the Laplacian of M . When $N = \mathbb{R}$, the $\Gamma_{ij}^k = 0$, and we see this is the same definition as the well-known harmonic functions, Definition A.33, thus we will call the critical points of the energy functional *harmonic maps*.

7.3. Relation to Volume

In the case of geodesics, showing the relation between the critical points of the length and energy functionals was simple; showing the relation between minimal sub-manifolds and harmonic maps, however, is more involved.

In Proposition 4.4, it was shown that if a curve had constant speed, then it minimizes the energy if and only if it minimizes the length functional. The following theorem is the higher-dimensional generalization, and it requires that the mapping be an isometry.

Theorem 7.6. *Let $f : (M^m, f^*g, \nabla^M) \hookrightarrow (N^n, g, \nabla^N)$ be an isometric immersion. Then f is harmonic if and only if $f(M)$ is minimal.*

Proof. Without loss of generality consider an arbitrary $p \in M$ with normal coordinates $\{\partial x^\alpha\}_{\alpha=1, \dots, m}$. Let the local coordinates on $f(M)$ be

$$e_\alpha = f_*(\partial x^\alpha) = \frac{\partial f^i}{\partial x^\alpha} \partial y^i,$$

where $(y^1, \dots, y^m)$ are the local coordinates in N around $f(p)$. Since f is an isometry and by the Gauss Formula, we remark that

$$\nabla_{e_\alpha}^N e_\beta = (\nabla_{e_\alpha}^N e_\beta)^\top + (\nabla_{e_\alpha}^N e_\beta)^\perp = f_*(\nabla_{\partial x^\alpha}^M \partial x^\beta) + (\nabla_{e_\alpha}^N e_\beta)^\perp = (\nabla_{e_\alpha}^N e_\beta)^\perp \quad (7.11)$$

and

$$\nabla_{e_\alpha}^N e_\beta = \nabla_{\frac{\partial f^i}{\partial x^\alpha} \partial y^i}^N \frac{\partial f^j}{\partial x^\beta} \partial y^j = \frac{\partial^2 f^k}{\partial x^\alpha \partial x^\beta} \partial y^k + \frac{\partial f^i}{\partial x^\alpha} \frac{\partial f^j}{\partial x^\beta} \Gamma_{ij}^k \partial y^k, \quad (7.12)$$

where Γ_{ij}^k are the Christoffel symbols for N . Thus, the mean curvature in these coordinates is

$$H_M = \sum_\alpha \mathbb{I}(e_\alpha, e_\alpha) = \sum_\alpha \frac{\partial^2 f^k}{(\partial x^\alpha)^2} \partial y^k + \frac{\partial f^i}{\partial x^\alpha} \frac{\partial f^j}{\partial x^\alpha} \Gamma_{ij}^k \partial y^k.$$

Thus $f(M)$ is a minimal surface if and only if for $k = 1, \dots, m$

$$\frac{\partial^2 f^k}{(\partial x^\alpha)^2} \partial y^k + \frac{\partial f^i}{\partial x^\alpha} \frac{\partial f^j}{\partial x^\alpha} \Gamma_{ij}^k \partial y^k = 0.$$

In arbitrary coordinates, Thus $f(M)$ is a minimal surface if and only if for $k = 1, \dots, m$

$$0 = \Delta_M f^k + (f^* g)^{\alpha\beta} \frac{\partial f^i}{\partial x^\alpha} \frac{\partial f^j}{\partial x^\beta} \Gamma_{ij}^k \partial y^k = \tau(f)^k$$

where Δ_M is the Laplacian of M . □

The restriction of f being an isometry can be loosened if we consider *conformal* maps on Riemann surfaces.

Definition 7.7. • We say that two metrics $\tilde{g}, g$ on a Riemannian manifold M are said to be of the *same conformal class* if there exists a function $\lambda : M \rightarrow \mathbb{R}$ such that

$$\tilde{g} = \lambda^2 g. \quad (7.13)$$

- We say that a smooth map $f : (M, g) \rightarrow (N, h)$ between Riemannian manifolds is *conformal* if there exists a function $\lambda : M \rightarrow \mathbb{R}$ such that

$$f^* h = \lambda^2 g. \quad (7.14)$$

Conformality has an interesting effect on the energy functional.

Theorem 7.8. Let $\dim M = 2$ and consider a harmonic map $f : (M, g) \rightarrow (N, h)$ be a harmonic map. The energy functional is conformally invariant : for g and $\tilde{g}$ metrics on M in the same conformal class, then

$$\mathcal{E}_g(f) = \mathcal{E}_{\tilde{g}}(f). \quad (7.15)$$

Proof. Let $\tilde{g}$ be in the same conformal class as g , such that there exists a $\lambda \in C^\infty(M)$ with $\tilde{g} = \lambda^2 g$.

- **Volume**

Let $d\mu_g$ and $d\mu_{\tilde{g}}$ be the volume forms for M , they are related as follows

$$\begin{aligned} d\mu_{\tilde{g}} &= \sqrt{\det(\tilde{g})} dx^1 dx^2 = \sqrt{\det(\lambda^2 g)} dx^1 dx^2 \\ &= \sqrt{\lambda^4 \det(g)} dx^1 dx^2 = \lambda^2 \sqrt{\det(g)} dx^1 dx^2 = \lambda^2 d\mu_g. \end{aligned}$$

- **Inner product**

Firstly, observe that since $(\tilde{g})_{\alpha\beta} = (\lambda^2 g)_{\alpha\beta}$, the induced tensors have opposite weight $(\tilde{g})^{\alpha\beta} = \lambda^{-2}(g)^{\alpha\beta}$. Now, we have

$$\|df\|_{\tilde{g}}^2 = \tilde{g}^{\alpha\beta} h_{ij} \frac{\partial f^i}{\partial x^\alpha} \frac{\partial f^j}{\partial x^\beta} = \lambda^{-2} g^{\alpha\beta} h_{ij} \frac{\partial f^i}{\partial x^\alpha} \frac{\partial f^j}{\partial x^\beta} = \lambda^{-2} \|df\|_g^2 \quad (7.16)$$

Putting everything together, we have

$$\mathcal{E}_{\tilde{g}}(f) = \frac{1}{2} \int_M \|df\|_{\tilde{g}}^2 d\mu_{\tilde{g}} = \frac{1}{2} \int_M \lambda^{-2} \|df\|_g^2 \lambda^2 d\mu_g = \frac{1}{2} \int_M \|df\|_g^2 d\mu_g = \mathcal{E}_g(f). \quad (7.17)$$

□

Remark 7.3.1. The above theorem explains the use of our notation, which stresses the metric.

Due to the canonical diffeomorphism $\mathbb{R}^2 \cong \mathbb{C}$, any surface has a canonical almost complex structure. Since conformality preserves angles, there is a one-to-one correspondence between an almost complex structure and a conformal structure. This correspondence will be crucial in chapter 8.

Definition 7.9. A *conformal structure* on a surface (M, g) is a smooth atlas with conformal coordinates: for a smooth chart $\varphi : U \rightarrow (\mathbb{R}^2, g^{\text{can}})$, there exists a function $\lambda : M \rightarrow \mathbb{R}$ such that $g^{\text{can}} = \lambda^2 g$.

From this, we have that surfaces are locally conformally flat, and the definition for a conformal map from a surface with a conformal structure becomes as follows

Definition 7.10. A map $f : (M, g) \rightarrow (N, h)$ from a surface M with conformal structure is *conformal* if

$$\left\| \frac{\partial f}{\partial x} \right\|^2 = \left\| \frac{\partial f}{\partial y} \right\|^2 \quad \text{and} \quad \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = 0. \quad (7.18)$$

In this way, we have the ‘relaxed’ version of Theorem 7.6.

Theorem 7.11. If $\dim M = 2$, for any smooth map $f : (M, g) \rightarrow (N, h)$, then $\mathcal{A}(f) \leq \mathcal{E}_g(f)$. With equality if and only if the map is conformal, as in Definition 7.10, in which case we call M a *parametric minimal surface*.

Proof. By the AM-GM inequality, we have

$$\sqrt{\left\|\frac{\partial f}{\partial x}\right\|^2 \left\|\frac{\partial f}{\partial x}\right\|^2 - \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle} \leq \sqrt{\left\|\frac{\partial f}{\partial x}\right\|^2 \left\|\frac{\partial f}{\partial x}\right\|^2} \leq \frac{1}{2} \left(\left\|\frac{\partial f}{\partial x}\right\|^2 + \left\|\frac{\partial f}{\partial x}\right\|^2 \right), \quad (7.19)$$

and by Theorem 7.8 we have $\mathcal{E}_g f, g) = \mathcal{E}_g f, g^{\text{can}})$, hence

$$\begin{aligned} \mathcal{A}(f) &= \int_M \sqrt{\left\|\frac{\partial f}{\partial x}\right\|^2 \left\|\frac{\partial f}{\partial x}\right\|^2 - \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle} dx dy \\ &\leq \int_M \frac{1}{2} \left(\left\|\frac{\partial f}{\partial x}\right\|^2 + \left\|\frac{\partial f}{\partial x}\right\|^2 \right) dx dy = \mathcal{E}_g f, g). \end{aligned} \quad (7.20)$$

If the mapping is conformal, then it is clear that $\mathcal{A}(f) = \mathcal{E}_g(f)$.

If $\mathcal{A}(g) = \mathcal{E}_g(f)$, then

$$2\sqrt{\det((f^*h)_{\alpha\beta})} = g^{\alpha\beta}(f^*h)_{\alpha\beta}\sqrt{\det(g_{\alpha\beta})}. \quad (7.21)$$

We can define $\lambda : M \rightarrow \mathbb{R}$ by

$$\frac{1}{2}\lambda^2 = \sqrt{\frac{\det((f^*h)_{\alpha\beta})}{\det(g_{\alpha\beta})}}$$

to obtain $(f^*h)_{\alpha\beta} = \lambda^2 g_{\alpha\beta}$. □

Corollary 7.11.1. *If $f : (M^2, g) \rightarrow (N, h)$ is a conformal harmonic map that minimizes $\mathcal{A}$, then it minimizes $\mathcal{E}_g$.*

Remark 7.3.2. It is important to note that a smooth map that minimizes $\mathcal{A}$ is minimal with respect to any metric of M . While a harmonic map that minimizes $\mathcal{E}$ is minimal with respect to the conformal class of the metric.

Remark 7.3.3. If M is a Riemannian surface (it has a complex structure), then the complex structure defines an equivalence conformal class for the Riemannian metric. Thus, the energy depends on the complex structure, and a minimizer is minimal with respect to the complex structure.

7.4. Hessian and Second Variation

We are now close to describing the Morse theory via harmonic maps of Sobolev spaces $H^k(M, N)$ ($2k > \dim M$).

Definition 7.12. The *Hessian* of the energy functional $\mathcal{E}_g : H^k(M, N) \rightarrow \mathbb{R}$ at a harmonic map $f : (M, g) \rightarrow (N, h)$,

$$\mathcal{E}_{**} : T_f H^k(M, N) \times T_f H^k(M, N) \rightarrow \mathbb{R},$$

is defined by the second variation of $\mathcal{E}_g$:

$$(W, V) \mapsto \left. \frac{\partial^2 \mathcal{E}_g(F_{s_1, s_2}(W, V))}{\partial s_1 \partial s_2} \right|_{(s_1, s_2) = (0, 0)}. \quad (7.22)$$

Theorem 7.13. *Let $f : (M, g, \nabla^M) \rightarrow (N, h, \nabla^N)$ be a smooth map between Riemannian manifolds. Consider a compactly supported two-parameter variation $F : (-\epsilon, \epsilon)^2 \times M \rightarrow N$ with variation fields*

$$V = \left. \frac{\partial F_{s_1, s_2}}{\partial s_1} \right|_{(s_1, s_2) = (0, 0)} \quad \text{and} \quad W = \left. \frac{\partial F_{s_1, s_2}}{\partial s_2} \right|_{s_1 = s_2 = 0}$$

Then the following equality holds

$$\begin{aligned} & \left. \frac{\partial^2}{\partial s_2 \partial s_1} \mathcal{E}_g(F_{s_1, s_2}) \right|_{s_1 = s_2 = 0} \\ &= - \int_M \left\langle \text{Tr}_g R^N(df, V, df), W \right\rangle_{f^*TN} d\mu_g + \int_M \left\langle \widetilde{\nabla} V, \widetilde{\nabla} W \right\rangle_{f^*TN} d\mu_g \\ & \quad - \int_M \left\langle \widetilde{\nabla}_{\partial s_2} V, \tau(f) \right\rangle_{f^*TN} d\mu_g \quad (7.23) \end{aligned}$$

where R^N is the curvature tensor of N , $\widetilde{\nabla}$ the pullback connection on f^*TN as in section 7.1 and $d\mu_g$ is the volume form of M .

If f is harmonic or if $\widetilde{\nabla}_{\partial s_2} V = 0$, then the second variation depends only on V and W and not on any higher derivatives of f with respect to s_1 and s_2 .

Proof. Let V and W be as in the statement, and let ∇ be the tensor connection on $\text{Hom}_f(TM, TN)$. By first considering the second partial derivative of the integrand, we see that

$$\begin{aligned} \frac{1}{2} \frac{\partial}{\partial s_2} \frac{\partial}{\partial s_1} \|df\|_{T^*M \otimes f^*TN}^2 &= \frac{1}{2} \frac{\partial}{\partial s_2} \frac{\partial}{\partial s_1} \left\langle \frac{\partial f}{\partial x^\alpha} dx^\alpha, \frac{\partial f}{\partial x^\beta} dx^\beta \right\rangle_{T^*M \otimes f^*TN} \\ &= \frac{\partial}{\partial s_2} \left\langle \nabla_{\partial s_1} \frac{\partial f}{\partial x^\alpha} dx^\alpha, \frac{\partial f}{\partial x^\beta} dx^\beta \right\rangle_{T^*M \otimes f^*TN} \\ &= \frac{\partial}{\partial s_2} \left\langle \nabla_{\partial x^\alpha} \frac{\partial f}{\partial s_1} dx^\alpha, \frac{\partial f}{\partial x^\beta} dx^\beta \right\rangle_{T^*M \otimes f^*TN}. \end{aligned}$$

By replacing $\frac{\partial}{\partial s_1}$ with V and using the fact that ∇ is compatible with the metric we have

$$\begin{aligned} &= \left\langle \widetilde{\nabla}_{\partial s_2} (\widetilde{\nabla}_{\partial x^\alpha} (V)) \otimes dx^\alpha, \frac{\partial f}{\partial x^\beta} dx^\beta \right\rangle_{T^*M \otimes f^*TN} \\ & \quad + \left\langle \widetilde{\nabla}_{\partial x^\alpha} (V) \otimes dx^\alpha, \nabla_{\partial s_2} \frac{\partial f}{\partial x^\beta} dx^\beta \right\rangle_{T^*M \otimes f^*TN}, \end{aligned}$$

further, via the symmetry of the connection and by replacing $\frac{\partial f}{\partial s_2}$ with W , the equality becomes

$$\begin{aligned} &= \left\langle \widetilde{\nabla}_{\partial s_2} (\widetilde{\nabla}_{\partial x^\alpha} (V)) \otimes dx^\alpha, \frac{\partial f}{\partial x^\beta} dx^\beta \right\rangle_{T^*M \otimes f^*TN} \\ & \quad + \left\langle \widetilde{\nabla}_{\partial x^\alpha} (V) \otimes dx^\alpha, \widetilde{\nabla}_{\partial x^\beta} (W) \otimes dx^\beta \right\rangle_{T^*M \otimes f^*TN}. \end{aligned}$$

By definition of the curvature tensor and R^N on N , we have

$$\begin{aligned}
&= \left\langle R^N \left(\frac{\partial f}{\partial s_2}, \frac{\partial f}{\partial x^\alpha}, V \right) \otimes dx^\alpha, \frac{\partial f}{\partial x^\beta} dx^\beta \right\rangle_{T^*M \otimes f^*TN} \\
&\quad + \left\langle \widetilde{\nabla}_{\partial x^\alpha} (\widetilde{\nabla}_{\partial s_2} (V)) \otimes dx^\alpha, \frac{\partial f}{\partial x^\beta} dx^\beta \right\rangle_{T^*M \otimes f^*TN} \\
&\quad + \left\langle \widetilde{\nabla}_{\partial x^\alpha} (V) \otimes dx^\alpha, \widetilde{\nabla}_{\partial x^\beta} (W) \otimes dx^\beta \right\rangle_{T^*M \otimes f^*TN},
\end{aligned}$$

which we simplify to

$$= - \left\langle \text{Tr}_g R^N(df, V, df), W \right\rangle_{f^*TN} + \left\langle \nabla(\widetilde{\nabla}_{\partial s_2} V), df \right\rangle_{T^*M \otimes f^*TN} + \left\langle \widetilde{\nabla} V, \widetilde{\nabla} W \right\rangle_{f^*TN} \quad (7.24)$$

by using the short hand $\nabla X = \nabla_{\partial x^\alpha} (X) \otimes dx^\alpha$ for $X \in \Gamma(f^*TN)$. Since ∇ is metric, the middle term can be simplified further

$$\begin{aligned}
\left\langle \nabla(\widetilde{\nabla}_{\partial s_2} V), df \right\rangle_{T^*M \otimes f^*TN} &= \left\langle \nabla_{\partial x^\alpha} (\widetilde{\nabla}_{\partial s_2} V) \otimes dx^\alpha, \frac{\partial f}{\partial x^\beta} dx^\beta \right\rangle_{T^*M \otimes f^*TN} \\
&= - \left\langle \widetilde{\nabla}_{\partial s_2} V \otimes dx^\alpha, \nabla_{\partial x^\alpha} \frac{\partial f}{\partial x^\beta} dx^\beta \right\rangle_{T^*M \otimes f^*TN} \\
&= - \left\langle \widetilde{\nabla}_{\partial s_2} V, \text{Tr}_g \nabla df \right\rangle_{f^*TN} \\
&= - \left\langle \widetilde{\nabla}_{\partial s_2} V, \tau(f) \right\rangle_{f^*TN}.
\end{aligned}$$

At last, we have obtained the desired equality

$$\begin{aligned}
&\frac{\partial^2}{\partial s_2 \partial s_1} \mathcal{E}_g(F_{s_1, s_2}) \Big|_{s_1=s_2=0} \\
&= - \int_M \left\langle \text{Tr}_g R^N(df, V, df), W \right\rangle + \int_M \left\langle \widetilde{\nabla} V, \widetilde{\nabla} W \right\rangle_{f^*TN} d\mu_g - \int_M \left\langle \widetilde{\nabla}_{\partial s_2} V, \tau(f) \right\rangle_{f^*TN} d\mu_g.
\end{aligned}$$

□

Corollary 7.13.1. *Under the assumption of Theorem 7.13 with a compactly supported one-parameter variation $F : (\epsilon, \epsilon) \times M \rightarrow N$, if f is harmonic or if $f(x, \cdot)$ is a geodesic for every x , then*

$$\frac{d^2}{ds^2} \mathcal{E}_g(F_s) \Big|_{s=0} = \int_M \|\widetilde{\nabla} V\|_{f^*TN}^2 - \int_M \left\langle \text{Tr}_g R^N(V, df, df), V \right\rangle_{f^*TN} d\mu_g. \quad (7.25)$$

Similarly to Corollary 4.6.1, from the formula of the second variation, it is easy to see that $\mathcal{E}_{**}$ is well-defined.

Lemma 7.14. *Let f, M, N, V and W be as in Theorem 7.13. The Hessian $\mathcal{E}_{**} : T_f H^k(M, N) \times T_f H^k(M, N) \rightarrow \mathbb{R}$ at a harmonic map $f : (M, g) \rightarrow (N, h)$ is a well-defined symmetric bilinear function.*

Corollary 7.14.1. *Under the assumptions of Theorem 7.13, if N has non-positive sectional curvature, then a harmonic map is a stable critical point of the energy functional in the sense that the second variation of the energy is non-negative.*

Proof. If N has non-positive curvature, then

$$\left\langle \text{Tr}_g R^N(X(p), df(v), df(v)), X(p) \right\rangle_{f^*TN} \leq 0$$

for all $p \in M$, $v \in T_p M$ and any $X \in \Gamma(f^*TN)$. Hence, we have the desired result. $\square$

This corollary shows that all harmonic maps to a Riemannian manifold with non-positive curvature are stable.

7.5. Null-space of the Energy Functional

As section 4.3, we wish to determine the Morse index of a harmonic map via Jacobi fields. So we first state the definitions for harmonic maps.

Jacobi Operator

The terms in the second variation can be rewritten as

$$- \int_M \left\langle \text{Tr}_g \left(R^N(V, df, df) + \widetilde{\nabla}^2 V \right), W \right\rangle_{f^*TN} d\mu_g. \quad (7.26)$$

where $\widetilde{\nabla}^2$ is the *second covariant derivative* from *Introduction to Riemannian manifolds* [15] Proposition 4.21. Using the notation of the *Rough Laplacian along the harmonic map* f $\Delta_f V = -\text{Tr}_g \widetilde{\nabla}^2 V$, we further have

$$- \int_M \left\langle \text{Tr}_g \left(R^N(V, df, df) \right) + \Delta_f V, W \right\rangle_{f^*TN} d\mu_g. \quad (7.27)$$

This is without doubt the higher-dimensional equivalent of the Jacobi operator.

Definition 7.15. A vector field J along a harmonic map $f : (M, g) \rightarrow (N, g)$ is called a *Jacobi field* if it is in the kernel of the *Jacobi Operator* $J_f : T_f H^k(M, N) \rightarrow T_f H^k(M, N)$ ¹: i.e

$$J_f(V) = \text{Tr}_g(R^N(V, df, df)) + \Delta_f V = 0. \quad (7.28)$$

Theorem 7.16. *The Jacobi operator is a second-order elliptic self-adjoint differential operator.*

Proof. Second-order elliptic differential operator is by “La formule de la variation seconde de l’énergie au voisinage d’une application harmonique” [21], while self-adjointness is because the operator $\mathcal{E}_{**}$ is symmetric. $\square$

¹Recall that in section 5.2 we showed that $T_f C^\infty(M, N) \cong \Gamma(f^*TN)$

Similarly to Jacobi fields along geodesics, the variation field of a *variation through harmonic maps* is a Jacobi field.

Definition 7.17. We say that a one-parameter compactly supported variation $F : (-\epsilon, \epsilon) \times M \rightarrow N$ of a harmonic map $f : (M, g) \rightarrow (N, h)$ is a *variation through harmonics* if the s -stages $F_s : M \rightarrow N$ are harmonic maps.

Theorem 7.18. *Let $f : (M, g) \rightarrow (N, h)$ be a harmonic map and $F : (-\epsilon, \epsilon) \times M \rightarrow N$ be a variation of f through harmonics. Then the variation field of F is a Jacobi field.*

Proof. Let $F : (-\epsilon, \epsilon) \times M \rightarrow N$ be a variation of f through harmonic maps and consider $V_s = \frac{\partial F_s}{\partial s}$. Since each s -stage is harmonic, the tension fields are all zero $\tau(F_s) = 0$. Therefore for each $s \in (-\epsilon, \epsilon)$,

$$\begin{aligned} 0 = \nabla_{V_s}(\tau(F_s)) &= \nabla_S(g^{\alpha\beta} \nabla_{\partial x^\alpha} \nabla_{\partial x^\beta} F_s^i) \\ &= g^{\alpha\beta} \nabla_S \nabla_{\partial x^\alpha} \nabla_{\partial x^\beta} F_s^i \\ &= g^{\alpha\beta} \left(\nabla_{\partial x^\alpha} \nabla_S \nabla_{\partial x^\beta} F_s^i + R^N \left(S, \frac{\partial F_s^i}{\partial x^\alpha}, \frac{\partial F_s^i}{\partial x^\beta} \right) \right). \end{aligned}$$

Specifically, this holds for $s = 0$ when V_0 is the variation vector field of F . $\square$

Like for geodesics, we can define *integrability* for Jacobi fields along a harmonic map.

Definition 7.19. A Jacobi field J along a harmonic map f is called *integrable* if there exists a compactly supported variation $F : (-\epsilon, \epsilon) \times M \rightarrow N$ of f through harmonics such that J is its variation field.

However, unlike geodesics, determining the integrability of a Jacobi field is not immediate, nor always true. Even in cases when the domain of a harmonic map is somewhat elementary, like the 2-Sphere, S^2 , one cannot make a universal statement about the integrability of Jacobi fields; see “Jacobi fields along harmonic 2-spheres in S^3 and S^4 are not all integrable”[17] and “Jacobi Fields Along Harmonic 2-Spheres in $\mathbb{C}P^2$ are Integrable” [18]. Nonetheless, attempts to understand when Jacobi fields are integrable are still being made. D. Adams and L. Simon proved in “Rates of asymptotic convergence near isolated singularities of geometric extrema” [1] the following theorem on the integrability of Jacobi fields.

Theorem 7.20. *Let $f : (M, g) \rightarrow (N, h)$ be a harmonic map between real-analytic manifolds². Then all Jacobi fields along f are integrable if and only if the space of harmonic maps $C^{2,\alpha}$ -close³ to f is a smooth manifold, whose tangent space at f is $\ker J_f$.*

The following pairs are the known real-analytic manifolds for which the above theorem holds:

- $M = S^1$ and N is a globally symmetric space,

²Differential manifolds such that the transition maps are real-analytic. Recall that an analytic function is locally represented by a convergent power series and that every real-analytic function is smooth, but not all real smooth functions are real-analytic.

³see the definition of the Hölder space in chapter 5.

- $M = S^1$ and N a manifold all of whose geodesics are closed and of the same length,
- N is a locally symmetric space of non-positive sectional curvature,
- $M = N = S^2$.

Remark 7.5.1. The aforementioned theorem can be readily extended to complex-analytic manifolds, leading to further investigation of the energy functional of harmonic maps between Kähler manifolds.

Interestingly, even though global integrability is an elusive subject, local integrability isn't. The following theorem, from "Stability of Harmonic Maps and Eigenvalues of the Laplacian" [29], shows that local integrability is always attained.

Theorem 7.21. *Every Jacobi vector field along a harmonic map $f : M \rightarrow N$ is locally the variation field of some one-parameter variation of f through harmonic maps.*

Not only is the integrability of Jacobi fields an open problem, but so is identifying a higher-dimensional counterpart to conjugate points. In chapter 4, we defined conjugate points to be the points on a curve where a Jacobi field vanishes. It is possible to find such points even in the case of harmonic maps. However, unlike geodesics, points alone do not affect minimality.

As conjugate points represent the compact hypersurfaces of the geodesics, it may be favorable for a higher-dimensional analogue of points along a geodesic to be represented by (compact) hypersurfaces – sub-manifolds of co-dimension one – of the image of a harmonic map. This generalization could be termed '*conjugate sub-manifolds*'. Another way to see conjugate points is as the locations on a geodesic curve where a family of geodesics arises. Thus, the idea for harmonic maps would be that conjugate sub-manifolds are manifolds that can be joined by a family of harmonic maps. However, since there is no 'global' integrability of Jacobi fields, this proposed definition may not be sensible, as it cannot be generalized to every harmonic map between Riemannian manifolds.

Null-space of Energy Functional

Finally, we can once again determine the nullity and the Morse index of the Hessian $\mathcal{E}_{**}$ via the Jacobi field.

A harmonic map $f \in H^k(M, N)$ is *non-degenerate* if the *null-space* of the Hessian $\mathcal{E}_{**}$,

$$\text{Nul}(\mathcal{E}_{**}) := \left\{ V \in T_f H^k(M, N) : \mathcal{E}_{**}(V, W) = 0 \quad \forall W \in T_f H^k(M, N) \right\}, \quad (7.29)$$

is trivial, equivalently if $\nu = \dim \text{Nul } \mathcal{E}_{**} = 0$. The Morse index of $\mathcal{E}_{**}$ at a non-degenerate harmonic map f is the dimension of the maximal subspace of $T_f H^k(M, N)$ such that $\mathcal{E}_{**}$ is negative definite.

Analogously to the case for geodesics, the null space of the Hessian $\mathcal{E}_{**}$ is closely related to the kernel of the Jacobi operator.

Theorem 7.22. *Let $\mathcal{E}_{**} : T_f H^k(M, N) \times T_f H^k(M, N) \rightarrow \mathbb{R}$ be the Hessian, then $V \in \text{Nul}(\mathcal{E}_{**})$ if and only if $J_f(V) = 0$. Furthermore, the Morse index of $\mathcal{E}_{**}$ is the sum of the multiplicities of the negative eigenvalues of the Jacobi operator.*

Proof. Let $W \in T_f H^k(M, N)$ be arbitrary then

$$\begin{aligned}\mathcal{E}_{**}(V, W) &= - \int_M \langle \text{Tr}_g(R^n(V, df, df)) + \Delta_f V, W \rangle_{f^*TN} d\mu_g = 0 \\ &\iff \langle \text{Tr}_g(R^n(V, df, df)) + \Delta_f V, W \rangle_{f^*TN} = 0 \\ &\iff \text{Tr}_g(R^n(V, df, df)) + \Delta_f V = J_f(V) = 0.\end{aligned}$$

□

In the case when the manifold M is compact, we know that the kernel of the Jacobi operator is finite, thus the nullity of $\mathcal{E}_{**}$ is finite.

7.6. Obstruction: Palais-Smale Condition

The following definitions and theorems are from *Infinite Dimensional Morse Theory and Multiple Solution Problems* [7]

Even though the calculation of the Morse index is burdensome in the case of harmonic maps, it does not compare to the far more significant interference for the Morse theory.

To define the CW-Structure of a Sobolev space $H^k(M, N)$, we need theorems ensuring the Handle-Body Decomposition, Theorem 2.10, as it was done in chapter 4. In chapter 4, as we were working with the Path Space – an object thoroughly studied in both topology and geometry – we could describe the topology of the Path Space by hand. However, there is an obstruction when dealing with arbitrary infinite-dimensional manifolds: the *Palais-Smale condition*.

Definition 7.23. Let M be a Banach manifold, $G : M \rightarrow \mathbb{R}$ a differentiable map, and $c \in \mathbb{R}$. We say that G satisfies the Palais-Smale condition at c , $(PS)_c$, if for any sequence $\{x_n\} \subset M$ such that $G(x_n) \rightarrow c$ and $dG(x_n) \rightarrow 0$ strongly possess a convergent subsequence. We say that G satisfies the Palais-Smale condition (PS) if it satisfies $(PS)_c$ for all c .

The Palais-Smale condition ensures the existence of a limit of a converging sequence of critical points and allows for the infinite-dimensional analogues to Theorem 2.8 and Corollary 2.9.1.

Lemma 7.24 (Deformation). *Let M be a Banach manifold and $G : M \rightarrow \mathbb{R}$ be a differentiable map satisfying $(PS)_c$ for $c \in [a, b]$ and K the set of critical points of G . If $K \cap G^{-1}[a, b] = \emptyset$, then M^a is a deformation retract of M^b .*

Theorem 7.25 (Handle-Body Theorem). *Suppose $G : M \rightarrow \mathbb{R}$ is a second-differentiable map satisfying the Palais-Smale condition, where M is a smooth Hilbert-Riemannian manifold, and c is an isolated critical value. If $K_c = \{z_1, z_2, \dots, z_l\}$ consists of only nondegenerate critical points with Morse indices $\{m_1, m_2, \dots, m_l\}$ respectively, then there exist $\epsilon > 0$ and homeomorphisms $h_i : B^{m_i} \rightarrow M$ for $i = 1, \dots, l$ such that*

$$M^{c+\epsilon} \simeq M^{c-\epsilon} \cup \bigcup_{i=1}^l h_i(B^{m_i}),$$

satisfying for each $i \in \{1, \dots, l\}$

$$M^{c-\epsilon} \cap h_i(B^{m_i}) = G^{-1}(c - \epsilon) \cap h_i(B^{m_i}) = h_i(\partial B^{m_i}), \quad (7.30)$$

where B^{m_i} is the m_i -dimensional ball, and the notation $\simeq$ means a deformation retract.

However, as the next theorem shows, one does not have the (PS) condition for the energy functional.

Theorem 7.26. *Let $f : (M, g) \rightarrow (N, h)$ be a smooth map between Riemannian manifolds. If $p > \dim M$, then the p -energy functional $\mathcal{E}^p(f, g) = \int_M \|df\|^p d\mu_g$ satisfies the Palais-Smale condition.*

For the proof, we refer to *Foundations of global non-linear analysis* [24].

Corollary 7.26.1. *Let $\mathcal{P}M = H^1(I, M)$ be the Path Space from chapter 4, then the energy functional satisfies the Palais-Smale condition.*

Remark 7.6.1. This can be promptly seen when (M, g) is a Riemann surface. As seen in the section relating minimal sub-manifolds to harmonic maps, the energy functional for maps from surfaces is conformal invariant.

Consider the charts for S^2 to be $S^2 \setminus \{(0, 0, 1)\}$ and $S^2 \setminus \{(0, 0, -1)\}$, and a sequence of harmonic maps $\{h_\lambda : S^2 \rightarrow S^2\}_\lambda$, such that in the chart of $S^2 \setminus N$, $h_\lambda(z) = \lambda z$. Given any conformal metric g , the energy function is non-zero $\mathcal{E}(h_\lambda) = c \neq 0$, and

$$\mathcal{E}(h_1) = \mathcal{E}(h_\lambda)$$

as $h_\lambda = \text{Id} \circ h_\lambda$ for $\lambda \neq 0$, Id trivially being a conformal map, and $h_1 = \text{Id}$. Observe that if $\lambda \rightarrow 0$, then we have point-wise convergence, in the $S^2 \setminus \{(0, 0, 1)\}$ chart, to $h_0 = 0$. So we have a sequence of holomorphic maps such $\mathcal{E}(h_\lambda) = c \neq 0$ and $d\mathcal{E}(h_\lambda) = 0$, but with

$$0 = \mathcal{E}(h_0) = \lim_{\lambda \rightarrow 0} \mathcal{E}(h_\lambda) = \lim_{\lambda \rightarrow 0} c \neq 0.$$

We sadly cannot define a CW-structure for the Sobolev space $H^k(M, N)$.

J. Sacks and K. Uhlenbeck considered a method to bypass the issue. They claim that for a Riemann surface M and α a real number greater than 1, the perturbed energy functional $\mathcal{E}_\alpha : H^k(M, N) \rightarrow \mathbb{R}$,

$$\mathcal{E}_\alpha(f) = \int_M (1 + \|df\|^2)^\alpha, \quad (7.31)$$

satisfies the Palais-Smale condition, and with critical points, the α -harmonic maps. With this, they showed that as α approaches 1, a sequence of α -harmonic maps converges to a harmonic map with the exchange of *bubbling*.

Theorem 7.27. *Let M be a compact Riemann surface and N a compact Riemannian manifold. Suppose $\alpha(i) > 1$ with $\alpha(i) \rightarrow 1$ as $i \rightarrow \infty$ and that there exist a set $\{f_{\alpha(i)} : M \rightarrow N\}$ of α_i -harmonic maps with bounded energy $\mathcal{E}_{\alpha_i}(f_{\alpha(i)}) < \mathcal{E}_{\max}$, for $\mathcal{E}_{\max} \in \mathbb{R}$.*

Then after passing to a subsequence $\{\alpha(i')\} \subset \{\alpha(i)\}$, there is a finite set $S = \{p_1, \dots, p_d\} \subset M$ and harmonic map $f : M \rightarrow N$ such that $f_{\alpha(i')} \rightarrow f$ on a compact set $M \setminus S$.

For each $p_j \in S$ there are non-constant harmonic maps $\varphi_j : S^2 \rightarrow N$ such that after a suitable sequence of rescalings, $f_{\alpha(i')}$ converges on compact subsets of $\mathbb{C} = S^2 \setminus \{\infty\}$ to φ_j . The energies $\mathcal{E}_{\alpha(i)}(f_{\alpha(i)})$ converge to a finite value and there is an energy inequality

$$\mathcal{E}(f) + \sum_{j=1}^d \mathcal{E}(\varphi_j) \leq \lim_{i \rightarrow \infty} \mathcal{E}_{\alpha(i)}(f_{\alpha(i)}) \leq \mathcal{E}_{\max}.$$

The maps φ_j are the ‘bubbles’ that are formed in the image of the harmonic maps. From the bubbling formation, we can see that if we consider Riemannian manifolds N such that $\pi_2(N) = 0$, that is, all maps $f : S^2 \rightarrow N$ are null-homotopic, we can bypass this obstruction.

Theorem 7.28. *Let M be a compact Riemann surface and N a Riemannian manifold such that $\pi_2(N) = 0$. Then the Handle-Body decomposition of Theorem 7.27 holds.*

Part III.

Moving Forward: Hamiltonian Formalism

8. Hamiltonian Formalism: Closed Geodesics

This chapter references *Morse Theory and Floer Homology*, *Summer Seminar on Floer Theory*[31], *Lectures on Floer Homology* [26], “Perturbed closed geodesics are periodic orbits: Index and transversality” [30], and *Morse Theory and Floer Homology* [4].

It is now clear that one cannot actually define a Morse theory for harmonic maps. To circumvent this issue, mathematicians have looked at new ways to approach harmonic maps. One such way is using the perturbed energy functional $\mathcal{E}_\alpha$ and its α -harmonic maps, which were considered at the end of the previous chapter. Another way is to consider *Floer homology*. This second approach, which we will call the ‘*Hamiltonian formalism*’ is what we will look at.

As has been the trend in this thesis, the study of the Hamiltonian formalism is well-established in the case of geodesics. So we begin by describing the base case of a symplectic manifold and 1-periodic Hamiltonian functions, which defines the symplectic action functional on the space of 1-periodic contractible loops $\mathcal{L}_0$. With this functional, we obtain Floer equations, which produce the cornerstone of the description of the Floer homology of $\mathcal{L}_0$

8.1. (Hamiltonian) Floer Homology and The Symplectic Action Functional

Floer Homology was constructed to be an infinite-dimensional analogue of Morse Homology for the Loop Space $\mathcal{L}N = H^1(S^1, N)$ of a symplectic manifold N ¹. In the case of Morse homology, we considered Morse functions on a smooth manifold; for Floer Homology, we shall consider the *Symplectic Action Functional*. The following table represents all the adaptations that will be discussed in this section.

¹the Loop Space can be seen to be the subspace of the path space $\mathcal{P}N$ where all the paths have the same beginning and end points

$$\mathcal{L}N = \{\gamma \in \mathcal{P}N = H^1(I, N) \mid \gamma(0) = \gamma(1)\}.$$

	Morse Theory	Floer Theory
Domain:	Finite dimensional manifolds	Loop space
Function :	Morse function	Symplectic action functional
Chain complex	Critical points of the Morse function	1-periodic orbits of the Hamiltonian flow
Generators :		
Index:	Morse index	Conley-Zehnder index
Boundary morphism:	Negative gradient flow lines	Floer trajectories

Symplectic Action Functional

Recall that for (N, ω) a symplectic manifold with symplectic form ω , there exist *Hamiltonian vector fields* X_H generated by *Hamiltonian functions* $H : N \rightarrow \mathbb{R} : X_H \lrcorner \omega = dH$. By definition, a Hamiltonian vector field X_H is zero at a point p if and only if p is a critical point of H , and we define the *Hamiltonian system* to be the system of differential equations associated with X_H to be $\dot{\gamma}(t) = X_H(\gamma)$, for γ a curve in N .

Remark 8.1.1. On the standard symplectic manifold $\mathbb{R}^{2n}$ with standard symplectic form $\omega_0 = \sum_i dp^i \wedge dq^i$ for coordinates $(q^1, \dots, q^n, p^1, \dots, p^n)$, the Hamiltonian system takes the form

$$\left(\dot{q}^i = \frac{\partial H}{\partial p^i}, \dot{p}^i = -\frac{\partial H}{\partial q^i} \right)$$

When on the standard symplectic manifold $(\mathbb{R}^{2n}, \omega_0)$, we consider the function

$$H(p, q) = \frac{1}{2} \|p\|^2 + V(q),$$

the associated Hamiltonian system is of the form

$$\left(\dot{q}^i = \frac{\partial H}{\partial p^i} = p, \dot{p}^i = -\frac{\partial H}{\partial q^i} = V(q) \right)$$

Remark 8.1.2. Observe that for a curve $\gamma : I \rightarrow N$, we can defined a Hamiltonian function $H(\gamma, \dot{\gamma}) = \frac{1}{2} \|\dot{\gamma}\|^2$, whose Hamiltonian system is equivalent to the the geodesic equation $\frac{D}{dt} \frac{d\gamma}{dt} = 0$:

$$\left(p = \frac{d\gamma}{dt}, \dot{p}^i = 0 \right).$$

With this in mind, consider a family of time-dependent 1-periodic Hamiltonian functions $\{H_t : N \rightarrow \mathbb{R}\}$ and associated Hamiltonian vector fields $X_t = X_{H_t}$, with Hamiltonian system $\dot{\gamma}(t) = X_t(\gamma)$.

The solutions generate a family of symplectomorphisms $\psi^t : M \rightarrow M$ via

$$\frac{d\psi^t}{dt} = X_{H_t} \circ \psi^t.$$

The fixed points of the ψ^1 are the one-periodic solutions of the Hamiltonian equation, forming a set which we will denote by

$$\mathbf{P}(H) := \{\gamma : S^1 \rightarrow N \mid \dot{\gamma}(t) = X_{H_t}(\gamma(t))\}.$$

Thus, from a family of 1-periodic Hamiltonian functions $\{H_t\}$ we can define a one form on the Tangent space of the loop space $\mathcal{L}N$,

$$\mathcal{A}_H : T\mathcal{L}N \rightarrow \mathbb{R} \tag{8.1}$$

$$(\gamma, \xi) \mapsto \mathcal{A}_H(\gamma, \xi) = \int_{S^1} \omega(\dot{\gamma}(t) - X_{H_t}(\gamma(t)), \xi) \tag{8.2}$$

which is called the *symplectic action functional* associated with $\{H_t\}$.

Definition 8.1. If $\pi_2(N) = 0$, then the symplectic action functional $\mathcal{A}_H$ on the space of contractible loops,

$$\mathcal{L}_0N := \{\gamma \in C^\infty(S^1, N) \mid \gamma \text{ is nulhomotopic}\}, \tag{8.3}$$

is defined as

$$\mathcal{A}_H : \mathcal{L}_0N \rightarrow \mathbb{R} \tag{8.4}$$

$$\gamma \mapsto \mathcal{A}_H(\gamma)(t) = - \int_{\mathbb{D}^2} u^* \omega - \int_0^1 H_t(\gamma(t)) \tag{8.5}$$

for $\gamma \in \mathcal{L}_0N$ where $u : \mathbb{D}^2 \rightarrow M$ is a smooth map such that $u(e^{2\pi it}) = \gamma(t)$ for $t \in \mathbb{R}$.

Remark 8.1.3. Consider the case for the standard symplectic manifold $\mathbb{R}^{2n}$ with the symplectic form $\omega_0 = \sum_i dp^i \wedge dq^i = d(\sum_i p^i dq^i)$. The symplectic action functional has the form

$$\mathcal{A}_H(\gamma) = \int_1^0 (H_t dt - pdq).$$

Floer Trajectories

The next step in defining Floer Homology is to identify the stable and unstable manifolds of the action functional. The following proposition gives us the set of critical points of the action functional.

Proposition 8.2. *The critical points of the symplectic action functional are the one-periodic orbits of the Hamiltonian flow: $\text{Crit}(\mathcal{A}_H) = \mathbf{P}(H)$.*

Proof. For $\gamma \in \mathcal{L}_0N$ and $\xi \in T_\gamma \mathcal{L}_0N$, we have a family of loops $\{\gamma_\tau\} \subset \mathcal{L}_0N$ such that $\frac{d}{d\tau} \gamma_\tau \Big|_{\tau=0} = \xi$ and $\gamma_0 = \gamma$, as well as a family of maps $\{u_\tau : \mathbb{D}^2 \rightarrow N\}$. Then

$$\begin{aligned} d\mathcal{A}_H(\gamma)(\xi) &= \frac{d}{d\tau} \Big|_{\tau=0} \left(- \int_{\mathbb{D}^2} u_\tau^* \omega - \int_{S^1} H_t(\gamma_\tau(t)) \right) (\xi) \\ &= - \int_{S^1} \omega(\xi(t), \dot{\gamma}(t)) dt - \int_{S^1} H_t(\gamma_\tau(t)) (\xi) \\ &= \int_{S^1} \omega(\dot{\gamma}(t), \xi(t)) - \omega(X_{H_t}(\gamma(t), \xi(t))) dt \\ &= \int_{S^1} \omega(\dot{\gamma}(t) - X_{H_t}(\gamma(t)), \xi) \end{aligned}$$

□

As in the case for the Morse Homology of a function, we look at the gradient of the symplectic action functional (with respect to the induced metric on $\mathcal{L}_0 N^2$), namely

$$\nabla \mathcal{A}_H(\gamma)(t) = J(\gamma(t))\dot{\gamma}(t) - \nabla H_t(\gamma(t)),$$

where J is the complex structure on N compatible with ω ³ and the gradient of H_t is with respect to the metric of N . The gradient flow lines of $\mathcal{A}_H$ are the solutions $u : \mathbb{R} \times S^1 \rightarrow N$ of $\frac{\partial u}{\partial s} + \nabla \mathcal{A}_H(u(s, \cdot)) = 0$, i.e. smooth maps $u : \mathbb{R} \times S^1 \rightarrow N$ solving the *Floer equation*

$$\frac{\partial u}{\partial s} + J(u) \frac{\partial u}{\partial t} - \nabla H_t(u) = 0. \quad (8.6)$$

The left-hand side of Equation 8.6 can be written as what we will call the ‘*Floer Map*’:

$$\mathcal{F} : H^1(\mathbb{R} \times S^1, N) \rightarrow L^2(\mathbb{R} \times S^1, N) \quad (8.7)$$

$$u \mapsto \partial_s u + J(u) \partial_t u - \nabla H_t(u) \quad (8.8)$$

where the solutions of Equation 8.6 are then the zeros of $\mathcal{F}$. With this nomenclature, we call the elements of the kernel of $\mathcal{F}$ *Floer Trajectories*.

Finally, from the Floer trajectories, we can define the Unstable and Stable sets, as it was done for Morse Homology with the gradient flow lines.

Definition 8.3. For $\gamma, \omega \in \text{Crit } \mathcal{A}_H$, the Stable and Unstable sets are the space of Floer trajectories :

$$\begin{aligned} W^s(\gamma) &:= \left\{ u \in W^{1,p}(\mathbb{R} \times S^1, N) : \partial_s u = -\nabla \mathcal{A}_H(u) \quad \lim_{s \rightarrow +\infty} u(s) = \gamma \right\} \\ W^u(\omega) &:= \left\{ u \in W^{1,p}(\mathbb{R} \times S^1, N) : \partial_s u = -\nabla \mathcal{A}_H(u) \quad \lim_{s \rightarrow -\infty} u(s) = \omega \right\} \end{aligned}$$

with

$$\mathcal{M}(\gamma, \omega) := \left\{ u \in W^{1,p}(\mathbb{R} \times S^1, N) : \partial_s u = -\nabla \mathcal{A}_H(u), \quad \lim_{s \rightarrow +\infty} u(s) = \gamma \text{ and } \lim_{s \rightarrow -\infty} u(s) = \omega \right\}. \quad (8.9)$$

Similarly to the Moduli spaces in Morse homology, the Moduli spaces $\mathcal{M}(\gamma, \omega)$ are also finite-dimensional manifolds.

Conley-Zahnder Index

The last component that we need is the Morse index equivalent. Since Floer homology is based on Morse homology, we want to define the index in terms of the dimensions of the moduli spaces.

²For g the metric on N , we define on $\mathcal{L}_0 N$ the induced metric

$$\langle \varphi, \psi \rangle = \int_0^1 g(\varphi(t), \psi(t)) dt.$$

³We say that a complex structure $J : TN \rightarrow TN$ is compatible to a symplectic structure on (N, g) if for any $X, Y \in \mathfrak{X}(N)$, $\omega(JX, JY) = \omega(X, Y)$ and $\omega(X, JY) = \langle X, Y \rangle_g$.

Theorem 8.4. *There exists a function $\mu_{CZ} : \text{Crit}(\mathcal{A}_H) \rightarrow \mathbb{Z}$ called the Conley-Zehnder index such that for all $\gamma, \omega \in \text{Crit}(\mathcal{A}_H)$*

$$\dim \mathcal{M}(\gamma, \omega) = \mu_{CZ}(\gamma) - \mu_{CZ}(\omega) \quad (8.10)$$

Corollary 8.4.1. *For $\gamma, \omega \in \text{Crit}(\mathcal{A}_H)$ the set $\widetilde{\mathcal{M}}(\gamma, \omega)$ is also a finite dimensional manifold of dimension*

$$\dim \widetilde{\mathcal{M}}(\gamma, \omega) = \mu_{CZ}(\gamma) - \mu_{CZ}(\omega) - 1$$

Due to Proposition 8.2, the Conley-Zehnder index is explicitly described in terms of the associated symplectomorphism of the Floer trajectories.

Definition 8.5. The Conley-Zehnder index of $\gamma \in \text{Crit}(\mathcal{A}_H) = \text{P}(H)$ is defined to be $\mu_{CZ}(\gamma) = \mu_{CZ}(\Psi)$ for ψ the associate symplectomorphism.

The Conley-Zehnder index of a path of symplectic matrices $\Psi : [0, 1] \rightarrow \text{Sp}(2n)$ starting at the identity $\Psi(0) = \mathbb{1}$ and ending in

$$\text{Sp}^*(2n) := \{A \in \text{Sp}(2n) \mid \text{not having } 1 \text{ as an eigenvalue}\}$$

is defined as

$$\mu_{CZ}(\Psi) = \frac{1}{2} \text{sign } \Gamma(\Psi, 0) + \sum_{t>0} \text{sign } \Gamma(\Psi, t) \quad (8.11)$$

summing over all crossings $0 < t < 1$ if and only if $\det(\mathbb{1} - \Psi(t)) = 0$ with corresponding crossing form

$$\Gamma(\Psi, t) : \ker(\mathbb{1} - \Psi(t)) \rightarrow \mathbb{R}$$

defined by

$$\xi_0 \mapsto \omega_0(\xi_0, \dot{\Psi}(t)\xi_0)$$

where ω_0 is the canonical symplectic form on R^{2n} .

The Floer Complex

Finally, we can describe the Floer complexes and homology.

Definition 8.6. The *Floer Chain groups* $\{C_k\}_{k \in \mathbb{N}}$ are defined as follows

$$C_k(\mathcal{A}_H) := \left\{ \sum_{i=1}^m a_i \gamma_i : m \in \mathbb{N}, a_i \in \mathbb{Z}_2, \gamma_i \in \text{Crit}(\mathcal{A}_H), \mu_{CZ}(\gamma_i) = k \right\}, \quad (8.12)$$

furthermore, the boundary operator $\partial_k : \text{Crit}_{k+1}(\mathcal{A}_H) \rightarrow \text{Crit}_k(\mathcal{A}_H)$ is the linear extension of the map

$$\partial_k(\gamma_i) = \sum_{\substack{y \in \text{Crit}(f) \\ \mu_{CZ}(\omega) = k}} \# \widetilde{\mathcal{M}}(\gamma, \omega) \omega \quad (8.13)$$

From which we define the *Floer Homology* as

$$FH_k(\mathcal{A}_H) = \frac{\ker \partial_k}{\text{im } \partial_{k+1}}$$

provided $\partial_k \circ \partial_{k+1} = 0$ holds.

8.2. The Conley-Zehnder Index, the Linearized Operator, and the Jacobi Operator

Let us look more in-depth at the Floer Map (Equation 8.8). By fixing a Floer trajectory $u : \mathbb{R} \times S^1 \rightarrow N$ we define the ‘*differential of the Floer Operator*’

$$\begin{aligned} (d\mathcal{F})_u &: H^1(\mathbb{R} \times S^1, u^*TN) \rightarrow L^2(\mathbb{R} \times S^1, u^*TN) \\ (d\mathcal{F})_u \xi &= \nabla_s \xi + J(u) \nabla_t \xi + \nabla_\xi J(u) \partial_t u - \nabla_\xi \nabla H_t(u) \end{aligned}$$

and by choosing an appropriate trivialization of u^*TN via isomorphisms $\Phi(s, t) : \mathbb{R}^{2n} \rightarrow T_{u(s,t)}N$ we obtain the ‘*Linearized Floer operator*’

$$\begin{aligned} D &: H^1(\mathbb{R} \times S^1, \mathbb{R}^{2n}) \rightarrow L^2(\mathbb{R} \times S^1, \mathbb{R}^{2n}) \\ \xi &\mapsto D(\xi) = \partial_s \xi + J_0 \partial_t \xi - S\xi \end{aligned}$$

Firstly, by Proposition 8.2, $\gamma \in \text{Crit}(\mathcal{A}_H)$ is a solution of the Hamiltonian system, $\dot{\gamma}(t) = X_t(\gamma)$. Thus, we can define family of symplectic diffeomorphisms ψ^t such that

$$\frac{d\psi^t}{dt} = X_t \circ \psi^t \text{ and } \psi^0 = \text{Id}.$$

With this system we say that a critical loop $\gamma \in \text{Crit}(\mathcal{A}_H)$ is *nondegenerate* if the differential of ψ^1 does not have eigenvalue 1:

$$\det(\text{Id} - d\psi^1(\gamma(0))) \neq 0.^4$$

We can relate this definition to the linearized Floer operator. In fact, associated with a symmetric matrix-valued function $S : \mathbb{R} \times S^1 \rightarrow \mathbb{R}^{2n \times 2n}$ is a symplectic matrix valued function $\Psi : \mathbb{R} \times \mathbb{R} \rightarrow \text{Sp}(2n)$ ⁵ by

$$J_0 \partial_t \Psi - S\Psi = 0$$

with $\Psi^\pm(t) := \lim_{s \rightarrow \pm\infty} \Psi(s, t)$. With this, we can use the linearized Floer operator to determine the dimensions of the moduli spaces for the Floer Homology of $\mathcal{L}_0 N$ as shown in *Lectures on Floer Homology* [26], given the loop is non-degenerate.

Theorem 8.7 ([26][Theorem 2.2]). *Suppose that $\det(\mathbb{1} - \Psi^\pm) \neq 0$. Then the linearized operator*

$$D : W^{1,p}(\mathbb{R} \times S^1; \mathbb{R}^{2n}) \rightarrow L^p(\mathbb{R} \times S^1; \mathbb{R}^{2n})$$

*is Fredholm for $1 < p < \infty$. Its Fredholm index, **index**, is given by the difference of the Conley-Zehnder indices, μ_{CZ} , of the limit matrices :*

$$\mathbf{index} D = \mu_{CZ}(\psi^+) - \mu_{CZ}(\psi^-).$$

⁴Here we use 0 to represent the base point.

⁵Recall that the group of symplectic matrices is defined as

$$\text{Sp}(2n) = \{A \in \mathbb{R}^{2n \times 2n} \mid A^T J_0 A = J_0\}$$

where $J_0 \in \mathbb{R}^{2n \times 2n}$ is the canonical complex structure.

The main components of the proof are the Conley-Zehnder index and the spectral flow. We will not go over either, but we refer to *The Conley-Zehnder index for a path of symplectic matrices* [12] and *Lectures on Floer Homology*[26] for in-depth explanations.

But how do we relate this back to geodesics and the Jacobi operator?

In Remark 8.1.2, we saw the relation between geodesics and a Hamiltonian system. Weber showed in [30] a one-to-one relation between the set of critical points of the energy functional and the Symplectic action functional:

Theorem 8.8. *There exists a bijection between the set of critical points of the energy functional and the symplectic action functional: $\text{Crit}(\mathcal{E}) \cong \text{Crit}(\mathcal{A}_H)$*

$$\begin{aligned}\gamma &\mapsto (\gamma, \dot{\gamma}) \\ \gamma &\leftarrow \pi(\gamma, \dot{\gamma})\end{aligned}$$

where we represent the elements of $\text{Crit}(A)_H$ as the solutions of the Hamiltonian system, and $\pi : T^*N \rightarrow N$ is the canonical projection.

Furthermore, it was shown that there is a direct relation between the Conley-Zehnder index and the Morse index.

Theorem 8.9 ([30][Theorem 1.2]). *Let $\gamma \in \text{Crit}(\mathcal{E})$ be nondegenerate, then*

$$\mu_{CZ}(\gamma) = \mu_M(\gamma)$$

*if the bundle $x^*TM \rightarrow S^1$ is trivial and*

$$\mu_{CZ}(\gamma) = \mu_M(\gamma) + 1$$

otherwise.

9. Hamiltonian Formalism: Minimal Surfaces

This chapter references *Generalizing symplectic topology from 1 to 2 dimensions* [6], *Fueter equations and the search for a higher dimensional Hamiltonian Floer theory I: analytical foundations and compactness*[3], *L^p -regularity of Cauchy-Riemann operators with non-constant C coefficients* [27], and “Quaternionic Analysis: Application to Boundary Value Problems” [11].

As it was seen in the last chapter, there exists a meaningful relation between the Morse index of geodesics and the Conley-Zehnder index of Floer trajectories. This relation, if confirmed, could provide new information about the Jacobi operator and the Morse index. To this end, we will consider current research on polysymplectic geometry and minimal surfaces, specifically smooth maps $f : S^1 \times S^1 \rightarrow N$ from the Torus to some Kähler manifold.

Remark 9.0.1. As we are focusing on smooth map from the torus $S^1 \times S^1$, we can consider a global frame $\{\partial_{t_1}, \partial_{t_2}\}$ of $T(S^1 \times S^1)$.

As the Torus is a Riemann surface, we will work with its complex structure and use the holomorphic and antiholomorphic derivatives :

$$\partial = \frac{1}{2} (\partial_{t_1} - i\partial_{t_2}) \text{ and } \bar{\partial} = \frac{1}{2} (\partial_{t_1} + i\partial_{t_2}).$$

With these complex derivatives, the ‘complex’ Laplacian on $S^1 \times S^1$ has the form

$$\Delta = -4\bar{\partial}\partial. \tag{9.1}$$

Furthermore, we can construct a Hamiltonian function as in Remark 8.1.1: $H : \mathbb{R}^{4n} \rightarrow \mathbb{R}$

$$H(q, p) = \frac{1}{2} \|p\|^2 + V(q) \tag{9.2}$$

such that

$$\nabla V(q) = \Delta(q).$$

To obtain the standard formula for the Hamiltonian, consider the mapping $(q, p) : S^1 \times S^1 \rightarrow \mathbb{R}^{4n}$, where $q = (q_1, q_2) : S^1 \times S^1 \rightarrow \mathbb{R}^{2n}$ is any map and $p = (p_1, p_2) : S^1 \times S^1 \rightarrow \mathbb{R}^{2n}$ is defined by $p = 2\partial q$:

$$p_1 = \partial_{t_1} q_1 + \partial_{t_2} q_2 \tag{9.3}$$

$$p_2 = \partial_{t_1} q_2 - \partial_{t_2} q_1. \tag{9.4}$$

To obtain the Laplace Δ , we require $-2\bar{\partial}p = \nabla V(q)$:

$$\frac{\partial V}{\partial q_1} = -\partial_{t_1}p_1 + \partial_{t_2}p_2 \quad (9.5)$$

$$\frac{\partial V}{\partial q_2} = -\partial_{t_1}p_2 - \partial_{t_2}p_1. \quad (9.6)$$

Following *Generalizing symplectic topology from 1 to 2 dimensions*[6] and *Fueter equations and the search for a higher dimensional Hamiltonian Floer theory I: analytical foundations and compactness* [3], by considering a polysymplectic form Ω , we can describe higher-dimensional analogues to the Hamiltonian system, Symplectic action functional, and the Linearized Floer operator.

Definition 9.1. An $\mathbb{R}^n$ -valued 2-form $\Omega \in \wedge^2 T^*N \otimes \mathbb{R}^n$ on a manifold N is called polysymplectic if it is closed and non-degenerate: $\Omega(X, \cdot) = 0 \implies X = 0$ for all $X \in \mathfrak{X}(N)$.

With the coordinates $Z = (q, p) : S^1 \times S^1 \rightarrow \mathbb{R}^{4n}$, we can construct a poly-symplectic form from two symplectic forms on $S^1 \times S^1$ to obtain the higher-dimensional equivalent of the Hamiltonian system of H . Indeed, if we consider symplectic forms

$$\omega_1 = dp_1 \wedge dq_1 + dp_2 \wedge dq_2 = d(p_1 dq_1 + p_2 dq_2) = d\theta_1$$

and

$$\omega_2 = dp_1 \wedge dq_2 - dp_2 \wedge dq_1 = d(p_1 dq_2 - p_2 dq_1) = d\theta_2,$$

then

$$\Omega = \omega_1 \otimes \partial_{t_1} + \omega_2 \otimes \partial_{t_2}$$

is a polysymplectic form with

$$dH(Z) = \omega_1(\cdot, \partial_{t_1}Z) + \omega_2(\cdot, \partial_{t_2}Z).$$

As in the geodesic case, the solutions of this system are the critical points of an action functional. This action functional will be called the *CRPS action functional*¹ and is defined by

$$\mathcal{A}_H(Z) = \int_{S^1 \times S^1} (\theta_1(\partial_{t_1}Z) + \theta_2(\partial_{t_2}Z) - H(Z)) dt_1 \wedge dt_2 \quad (9.7)$$

where θ_i are the primitives of the symplectic forms ω_i , $i = 1, 2$ and $Z = (q, p) : S^1 \times S^1 \rightarrow T^*N$ with q the standard coordinates and N a Kähler manifold.

The gradient of the CRPS action functional gives the two-dimensional analogue of the Floer equation, Equation 8.6:

$$\frac{\partial u}{\partial s} + J(u) \frac{\partial u}{\partial t_1} + K(u) \frac{\partial u}{\partial t_2} = \nabla H(u). \quad (9.8)$$

whose solutions are smooth maps $u : \mathbb{R} \times S^1 \times S^1 \rightarrow T^*N$. In this ‘two-dimensional Floer equation’, J and K are anti-commutative ($JK = -KJ$) complex structures on N compatible with the polysymplectic form Ω .

¹Complex-Regularized Polysymplectic

As in the previous section, Equation 9.8 can be written as the ‘two-dimensional Floer Map’:

$$\begin{aligned}\mathcal{F} : H^1(\mathbb{R} \times S^1 \times S^1, N) &\rightarrow L^2(\mathbb{R} \times S^1 \times S^1, N) \\ u &\mapsto \mathcal{F}_H(u) = \partial_s u + J(u)\partial_{t_1} u + K(u)\partial_{t_2} u - \nabla H_t(u),\end{aligned}$$

which under the appropriate trivialization via isomorphisms $\Phi(s, t_1, t_2) : \mathbb{R}^{4n} \rightarrow T_{u(s, t_1, t_2)}N$, gives us the ‘two-dimensional Linearized Floer Operator’:

$$D : H^1(\mathbb{R} \times S^1 \times S^1, \mathbb{R}^{4n}) \rightarrow L^2(\mathbb{R} \times S^1 \times S^1, \mathbb{R}^{4n}) \quad (9.9)$$

$$\xi \mapsto D\xi = \partial_s \xi + J_0 \partial_{t_1} \xi + K_0 \partial_{t_2} \xi - S\xi. \quad (9.10)$$

We would like to generalize Theorem 8.7 and Theorem 8.9 to two dimensions.

Some parts of the Theorem 8.7 and Theorem 8.9 can be generalized. For instance, using a similar procedure as in *Morse Theory and Floer Homology*[4], one also sees that the limit matrices $S^\pm = \lim_{s \rightarrow \pm\infty} S(s, t)$ are symmetric, with further generalizations in section 9.1. Furthermore, one also has the existence of a bijection between the critical points of the CRPS action functional and harmonic maps, with a further generalization in section 9.2. In both theorems, the obstacle is the generalization of symplectic matrices to ‘polysymplectic matrices’; a lot is still unknown about polysymplectic structures. Unlike the case of symplectic manifolds, Hamiltonian functions do not provide the same information in the polysymplectic case. From [6, Definition 1.2], the connection between Hamiltonian functions and Hamiltonian vector fields is replaced by the *current*, a smooth map $F : (M, \Omega) \rightarrow \mathbb{R}^2$ from a polysymplectic manifold with polysymplectic form such that there exists a vector field X_F such that $\Omega(X_F, -) = dF$.

9.1. Fredholm Property

Before we can discuss the Fredholm index of D , we first need to show that D is Fredholm. The goal of this subsection is to prove the following theorem.

Theorem 9.2. *The two-dimensional linearized Floer operator:*

$$\begin{aligned}D : H^1(\mathbb{R} \times S^1 \times S^1, \mathbb{R}^{4n}) &\rightarrow L^2(\mathbb{R} \times S^1 \times S^1, \mathbb{R}^{4n}) \\ \xi &\mapsto D\xi = \partial_s \xi + J_0 \partial_{t_1} \xi + K_0 \partial_{t_2} \xi - S\xi\end{aligned}$$

is Fredholm.

First, we recall the definition of a Fredholm operator.

Definition 9.3. A linear operator $T : H \rightarrow H$ between Hilbert spaces is called *Fredholm* if both $\ker(T)$ and $\ker(T^*)$ are finite-dimensional and $\text{ran}(T)$ is closed.

Rather than manually determining the above requirements, we will use the following theorem.

Theorem 9.4 (Elliptic Estimate [26][Exercise 2.1]). *Let X, Y, Z be Banach spaces, $D : X \rightarrow Y$ be a linear operator, and $K : X \rightarrow Z$ be a compact linear operator. Suppose that there exists a constant $C > 0$ such that the following inequality holds for all $x \in X$*

$$\|x\|_X \leq C (\|Dx\|_Y + \|Kx\|_Z). \quad (9.11)$$

Then D has a closed range and finite-dimensional kernel.

Corollary 9.4.1. *Let $T : X \rightarrow Y$ be a linear operator. If T and T^* satisfy the elliptic estimate, then T is a Fredholm operator.*

Thus, we will show that $D = \partial_s + J_0 \partial_{t_1} + K_0 \partial_{t_2} - S$ and its adjoint $D^* = -\partial_s + J_0 \partial_{t_1} + K_0 \partial_{t_2} - S^t$ satisfy the elliptic estimate, when the compact operator is the embedding $H^1 \hookrightarrow L^2$ from Theorem 5.9.

Remark 9.1.1. By the properties of the adjoint, we have

$$\begin{aligned} D^* &= (\partial_s + J_0 \partial_{t_1} + K_0 \partial_{t_2} - S)^* \\ &= (\partial_s)^* + (J_0 \partial_{t_1})^* + (K_0 \partial_{t_2})^* - S^* \\ &= -\partial_s + (-J_0)(-\partial_{t_1}) + (-K_0)(-\partial_{t_2}) - S^t \\ &= -\partial_s + J_0 \partial_{t_1} + K_0 \partial_{t_2} - S^t. \end{aligned}$$

It suffices to show that D satisfies the elliptic estimate. To do so, we will consider the two-dimensional linearized Floer operator as the sum of two components:

$$\partial_s + J_0 \partial_{t_1} + K_0 \partial_{t_2} - S. \quad (9.12)$$

The first component can be seen as the reduced quaternionic Cauchy-Riemann operator, which is unbounded as an operator on L^2 , while the second S is, by construction, a bounded operator on L^2 . Hence we will show that $\partial_s + J_0 \partial_{t_1} + K_0 \partial_{t_2}$ satisfies the elliptic estimate with the embedding $H^1 \hookrightarrow L^2$.

Remark 9.1.2. We will use the symbol $\lesssim$ for $\leq C$ where C is some constant.

Quaternionic Cauchy-Riemann Operator

We will show that the *standard (left) quaternionic Cauchy-Riemann operator*

$$\bar{\partial}_l := \partial_s + i \partial_{t_1} + j \partial_{t_2} + k \partial_{t_3} : H^1(\Omega, \mathbb{R}^4) \rightarrow L^2(\Omega, \mathbb{R}^4) \quad (9.13)$$

satisfies the elliptic estimate on a bounded open subset $\Omega \subseteq \mathbb{R}^4$. From which we can deduce that the elliptic estimate will also hold for the reduced (left) quaternionic Cauchy-Riemann operator: $\bar{\partial}_l^r = \partial_s + j \partial_{t_2} + k \partial_{t_3}$.

Firstly quaternions $\mathbb{H}$ form a four-dimensional algebra over the real field $\mathbb{R}$, with an identity element denoted by 1. A typical basis is $\{1, i, j, k\}$ and it satisfies

$$i^2 = j^2 = k^2 = -1 \quad (9.14)$$

$$ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j \quad (9.15)$$

Given a quaternionic element $q = s + it_1 + jt_2 + kt_3$, a quaternionic conjugate $\bar{q} = s - it_1 - jt_2 - kt_3$ and a quaternionic norm $|q| = \sqrt{q\bar{q}}$, one can define the quaternionic inverse $q^{-1} = \frac{\bar{q}}{|q|^2}$.

Remark 9.1.3. One can also see a quaternionic space as a two-dimensional algebra over the complex field $\mathbb{C}$. In this case, the basis is $\{1, j\}$ such that $ji = k$ and quaternionic elements are of the form $q = u + jv$ for $u, v \in \mathbb{C}$:

$$q = u + jv = (s + it_1) + j(t_2 + it_3) = s + it_1 + jt_2 + j it_3.$$

Remark 9.1.4. Following the notation of the linearized Floer operator Equation 9.10, we will refer the 3-dimensional subspace with basis $\{1, j, k\}$ satisfying $-1 = j^2 = k^2$ and $jk = -kj$ as the *reduced quaternions*.

From “Quaternionic Analysis: Application to Boundary Value Problems”[11], the fundamental solution of the (left) quaternionic Cauchy-Riemann equation is the map

$$\begin{aligned} K : \mathbb{H} \setminus \{0\} &\rightarrow \mathbb{H} \\ q &\mapsto \frac{1}{2\pi^2} q^{-1}, \end{aligned}$$

which is called the (quaternionic) Cauchy Kernel. Observe that when $0 \in \Omega \subseteq \mathbb{R}^4$, $K \in L^2(\Omega)$.

Remark 9.1.5. We have that the Cauchy kernel will look the same for the reduced quaternionic operator, and it will be defined on $\mathbb{H} \setminus \{\text{Span}_{\mathbb{R}}\{i\}, 0\}$.

The Cauchy Kernel will play a fundamental role in showing that $\bar{\partial}_l$ satisfies the elliptic estimate, one of its roles is in the following formula. For a bounded subset $\Omega \subseteq \mathbb{R}^4$ with boundary $\partial\Omega = \Gamma$, and for all $\varphi \in C^1(\bar{\Omega})$, we have the *Borel-Pompeiu Formula*:

$$\int_{\Gamma} K(y-x)\varphi(y)d\Gamma_y - \int_{\Omega} K(y-x)(\bar{\partial}_l\varphi)(y)d\Omega_y = \begin{cases} \varphi(x) & x \in \Omega \\ 0 & x \in \mathbb{R}^4 \setminus \Omega \end{cases} \quad (9.16)$$

where $d\Gamma_y$ and $d\Omega_y$ are the Lebesgue measures on the respective spaces.

Now that we have an understanding of the space we are working with, we can begin laying the foundations for the proof.

Remark 9.1.6. We will be working with a bounded open subset $\Omega \subseteq \mathbb{R}^4$.

Furthermore, we will use the notation C_0^∞ and H_0^1 for compactly supported smooth maps and maps of class H^1 , respectively.

Given a quaternionic continuous map $\varphi : \alpha + i\beta + j\gamma + k\delta : \Omega \rightarrow \mathbb{R}^4$, we have that

$$|\bar{\partial}_l\varphi|^2 = |\partial_s\varphi|^2 + |\partial_{t_1}\varphi|^2 + |\partial_{t_2}\varphi|^2 + |\partial_{t_3}\varphi|^2 + 2iA_i + 2jA_j + 2kA_k, \quad (9.17)$$

for A_i, A_j, A_k appropriate linear combinations of the partial derivatives $\partial_s\varphi, \partial_s\bar{\varphi}, \partial_{t_l}\varphi, \partial_{t_l}\bar{\varphi}$ for $l = 1, 2, 3$. Therefore, the L^2 norm of φ is

$$\|\bar{\partial}_l\varphi\|_{L^2(\Omega)}^2 = \|\nabla\varphi\|_{L^2(\Omega)}^2 + 2i \int_{\partial\Omega} A_i + 2j \int_{\partial\Omega} A_j + 2k \int_{\partial\Omega} A_k, \quad (9.18)$$

Thus if $\varphi \in H_0^1(\Omega, \mathbb{R}^4)$, then $\|\bar{\partial}_l\varphi\|_{L^2(\Omega)}^2 = \|\nabla\varphi\|_{L^2(\Omega)}^2$. Consider $f \in L^2(\Omega, \mathbb{R}^4)$ and define the operator $C : L^2(\Omega, \mathbb{R}^4) \rightarrow L^2(\Omega, \mathbb{R}^4)$ defined by convolution of the Cauchy kernel

$$Cf = -K * f. \quad (9.19)$$

From Equation 9.16, we have the following theorem and lemma.

Theorem 9.5. *If $\varphi \in C_0^\infty(\Omega, \mathbb{R}^4)$, then $\varphi = C(\bar{\partial}_l \varphi)$.*

Lemma 9.6. *If $\varphi \in C_0^\infty(\Omega, \mathbb{R}^4)$, then $C\varphi \in C_0^\infty(\Omega, \mathbb{R}^4)$ and $\bar{\partial}_l C\varphi = \varphi$.*

Proof. $C\varphi = (-K * \varphi)|_\Omega$, hence $\bar{\partial}_l C\varphi = \bar{\partial}_l (-K * \varphi)|_\Omega = (-K * \bar{\partial}_l \varphi)|_\Omega = C(\bar{\partial}_l \varphi) = \varphi$. $\square$

From Theorem 5.5, the above theorem holds also for H^1 .

Proposition 9.7. *If $\varphi \in H_0^1(\Omega, \mathbb{R}^4)$, then $\varphi = C\bar{\partial}_l \varphi$.*

The convolution C is bounded.

Theorem 9.8. *Let $\Omega \subset \mathbb{R}^4$ be a bounded subset. Then for $\varphi \in H_0^1(\Omega, \mathbb{R}^4)$*

$$\|\nabla(C\varphi)\|_{L^2(\Omega)} \leq \|\varphi\|_{L^2(\Omega)}. \quad (9.20)$$

Proof. The proof is via a dense sequence of smooth maps, due to Theorem 5.5. Let $\varphi \in C_0^\infty(\Omega, \mathbb{R}^4)$ and consider $C\varphi = \alpha + i\beta + j\gamma + k\delta$ for $\alpha, \beta, \gamma, \delta \in C^\infty(\mathbb{R}^4)$. We have that $\partial_s(C\varphi) = C(\partial_s \varphi)$ and $\partial_{t_i}(C\varphi) = C(\partial_{t_i} \varphi)$ for $i = 1, 2, 3$. Since Ω is bounded, there exists a ball B_R of radius R containing Ω , hence the L^2 norm of φ is

$$\|\varphi\|_{L^2(\Omega)}^2 = \|\bar{\partial}_l C\varphi\|_{L^2(B_R)}^2 = \|\nabla(C\varphi)\|_{L^2(B_R)}^2 + 2i \int_{\partial B_R} A_i + 2j \int_{\partial B_R} A_j + 2k \int_{\partial B_R} A_k.$$

Observe that for an arbitrary $\psi \in C_0^\infty(\mathbb{R}^4)$ and $|q| \gg 1$, the convolution $C\psi$ is bounded:

$$\begin{aligned} |C\psi(q)| &= |K * \psi(q)| = \left| \int_{\text{supp } \psi} \frac{\psi(\tilde{q})}{2\pi^2 |\tilde{q} - q|} d\tilde{q} \right| \\ &\leq \frac{1}{2\pi^2} \int_{\text{supp } \psi} \frac{|\psi(\tilde{q})|}{|\tilde{q} - q|} d\tilde{q} \\ &\leq \frac{\|\psi_1\|}{2\pi^2} \frac{1}{\min\{|\tilde{q} - q| : \tilde{q} \in \text{supp } \psi\}} \\ &\lesssim \frac{1}{2\pi^2 |q|}. \end{aligned}$$

Thus, applying this to $\varphi, \partial_s \varphi, \partial_{t_l} \varphi$ for $l = 1, 2, 3$, we have

$$\begin{aligned} |\alpha(q)| &\leq |C\varphi(q)| \lesssim \frac{1}{q}, & |\partial_s \beta(q)| &\leq |C\partial_s \varphi(q)| \lesssim \frac{1}{q}, \\ |\partial_s \gamma(q)| &\leq |C\partial_s \varphi(q)| \lesssim \frac{1}{q}, & |\partial_s \delta(q)| &\leq |C\partial_s \varphi(q)| \lesssim \frac{1}{q}, \\ |\partial_{t_l} \beta(q)| &\leq |C\partial_{t_l} \varphi(q)| \lesssim \frac{1}{q}, & |\partial_{t_l} \gamma(q)| &\leq |C\partial_{t_l} \varphi(q)| \lesssim \frac{1}{q}, \\ |\partial_{t_l} \delta(q)| &\leq |C\partial_{t_l} \varphi(q)| \lesssim \frac{1}{q}. \end{aligned}$$

Furthermore, as A_i, A_j , and A_k are linear combinations of the partial derivatives $\partial_s \varphi$, $\partial_s \bar{\varphi}$, $\partial_{t_l} \varphi$, $\partial_{t_l} \bar{\varphi}$ for $l = 1, 2, 3$, we can use the above bounds as well as the triangle inequality to bound A_i, A_j , and A_k : for $\star = i, j, k$ we have

$$\left| \int_{B_R} A_\star \right| \leq \int_{\partial B_R} |A_i| \lesssim \frac{|\partial B_R|}{R^2} \lesssim \frac{1}{R}.$$

Thus, each term A_i, A_j , and A_k tends to 0 as $R \rightarrow \infty$ and $\|\nabla(C\varphi)\|_{L^2(\Omega)} \leq \|\varphi\|_{L^2(\Omega)}$ for $\varphi \in C_0^\infty(\Omega, \mathbb{R}^4)$. $\square$

Following the Theorem 9.5 and Theorem 9.8, we have the following bound.

Corollary 9.8.1. *If $\varphi \in W_0^{1,p}(\Omega, \mathbb{R}^4)$, then*

$$\|\nabla \varphi\|_{L^2(\Omega)} \lesssim \|\bar{\partial}_l \varphi\|_{L^2(\Omega)}.$$

Proof. Since $\varphi = C\bar{\partial}_l \varphi$, the above theorem gives us the desired bound: $\|\nabla \varphi\|_{L^2(\Omega)} = \|\nabla(C\bar{\partial}_l \varphi)\|_{L^2(\Omega)} \leq \|\bar{\partial}_l \varphi\|_{L^2(\Omega)}$. $\square$

Finally, we have the desired elliptic estimate for the (left) quaternionic Cauchy-Riemann operator.

Proposition 9.9. *For $\varphi \in H^1(\Omega, \mathbb{R}^4)$ and $U \subset \Omega$*

$$\|\varphi\|_{H^1(U)} \lesssim \|\bar{\partial}_l \varphi\|_{L^2(\Omega)} + \|\varphi\|_{L^2(\Omega)}.$$

Proof. Let $\eta \in C_0^\infty(\Omega)$ such that $\eta|_U = 1$. Then $\eta\varphi \in H_0^1(\Omega, \mathbb{R}^4)$ and

$$\begin{aligned} \|\varphi\|_{H^1(U)} &= \|\eta\varphi\|_{H^1(U)} \leq \|\eta\varphi\|_{H^1(\Omega)} \\ &\lesssim \|\nabla(\eta\varphi)\|_{L^2(\Omega)} + \|\eta\varphi\|_{L^2(\Omega)} \\ &\lesssim \|\bar{\partial}_l(\eta\varphi)\|_{L^2(\Omega)} + \|\varphi\|_{L^2(\Omega)} \\ &= \|\bar{\partial}_l(\eta)\varphi + \eta\bar{\partial}_l(\varphi)\|_{L^2(\Omega)} + \|\varphi\|_{L^2(\Omega)} \\ &\leq \|\bar{\partial}_l(\eta)\|_\infty \|\varphi\|_{L^2(\Omega)} + \|\eta\|_\infty \|\bar{\partial}_l(\varphi)\|_{L^2(\Omega)} + \|\varphi\|_{L^2(\Omega)} \\ &\lesssim \|\varphi\|_{L^2(\Omega)} + \|\bar{\partial}_l(\varphi)\|_{L^2(\Omega)}. \end{aligned}$$

$\square$

Now that we know that on the quaternions $\mathbb{H}$ the standard (left) quaternionic Cauchy-Riemann operator satisfies the elliptic estimate, we can check whether the same holds for a manifold M^{4n} with an almost quaternionic structure.

Definition 9.10. Let M be a smooth $4n$ -manifold, for $n \geq 1$.

We call a triple of non-commutative endomorphisms $\mathcal{Q} = (I, J, K) \subseteq \text{End}(TM)$ a *quaternionic structure* if for each $p \in M$

$$I(p)^2 = J(p)^2 = K(p)^2 = -\text{Id}_{T_p M} = I(p)J(p)K(p).$$

The *standard quaternionic structure* on $\mathbb{H}^n$ is $\mathcal{Q}_0 = (I_0, J_0, K_0)$ where

$$I_0 = \begin{pmatrix} 0 & -\mathbb{1}_n & 0 & 0 \\ \mathbb{1}_n & 0 & 0 & 0 \\ 0 & 0 & 0 & -\mathbb{1}_n \\ 0 & 0 & \mathbb{1}_n & 0 \end{pmatrix}, J_0 = \begin{pmatrix} 0 & 0 & -\mathbb{1}_n & 0 \\ 0 & 0 & 0 & \mathbb{1}_n \\ \mathbb{1}_n & 0 & 0 & 0 \\ 0 & -\mathbb{1}_n & 0 & 0 \end{pmatrix}, K_0 = \begin{pmatrix} 0 & 0 & 0 & -\mathbb{1}_n \\ 0 & 0 & -\mathbb{1}_n & 0 \\ 0 & \mathbb{1}_n & 0 & 0 \\ \mathbb{1}_n & 0 & 0 & 0 \end{pmatrix}.$$

A smooth $4n$ -manifold M^{4n} is said to be *almost quaternionic manifold* if it has a *quaternionic structure* $\mathcal{Q}$. We say that M is a *quaternionic manifold* if there exists a torsion-free connection that preserves $\mathcal{Q}$: $\nabla \Gamma(\mathcal{Q}) \subseteq \Gamma(\mathcal{Q})$.

Remark 9.1.7. As in Remark 9.1.4, we call the pair $\mathcal{Q}_{\text{red}} = (J, K)$ a *reduced quaternionic structure* if for each $p \in M$

$$J(p)^2 = K(p)^2 = -\text{Id}_{T_p M} \quad \text{and} \quad J(p)K(p) = -K(p)J(p).$$

Hence a $3n$ -dimensional smooth manifold is called an *almost reduced quaternionic manifold* if it has a reduced quaternionic structure $\mathcal{Q}_{\text{red}}$.

Now we can define the (left) quaternionic Cauchy-Riemann operator on an almost quaternionic manifold M with quaternionic structure $\mathcal{Q} = (I, J, K)$:

$$\bar{\partial}_l := \partial_s + I\partial_{t_1} + J\partial_{t_2} + K\partial_{t_3} : H^1(M, \mathbb{R}^{4n}) \rightarrow L^2(M, \mathbb{R}^{4n}). \quad (9.21)$$

Given a smooth manifold M with a standard quaternionic structure, the (left) quaternionic Cauchy-Riemann operator reduces to its standard (left) quaternionic Cauchy-Riemann operator. Hence, it satisfies the elliptic estimate.

—Proof of Theorem 9.2—

Proof of Theorem 9.2. Let D be the (two-dimensional) linearized Floer operator.

- Claim 1: $D := \bar{\partial}_l^r - S$, where $\bar{\partial}_l^r := \partial_s + J_0\partial_{t_1} + K_0\partial_{t_2}$ is the reduced (left) quaternionic Cauchy-Riemann operator.

From the two-dimensional Floer equation Equation 9.8, the complex structures J and K , define a reduced quaternionic structure on $\mathbb{R} \times S^1 \times S^1$. It can then be observed that the linearized Floer operator can be expressed as the sum of two components:

$$\underline{\partial_s + J_0\partial_{t_1} + K_0\partial_{t_2}} - S.$$

The first component is the reduced (left) quaternionic Cauchy-Riemann operator $\bar{\partial}_l^r$ and the second is the Hessian S .

- Claim 2: D satisfies the elliptic estimate.

Let $\xi \in H^1(\mathbb{R} \times S^1 \times S^1)$ and consider D as the sum $\bar{\partial}_l^r - S$. As seen above, $\bar{\partial}_l^r$ satisfies the elliptic-regularity. Furthermore, by construction $S : L^2 \rightarrow L^2$ is a bounded operator. Hence, we have the desired bound:

$$\begin{aligned} \|\xi\|_{H^1} &\lesssim \|\bar{\partial}_l^r \xi\|_{L^2} + \|\xi\|_{L^2} \leq \|\bar{\partial}_l^r \xi - S\xi\|_{L^2} + \|S\xi\|_{L^2} + \|\xi\|_{L^2} \\ &\lesssim \|\bar{\partial}_l^r \xi - S\xi\|_{L^2} + \|\xi\|_{L^2}. \end{aligned}$$

- Claim 3: D is Fredholm.

As the adjoint is $D^* = -\partial_s + J_0\partial_{t_1} + K_0\partial_{t_2} - S^t$, similar proofs show that $-\partial_s + J_0\partial_{t_1} + K_0\partial_{t_2}$ satisfies the elliptic estimate and so does D^* . Since D and D^* satisfy the elliptic estimate, Corollary 9.4.1 states that D is a Fredholm operator. □

However, we cannot go further than this. In the case of the geodesics, symplectic matrices are essential in determining the Conley-Zehnder index. Very little is known about polysymplectic forms and their matrix counterpart. For further progress to be made, more research is needed on ‘polysymplectic matrices’.

9.2. Relation between Energy functional and CRPS Functional

As in the symplectic case, the critical points of the CRPS action functional are the smooth maps $Z : S^1 \times S^1 \rightarrow T^*N$ that solve the polysymplectic equation $dH(Z) = \omega_1(\cdot, \partial_{t_1} Z) + \omega_2(\cdot, \partial_{t_2} Z)$. Therefore, we can express these critical points as the pairs $(q, 2\partial q)$. In [3], a bijection between harmonic maps from the torus to a Kähler manifold and these solutions is proved.

Theorem 9.11 ([3][Theorem 2.4]). *There exists a bijection between the set of critical points of the energy functional and the CRPS action functional: $\text{Crit}(\mathcal{E}) \cong \text{Crit}(\mathcal{A}_H)$*

$$q \mapsto (q, 2\partial q) \tag{9.22}$$

$$q \leftarrow \pi(q, 2\partial q) \tag{9.23}$$

where we represent the elements of $\text{Crit}(\mathcal{A}_H)$ as the solutions of the polysymplectic Hamiltonian system, and $\pi : T^*N \rightarrow N$ is the canonical projection.

Thus critical points of the CRPS action functional $(q, p = 2\partial q) \in \text{Crit}(\mathcal{A}_H)$ satisfy

$$\begin{pmatrix} 2\partial q \\ 2\bar{\partial} p \end{pmatrix} = \begin{pmatrix} p \\ 0 \end{pmatrix}. \tag{9.24}$$

By linearizing the above at a critical point (q, p) as was done in “Perturbed closed geodesics are periodic orbits: Index and transversality” [30], we have that the hessian $-S$ of $\mathcal{A}_H$ is of the form:

$$-S \begin{pmatrix} \xi \\ \eta \end{pmatrix} = \begin{pmatrix} -\partial\eta - \bar{\partial}\bar{\partial}\xi - 2R^N(\xi, \partial_t\phi, \bar{\partial}\phi) \\ 2\partial\xi - \eta \end{pmatrix}. \quad (9.25)$$

From “La formule de la variation seconde de l’énergie au voisinage d’une application harmonique”, chapter 2: Etude du cas Kählerien [21], the Jacobi operator for the energy functional of harmonic maps $f : S^1 \times S^1 \rightarrow N$ in complex coordinates has the form:

$$J(V) = -\partial\bar{\partial}V - \bar{\partial}\partial V - 2R^N(V, \partial\phi, \bar{\partial}\phi). \quad (9.26)$$

It is clear to see that $(\xi, 2\partial\xi) \in \ker(S)$ if and only if $\xi \in \ker(J)$. Thus, a relation between the Morse index of the Jacobi operator and the ‘Conley-Zehnder’ index of the linearized Floer operator can exist. However, to actually determine and prove such a relation as in Theorem 8.9, further research needs to be done. Once again, symplectic matrices were crucial in showing the bijection in the proof of Theorem 8.9, and it is safe to assume that ‘polysymplectic matrices’ will play that role in our case.

Remark 9.2.1. We use the *Conley-Zehnder* terminology in the spirit of consistency. It is important to note that, as of yet, an index for CRPS has not been described. As the Conley-Zehnder index was established in relation to symplectic matrices, we hypothesise that an index of the critical points of the CRPS action can be described via the polysymplectic form.

10. Conclusion

Finally, we have completed our journey. The goal of this thesis was to consider harmonic maps as tools to understand the topology of certain infinite-dimensional mapping spaces, Sobolev spaces H^k . We have done just that, by first considering them in the context of Morse theory and then in the context of Floer theory.

We began by considering the cornerstones of this field: geodesics. We studied the crucial components for Morse theory on geodesics – the energy functional, the Jacobi operator, and conjugate points. When we generalized these concepts to arbitrary harmonic maps, the immediate generalization of geodesics and the critical points of the energy functional, we came across some interesting behaviors. The energy functional provided us with many obstacles. Firstly, one needs to consider specific mapping spaces to ensure smoothness and continuity, which restricts the type of maps to consider as well as the domains of these maps. Secondly, even though the Hessian of the energy functional produced the Jacobi operator and its relation to the Morse index, it could not provide equivalent information to the geodesic case. Indeed, the Jacobi fields do not have the same properties as their ‘geodesic’ counterparts in terms of integrability and of ‘conjugate points’. The most glaring of obstacles is that the energy functional could not aid in the decomposition of these mapping spaces when the dimension of the domain of the smooth maps is greater than 1. Thus, we were forced to consider a different approach to study harmonic maps and mapping spaces.

In this thesis, we chose to consider the Hamiltonian approach. As the immediate generalization of geodesics are two-dimensional manifolds – surfaces – complex structures immediately arise. The ‘Hamiltonian approach’ allows us to use this complex structure, which is virtually untouched in Morse theory. When switching our view from standard Euclidean to complex, we are back to working with one-dimensional objects, albeit complex ones. Furthermore, this method allows us to not depend on the Sobolev embedding of $H^k(M, N) \hookrightarrow C^0(M, N)$ and simply use H^1 again. When we looked at the energy functional and its harmonic maps, it came from the knowledge of geodesics. This is still the case with the Hamiltonian formalism, but with the starting point being the symplectic action functional. As so much has already been studied for geodesics, the relation between the Conley-Zehnder index and the Morse index may not be revolutionary. However, the polysymplectic case with its harmonic maps is a different case. Morse theory fails to work in higher dimensions, so a connection between the Morse index of a harmonic map and the “Conley-Zehnder” index would provide new and unique information. To achieve this, further research is required into the analogue of the Conley-Zehnder index and poly-symplectic matrices. Thus, it follows that this approach and the findings that it may provide could lead to new innovative works.

We conclude this thesis with a list of open problems.

- How can we describe ‘poly-symplectic matrices’? In the construction of the CRPS action functional, the poly-symplectic form was defined from two symplectic forms. Can there be a relation between some poly-symplectic matrices and symplectic matrices?
- What is the polysymplectic equivalent of the Coneley-Zehnder index for a path of symplectic matrices?
- Could there even exist “conjugate sub-manifolds,” and what would we need to look for?
- Conjugate points arose, also, as the critical points of the exponential map, so what could be the equivalent in higher dimensions?
- One could further investigate the already known cases where integrability holds, like the case of maps from the 2-sphere and $\mathbb{C}P^2$, and check if we can say anything about “conjugate boundaries”.

11. Summary

Harmonic mappings between Riemannian manifolds have been studied in mathematics, physics, and engineering; they act as a bridge between geometric and physical problems. Harmonic maps are both the higher-dimensional analogues of geodesics, as well as generalizations of classical harmonic functions. In mathematics, harmonic maps are used to analyze the moduli spaces of Riemann surfaces and to investigate rigidity problems on Riemannian manifolds. In physics, they are used in quantum field theory, gauge theory, and general relativity.

In this thesis, we focus on harmonic maps as tools for describing the topology of Sobolev manifolds, infinite-dimensional manifolds of maps between smooth Riemannian manifolds. We have divided it into three parts. The first part considers the classic case of Morse theory through geodesics. It allows us to familiarize ourselves with the methodology and provides a blueprint for higher-dimensional analogues. The second part considers the higher-dimensional generalization of geodesics. It explores the analogies of geometric objects and operators found in the first part, as well as discusses the limitations of using Morse theory in higher dimensions. Finally, the third part examines a different method specifically for minimal surfaces. This method aims to describe the topology of Sobolev spaces via Floer theory to counteract the failings of Morse theory.

In ‘The One-Dimensional Case: Geodesics and the Path Space’, we use Morse theory to describe the Sobolev space of all paths in a Riemannian manifold M , $H^1(I, M)$, via the critical points of the energy functional. We follow the method for describing the Morse theory of a smooth manifold via a function, detailed in chapter 2. That is, we determine the Morse index of the non-degenerate critical points of a functional. As the study of geodesics stems from the study of length-minimizing curves, we look at the length functional

$$L_g(\gamma) = \int_I \left\| \frac{d\gamma}{dt} \right\|_g dt \quad , \gamma \in H^1(I, M)$$

and the energy functional

$$\mathcal{E}_g(\gamma) = \frac{1}{2} \int_I \left\| \frac{d\gamma}{dt} \right\|_g^2 dt \quad \forall \gamma \in \mathcal{PM} \quad , \gamma \in H^1(I, M).$$

Via variations of paths, we define the ‘derivatives’ of the length and energy functionals as the first variation formula. Via this formula, we give an explicit description of what the critical points are for these two functionals. We show that under specific conditions – when a path has unit speed – the critical points of the energy functional and the length functional coincide, and we are free to consider geodesics as the critical points of the energy functional. With the functional and its critical points specified, we turn to

determining the Morse index of non-degenerate geodesics. To do so, we define the Hessian of the energy functional via a 2-parameter variation. From this formula, we define the Jacobi operator

$$J(W) = \frac{D^2}{dt^2}W + R\left(W, \frac{d\gamma}{dt}, \frac{d\gamma}{dt}\right).$$

This operator and specific points along a geodesic, called conjugate points, are then used to determine when a geodesic is non-degenerate and what its Morse index is. With all this defined, we describe the topology of $H^1(I, M)$ via the geodesics. At the end, in chapter 5, we finally describe Sobolev spaces on Riemannian manifolds M and N , to be Hilbert manifolds $W^{k,p}(M, N)$ modeled on the Sobolev spaces $W^{k,p}(U, V)$, for U, V coordinates neighborhoods of M and N , respectively. For $k \in \mathbb{N}$, $1 \leq p \leq \infty$ and an open subset $U \subseteq \mathbb{R}^m$, we define a Sobolev space $W^{k,p}(U, \mathbb{R}^n)$ as the Banach space of all Lebesgue p -integrable maps such that the α th-weak derivatives, for $|\alpha| \leq k$, are p -integrable:

$$W^{k,p}(U, \mathbb{R}^n) := \{u \in L^p(U) : \forall |\alpha| \leq k, D^\alpha u \in L^p(U)\},$$

and use the notation H^1 for $W^{1,2}$.

As the study of geodesics began by studying the shortest path between two points on a manifold, in ‘Generalizing Geodesics to Higher Dimensions: Minimal Sub-manifolds and Harmonic Maps’, we study the higher-dimensional generalization of geodesics, i.e., harmonic maps, by first considering volume-minimizing sub-manifolds. When considering variations of anisometric embeddings, we obtain the first variation formula of the volume functional

$$\text{Vol}(f) = \int_M d\mu_g.$$

This formula gives an explicit description of a minimal sub-manifold. Following the 1-dimensional case, we move on to considering the energy functional of a smooth map $f : (M, g) \rightarrow (N, h)$ instead,

$$\mathcal{E}_g(f) = \frac{1}{2} \int_M \|df\|_{\text{Hom}_f(TM, TN)}^2 d\mu_g.$$

Similarly, via the first variation formula, we have an explicit formula for its critical points, the harmonic maps. Again, under specific conditions – when the smooth map is an isometric immersion or when it is a conformal map from a surface – the critical points of the energy functional and the volume functional coincide. Even though some details are lost, focusing on the energy functional allows for easier calculations of non-degeneracy and of the Morse index. Nonetheless, one can find in section 6.3 the formula for the second variation of the volume functional. Like in the geodesic case, the second variation of the energy functional produces a Jacobi operator which is again related to non-degeneracy and the Morse index. However, we are faced with the main obstruction for a Morse theory to be defined via harmonic maps: the Palais-Smale condition.

By the end of the previous part, we know that Morse theory is not the right tactic. In ‘Moving Forward: Hamiltonian Formalism,’ we describe how Floer theory could be a better approach. When restricting to geodesics on a symplectic manifold, a third

functional can be constructed, the Symplectic action functional. With this functional, one can describe an infinite-dimensional version of Morse homology, where we refer to ?? for the background information. In the case of geodesics, it has been shown by Weber in [30] the existence of a connection between geodesics and the critical points of the symplectic action functional. This result is what we aim to achieve for harmonic maps $Z : \mathbb{T}^2 \rightarrow N$, from the torus $\mathbb{T}^2$ to some Kähler manifolds of dimension $4n$. This idea is backed by recent research by O.Fabert and R.Brilleslijper, [6] [3], on the two-dimensional analogue of the symplectic action function, the CRPS action functional.

Part IV.

Appendix

A. Riemannian Geometry: a Summary

This chapter will revisit some knowledge of connections, covariant derivatives, curvature, and the length of curves on a Riemannian manifold. It references *Introduction to Riemannian manifolds*[15], *Morse Theory*[22] and “Mastermath Differential Notes Fall 2024”[9].

A.1. Connections on Tangent Bundles

Definition A.1. A *connection* on a vector bundle E is a bilinear map

$$\mathfrak{X}(M) \times \Gamma(E) \rightarrow \Gamma(E) \quad (X, s) \mapsto \nabla_X(s), \quad (\text{A.1})$$

satisfying

$$\nabla_{fX}(s) = f\nabla_X(s) \quad \text{and} \quad \nabla_X(fs) = f\nabla_X(s) + X(f)s. \quad (\text{A.2})$$

Remark A.1.1. When $E = TM$, then one has $\mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M) \quad (X, Y) \mapsto \nabla_X(Y)$, and we will say that the *connection is on M* .

Following the remark, we can also specify what a connection on tensor bundles is:

Proposition A.2 (Connection on Tensor Bundles). *Let M be a smooth manifold with or without boundary, and let ∇ be a connection in TM . Then ∇ uniquely determines a connection in each tensor bundle $T^{(k,l)}TM$, also denoted by ∇ , such that the following four conditions are satisfied:*

1. $T^{(1,0)}TM = TM$, i.e ∇ agrees with the given connection.
2. In $T^{(0,0)}TM = M \times \mathbb{R}$, $\nabla_X f = Xf$.
3. $\nabla_X(F \otimes G) = (\nabla_X F) \otimes G + F \otimes (\nabla_X G)$.
4. ∇ commutes with all contractions: if “ Tr ” denotes a trace on any pair of indices, one covariant and one contravariant, then $\nabla_X(\text{Tr } F) = \text{Tr}(\nabla_X F)$.
5. ∇ obeys the following product rule with respect to the natural pairing between a co-vector field η and a vector field Y : $\nabla_X(\eta, Y) = (\nabla_X \eta, Y) + (\eta, \nabla_X Y)$.
6. For all $F \in T^{(k,l)}TM$, smooth 1-forms $\eta_1, \dots, \eta_k$, and smooth vector fields $Y_1, \dots, Y_l$:

$$\begin{aligned} \nabla_X F(\eta_1, \dots, \eta_k, Y_1, \dots, Y_l) &= X(F(\eta_1, \dots, \eta_k, Y_1, \dots, Y_l)) \\ &- \sum_{i=1}^k F(\eta_1, \dots, \nabla_X \eta_i, \dots, \eta_k, Y_1, \dots, Y_l) - \sum_{j=1}^l F(\eta_1, \dots, \eta_k, Y_1, \dots, \nabla_X Y_j, \dots, Y_l) \end{aligned}$$

For the proof we refer to [15, Proposition 4.15].

When considering the tangent bundle TM as the vector bundle over M , we can describe the connection in local coordinates with specific smooth functions called the *Christoffel symbols*.

Consider local coordinates $\{u^i\}_{i=1}^n$ on some coordinate neighborhood $U \stackrel{o}{\subseteq} M$ and let $\partial_i := \partial_{u^i}$, then any vector field $X \in \mathfrak{X}(U)$ can be expressed as $X = \sum_{i=1}^n x^i \partial_i$ for some $x^i \in C^\infty(U, \mathbb{R})$. Furthermore, for any i and j , the vector field $\nabla_{\partial_i} \partial_j$ can be expressed in terms of the *Christoffel symbols* Γ_{ij}^k

$$\nabla_{\partial_i} \partial_j = \sum_k \Gamma_{i,j}^k \partial_k \quad (\text{A.3})$$

Definition A.3. Consider local coordinates $\{u^i\}_{i=1}^n$ on some open coordinate neighborhood $U \stackrel{o}{\subseteq} M$ and let $\partial_i := \partial_{u^i}$. For any $X, Y \in \mathfrak{X}(U)$ with $X = \sum_{i=1}^n x^i \partial_i$ and $Y = \sum_{i=1}^n y^i \partial_i$, the *covariant derivative* of Y in the direction of X can be expressed locally as

$$\nabla_X Y = \sum_k \left(\sum_i x^i \left(\partial_i y^k + \sum_j \Gamma_{ij}^k y^j \right) \right) \partial_k \quad (\text{A.4})$$

Definition A.4. Let E be a vector bundle over M . Given a path $\gamma : I \rightarrow M$, defined over some interval $I \subseteq \mathbb{R}$, we call a *path in E above γ* any path $u : I \rightarrow E$ with the property that $u(t) \in E_{\gamma(t)}$ for all t . We will denote the vector space of such vector fields as $\mathfrak{X}(\gamma) \subset \mathfrak{X}$.

Definition A.5 (Vector fields along curves). In the case $E = TM$, a path $V : I \rightarrow TM$ in TM above γ will be called a *vector field along γ* and the property is $V_t \in T_{\gamma(t)}M$ for all t .

Theorem A.6 (Covariant Derivative along curves). *Consider M a smooth manifold with a connection $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$. For each curve $\gamma : I \subseteq \mathbb{R} \rightarrow M$, the connection determines a unique operator $\frac{D}{dt} : \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$, called the *covariant derivative along a curve* and it satisfies the following properties:*

- **Linearity:** For $V, W \in \mathfrak{X}(M)$

$$\frac{D(V+W)}{dt} = \frac{D(V)}{dt} + \frac{D(W)}{dt}.$$

- **Product Rule:** If $f \in C^\infty(I)$ then

$$\frac{D(fV)}{dt} = \frac{df}{dt} V + f \frac{D(V)}{dt}.$$

- **Induced Vector fields:** If V is induced by a vector field Y , $V_t = Y_{\gamma(t)} \forall t$ then

$$\frac{D(V)}{dt} = \nabla_{\frac{d\gamma}{dt}} Y.$$

Proof. Choose a local coordinate system for M , and let $u^1(t), \dots, u^n(t)$ denote the coordinates of the point $\gamma(y)$. The vector field V can be expressed uniquely in the form $V = \sum_i v^i \partial_i$ where $\{v^i\}$ are real valued functions on some open subset of $\mathbb{R}$ and $\{\partial_i\}$ are the standard vector fields. From the properties in the theorem, one has

$$\begin{aligned}
\frac{DV}{dt} &= \sum_i \frac{Dv^i \partial_i}{dt} = \sum_i \left(\frac{dv^i}{dt} \partial_i + v^i \frac{D(\partial_i)}{dt} \right) \\
&= \sum_i \left(\frac{dv^i}{dt} \partial_i + v^i \nabla_{\frac{d\gamma}{dt}} \partial_i \right) \\
&= \sum_i \left(\frac{dv^i}{dt} \partial_i + v^i \left(\sum_j \frac{du^j}{dt} \nabla_{\partial_j} \partial_i \right) \right) \\
&= \sum_k \left(\frac{dv^k}{dt} + \left(\sum_{i,j} \frac{du^j}{dt} \Gamma_{ij}^k v^i \right) \right) \partial_k
\end{aligned} \tag{A.5}$$

where in the second-to-last equality is because one has $\frac{d\gamma}{dt} = \sum_j \frac{du^j}{dt} \partial_j$ and in the last equality one uses Equation A.4. Conversely, defining $\frac{DV}{dt}$ by this formula, it is simple to see that the three properties are satisfied. $\square$

Definition A.7. A vector field $V \in \mathfrak{X}(M)$ along $\gamma : I \rightarrow M$ is said to be *parallel* if $\frac{DV}{dt} \equiv 0$.

Theorem A.8. Given a curve γ and a tangent vector V_0 at the point $\gamma(0)$, there is one and only one parallel vector field V along γ which extends V_0

Definition A.9. A connection ∇ on a smooth manifold M is *compatible* with the Riemannian metric if for any $X, Y, Z \in \mathfrak{X}(M)$ the following equality holds

$$\nabla_X \langle Y, X \rangle_g = \langle \nabla_X Y, X \rangle_g + \langle Y, \nabla_X X \rangle_g \tag{A.6}$$

Theorem A.10 (Connections compatible with inner products). *Let (M, g) be a Riemannian manifold with connection $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$. The following are equivalent:*

1. ∇ is compatible with g .
2. If V and W are parallel vector fields along a curve γ , then $\langle V, W \rangle_g = c$ for some constant $c \in \mathbb{R}$.
3. If V and W are vector fields along a curve γ , then

$$\frac{d}{dt} \langle V, W \rangle_g = \left\langle \frac{DV}{dt}, W \right\rangle + \left\langle V, \frac{DW}{dt} \right\rangle. \tag{A.7}$$

For the proof, we refer to *Introduction to Riemannian manifolds*, p119-121 [15].

Definition A.11. A connection $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ is said to be *symmetric* if for any $X, Y \in \mathfrak{X}(M)$ the following equality holds

$$\nabla_X Y - \nabla_Y X = [X, Y]. \tag{A.8}$$

Since for i, j we have $[\partial_i, \partial_j] = 0$, the Christoffel symbols for a symmetric connection are symmetric: $\Gamma_{ij}^k = \Gamma_{ji}^k$.

Another way of describing the symmetry of a connection is via the *torsion tensor*:

$$\tau : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M) \quad \tau(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y] \quad (\text{A.9})$$

A symmetric connection is also called *torsion-free*.

Theorem A.12 (Fundamental Theorem of Riemannian Geometry). *Let (M, g) be a Riemannian manifold. There exists a unique symmetric connection on TM that is compatible with the Riemannian metric g , called the Levi-Civita connection.*

For the proof we refer to [15, p.122-123] or [22, Lemma 8.6] .

Corollary A.12.1. *Let ∇ be the Levi-Civita connection on (M, g) , then for any vector fields $V, W, X \in \mathfrak{X}(M)$:*

$$X \langle V, W \rangle_g = \langle \nabla_X V, W \rangle_g + \langle V, \nabla_X W \rangle_g \quad (\text{A.10})$$

Remark A.1.2. From now on, we will assume that the connection we are working with is the Levi-Civita connection.

An aid to understanding symmetry is examining surfaces on a manifold and the vector fields along them. As in Definition A.5, one can define a vector field along a curve $\gamma : I \rightarrow M$. One can do the same for a parametrized surface in M : smooth maps $M : \mathbb{R}^2 \rightarrow M$.

Definition A.13. Let M be as defined above. We say $V : \mathbb{R}^2 \rightarrow TM$ is a *vector field along M* if $V_{(x,y)} \in T_{M(x,y)}M$ for all $(x, y) \in \mathbb{R}^2$.

Alike to the case of a vector field along a curve, one can define covariant derivatives for a vector field along a surface

$$\frac{DV}{\partial x} \quad \text{and} \quad \frac{DV}{\partial y},$$

which satisfy a equivalent properties as in Theorem A.6, specifically

$$\frac{D}{\partial x} = \nabla_{\frac{\partial M}{\partial x}} \quad \text{and} \quad \frac{D}{\partial y} = \nabla_{\frac{\partial M}{\partial y}} \quad (\text{A.11})$$

Similarly, when working with a connection compatible with a metric g on M , one has equivalent equations to Equation A.7:

$$\frac{\partial}{\partial x} \langle V, W \rangle_g = \left\langle \nabla_{\frac{\partial}{\partial x}} V, W \right\rangle + \left\langle V, \nabla_{\frac{\partial}{\partial x}} W \right\rangle \quad (\text{A.12})$$

$$\frac{\partial}{\partial y} \langle V, W \rangle_g = \left\langle \nabla_{\frac{\partial}{\partial y}} V, W \right\rangle + \left\langle V, \nabla_{\frac{\partial}{\partial y}} W \right\rangle. \quad (\text{A.13})$$

The way to define them is by keeping one parameter constant for the surface and obtaining a curve with a vector field along it, and then using the covariant derivative for such a vector field.

For the case $\frac{DV}{\partial x}$, fix an arbitrary y_0 , then M restricts to a curve $x \mapsto M(x, y_0)$ from which we obtain a vector field along it, $V_{(x, y_0)}$. The covariant derivative with respect to x is defined to be $\frac{DV}{\partial x}\big|_{(x, y_0)}$. This defines $\frac{DV}{\partial x}$ along the entire surface.

As one has the velocity and acceleration vectors for a curve γ , $\frac{d\gamma}{dt}$ and $\frac{D}{dt}\frac{d\gamma}{dt}$, equivalent definitions exist for surfaces M :

- The *velocity vector fields* of M are $\frac{\partial M}{\partial x}$ and $\frac{\partial M}{\partial y}$,
- and the *acceleration vector fields* of M are $\frac{D}{\partial x}\frac{\partial M}{\partial x}$ and $\frac{D}{\partial y}\frac{\partial M}{\partial y}$.

The mixed derivatives $\frac{D}{\partial x}\frac{\partial M}{\partial y}$ and $\frac{D}{\partial y}\frac{\partial M}{\partial x}$ behave nicely, when symmetry is assumed.

Lemma A.14. *Let ∇ be a symmetric connection on a Riemannian manifold (M, g) . Then for any surface M ,*

$$\frac{D}{\partial x}\frac{\partial M}{\partial y} = \frac{D}{\partial y}\frac{\partial M}{\partial x} \quad (\text{A.14})$$

Proof. Consider local coordinates $\{u^i\}$ with $\partial_i = \partial_{u^i}$. In local coordinates, we have velocity vector fields $\frac{\partial M}{\partial x} = \sum_k \frac{\partial u^k}{\partial x} \partial_k$, and $\frac{\partial M}{\partial y} = \sum_k \frac{\partial u^k}{\partial y} \partial_k$, with acceleration fields $\frac{D}{\partial x}\frac{\partial M}{\partial y} = \sum_k \left(\frac{\partial^2 u^k}{\partial x \partial y} + \sum_{ij} \frac{\partial u^i}{\partial y} \frac{\partial u^j}{\partial x} \Gamma_{ij}^k \right) \partial_k$ and $\frac{D}{\partial y}\frac{\partial M}{\partial x} = \sum_k \left(\frac{\partial^2 u^k}{\partial y \partial x} + \sum_{ij} \frac{\partial u^i}{\partial x} \frac{\partial u^j}{\partial y} \Gamma_{ji}^k \right) \partial_k$. Due to symmetry, $\Gamma_{ij}^k = \Gamma_{ji}^k$, by interchanging $i \rightleftharpoons j$, we have the desired equality. $\square$

A.2. Curvature Tensor

A Riemannian metric is said to be *flat* if it is locally isometric to Euclidean space. The Euclidean connection ∇^{can} on $\mathbb{R}^n$ satisfies the following equation for any vector fields $X, Y, Z \in \mathfrak{X}(\mathbb{R}^n)$:

$$\nabla_X^{\text{can}} \nabla_Y^{\text{can}} Z - \nabla_Y^{\text{can}} \nabla_X^{\text{can}} Z = \nabla_{[X, Y]}^{\text{can}} Z.$$

From this we can call a connection ∇ on M *flat* if

$$\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z = \nabla_{[X, Y]} Z.$$

Proposition A.15. *The map $R : \mathfrak{X}(M) \times \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ defined by*

$$R(X, Y, Z) = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z \quad (\text{A.15})$$

is $(1, 3)$ -tensor field on M

For the proof, we refer to [15] or [22].

Remark A.2.1. In local coordinates, the curvature tensor takes the form

$$R(\partial_i, \partial_j, \partial_k) = R_{ijk}^l \partial_l \quad (\text{A.16})$$

$$R_{ijk}^l = \partial_i \Gamma_{jk}^l - \partial_j \Gamma_{ik}^l + \Gamma_{jk}^m \Gamma_{im}^l - \Gamma_{ik}^m \Gamma_{jm}^l. \quad (\text{A.17})$$

Lemma A.16. Let $M : \mathbb{R}^2 \rightarrow M$ be a parametrized surface and V a vector field along M , then

$$R\left(\frac{\partial M}{\partial x}, \frac{\partial M}{\partial y}, V\right) = \frac{D}{\partial x} \frac{D}{\partial y} V - \frac{D}{\partial y} \frac{D}{\partial x} V \quad (\text{A.18})$$

Proof. Consider the surface in local coordinates $M(x, y) = u^1(x, y), \dots, u^n(x, y)$ and the vector field $V = \sum_j v^j \partial_j$ with velocity vectors

$$\frac{D}{\partial x} V = \frac{D}{\partial x} \left(\sum_j v^j \partial_j \right) = \sum_j \left(\frac{\partial v^j}{\partial x} \partial_j + v^j \frac{D}{\partial x} \partial_j \right) \quad (\text{A.19})$$

$$\frac{D}{\partial y} V = \frac{D}{\partial y} \left(\sum_j v^j \partial_j \right) = \sum_j \left(\frac{\partial v^j}{\partial y} \partial_j + v^j \frac{D}{\partial y} \partial_j \right), \quad (\text{A.20})$$

and mixed derivatives

$$\begin{aligned} \frac{D}{\partial x} \frac{D}{\partial y} V &= \frac{D}{\partial x} \left(\sum_j \left(\frac{\partial v^j}{\partial y} \partial_j + v^j \frac{D}{\partial y} \partial_j \right) \right) \\ &= \sum_j \left(\frac{\partial^2 v^j}{\partial x \partial y} \partial_j + \frac{\partial v^j}{\partial y} \frac{D}{\partial x} \partial_j + \frac{\partial v^j}{\partial x} \frac{D}{\partial y} \partial_j + v^j \frac{D}{\partial x} \frac{D}{\partial y} \partial_j \right), \end{aligned}$$

and

$$\frac{D}{\partial y} \frac{D}{\partial x} V = \sum_j \left(\frac{\partial^2 v^j}{\partial x \partial y} \partial_j + \frac{\partial v^j}{\partial x} \frac{D}{\partial y} \partial_j + \frac{\partial v^j}{\partial y} \frac{D}{\partial x} \partial_j + v^j \frac{D}{\partial y} \frac{D}{\partial x} \partial_j \right).$$

Subtracting the two mixed derivatives gives us,

$$\frac{D}{\partial y} \frac{D}{\partial x} V - \frac{D}{\partial x} \frac{D}{\partial y} V = \sum_j v^j \left(\frac{D}{\partial y} \frac{D}{\partial x} \partial_j - \frac{D}{\partial x} \frac{D}{\partial y} \partial_j \right). \quad (\text{A.21})$$

Recall that $\frac{\partial M}{\partial x} = \sum \frac{\partial u^k}{\partial x} \partial_k$ and $\frac{\partial M}{\partial y} = \sum \frac{\partial u^i}{\partial y} \partial_i$, so $\frac{D}{\partial x} \partial_j = \nabla_{\frac{\partial M}{\partial x}} \partial_j = \sum_k \frac{\partial u^k}{\partial x} \nabla_{\partial_k} \partial_j$, and the mixed derivatives can be written out as

$$\begin{aligned} \frac{D}{\partial y} \frac{D}{\partial x} \partial_j &= \frac{D}{\partial y} \left(\sum_k \frac{\partial u^k}{\partial x} \nabla_{\partial_k} \partial_j \right) = \sum_k \left(\frac{\partial^2 u^k}{\partial y \partial x} \nabla_{\partial_k} \partial_j + \frac{\partial u^k}{\partial x} \nabla_{\frac{\partial M}{\partial y}} \nabla_{\partial_k} \partial_j \right) \\ &= \sum_k \left(\frac{\partial^2 u^k}{\partial y \partial x} \nabla_{\partial_k} \partial_j + \sum_i \left(\frac{\partial u^k}{\partial x} \frac{\partial u^i}{\partial y} \nabla_{\partial_i} \nabla_{\partial_k} \partial_j \right) \right), \end{aligned}$$

and,

$$\frac{D}{\partial x} \frac{D}{\partial y} \partial_j = \sum_i \left(\frac{\partial^2 u^i}{\partial x \partial y} \nabla_{\partial_i} \partial_j + \sum_k \left(\frac{\partial u^i}{\partial y} \frac{\partial u^k}{\partial x} \nabla_{\partial_k} \nabla_{\partial_i} \partial_j \right) \right).$$

Lastly, by interchanging $i \rightleftharpoons k$, and $x \rightleftharpoons y$, the right-hand-side of Equation A.21 becomes

$$\begin{aligned} \frac{D}{\partial x} \frac{D}{\partial y} \partial_j - \frac{D}{\partial y} \frac{D}{\partial x} \partial_j &= \sum_{i,k} \frac{\partial u^i}{\partial y} \frac{\partial u^k}{\partial x} (\nabla_{\partial_k} \nabla_{\partial_i} \partial_j - \nabla_{\partial_i} \nabla_{\partial_k} \partial_j) \\ &= \sum_{i,k} \frac{\partial u^i}{\partial y} \frac{\partial u^k}{\partial x} (R(\partial_k, \partial_i, \partial_j)) = R\left(\frac{\partial M}{\partial x}, \frac{\partial M}{\partial y}, \partial_j\right), \quad (\text{A.22}) \end{aligned}$$

and

$$\frac{D}{\partial y} \frac{D}{\partial x} V - \frac{D}{\partial x} \frac{D}{\partial y} V = \sum_j v^j R \left(\frac{\partial M}{\partial x}, \frac{\partial M}{\partial y}, \partial_j \right) = R \left(\frac{\partial M}{\partial x}, \frac{\partial M}{\partial y}, V \right). \quad (\text{A.23})$$

□

Lemma A.17 (Properties of the Curvature Tensor). *The Curvature tensor of a Riemannian Manifold (M, g) satisfies the following for any $X, Y, Z, W \in \mathfrak{X}(M)$:*

1. $R(X, Y, Z) + R(Y, X, Z) = 0$
2. $R(X, Y, Z) + R(Y, Z, X) + R(Z, X, Y) = 0$
3. $\langle R(X, Y, Z), W \rangle_g + \langle R(X, Y, W), Z \rangle_g = 0$
4. $\langle R(X, Y, Z), W \rangle_g = \langle R(Z, W, X), Y \rangle_g$

Proof. 1. Let X, Y, Z be as in the statement.

$$\begin{aligned} R(X, Y, Z) + R(Y, X, Z) &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z + \nabla_Y \nabla_X Z - \nabla_X \nabla_Y Z - \nabla_{[Y, X]} Z \\ &= -\nabla_{[X, Y]} Z - \nabla_{[Y, X]} Z \\ &= -\nabla_{[X, Y]} Z + \nabla_{[X, Y]} Z = 0. \end{aligned}$$

2. Recall that the connection is symmetric : $\forall V, W \in \mathfrak{X}(M) \quad \nabla_V W - \nabla_W V = [V, W]$

$$\begin{aligned} R(X, Y, Z) + R(Y, Z, X) + R(Z, X, Y) &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z + \nabla_Y \nabla_Z X - \nabla_Z \nabla_Y X - \nabla_{[Y, Z]} X \\ &\quad + \nabla_Z \nabla_X Y - \nabla_X \nabla_Z Y - \nabla_{[Z, X]} Y \\ &= \nabla_X (\nabla_Y Z - \nabla_Z Y) - \nabla_{[X, Y]} Z + \nabla_Y (\nabla_Z X - \nabla_X Z) - \nabla_{[Y, Z]} X \\ &\quad + \nabla_Z (\nabla_X Y - \nabla_Y X) - \nabla_{[Z, X]} Y \\ &= \nabla_X [Y, Z] - \nabla_{[X, Y]} Z + \nabla_Y [Z, X] - \nabla_{[Y, Z]} X + \nabla_Z [X, Y] - \nabla_{[Z, X]} Y \\ &= (\nabla_X [Y, Z] - \nabla_{[Y, Z]} X) + (\nabla_Y [Z, X] - \nabla_{[Z, X]} Y) + (\nabla_Z [X, Y] - \nabla_{[X, Y]} Z) \\ &= [[Y, Z], X] + [[Z, X], Y] + [[X, Y], Z] = 0. \end{aligned}$$

3. Showing that $\langle R(X, Y, Z), W \rangle$ is skew-symmetric in Z and W is equivalent to showing

$$\langle R(X, Y, Z), Z \rangle_g = \langle \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z, Z \rangle_g = 0.$$

Recall the compatibility for vector fields $A, B, C \in \mathfrak{X}(M)$: $A \langle B, C \rangle_g = \langle \nabla_A B, C \rangle_g +$

$\langle B, \nabla_A C \rangle_g$. So we have

$$\begin{aligned}
[X, Y] \langle Z, Z \rangle_g &= \langle \nabla_{[X, Y]} Z, Z \rangle_g + \langle Z, \nabla_{[X, Y]} Z \rangle_g = 2 \langle \nabla_{[X, Y]} Z, Z \rangle_g \\
X \langle Z, Z \rangle_g &= \langle \nabla_X Z, Z \rangle_g + \langle Z, \nabla_X Z \rangle_g = 2 \langle \nabla_X Z, Z \rangle_g \\
Y \langle Z, Z \rangle_g &= \langle \nabla_Y Z, Z \rangle_g + \langle Z, \nabla_Y Z \rangle_g = 2 \langle \nabla_Y Z, Z \rangle_g \\
XY \langle Z, Z \rangle_g &= X (2 \langle \nabla_Y Z, Z \rangle_g) = 2 \langle \nabla_X \nabla_Y Z, Z \rangle_g + 2 \langle \nabla_Y Z, \nabla_X Z \rangle_g \\
YX \langle Z, Z \rangle_g &= Y (2 \langle \nabla_X Z, Z \rangle_g) = 2 \langle \nabla_Y \nabla_X Z, Z \rangle_g + 2 \langle \nabla_X Z, \nabla_Y Z \rangle_g
\end{aligned}$$

so then

$$\begin{aligned}
(YX - XY + [X, Y]) \langle Z, Z \rangle_g &= 2 \langle \nabla_Y \nabla_X Z, Z \rangle_g + 2 \langle \nabla_X Z, \nabla_Y Z \rangle_g \\
&\quad + 2 \langle \nabla_X \nabla_Y Z, Z \rangle_g + 2 \langle \nabla_Y Z, \nabla_X Z \rangle_g + 2 \langle \nabla_{[X, Y]} Z, Z \rangle_g \\
&= 2 \langle \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z, Z \rangle_g.
\end{aligned}$$

But $[X, Y] = XY - YX$, so $\langle \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z, Z \rangle_g = 0$.

4. From property 2, we have the following 4 equations

$$\begin{aligned}
\langle R(X, Y, Z), W \rangle_g + \langle R(Y, Z, X), W \rangle_g + \langle R(Z, X, Y), W \rangle_g &= 0 \\
\langle R(Y, Z, W), X \rangle_g + \langle R(Z, W, Y), X \rangle_g + \langle R(W, Y, Z), X \rangle_g &= 0 \\
\langle R(Z, W, X), Y \rangle_g + \langle R(W, X, Z), Y \rangle_g + \langle R(X, Z, W), Y \rangle_g &= 0 \\
\langle R(W, X, Y), Z \rangle_g + \langle R(X, Y, W), Z \rangle_g + \langle R(Y, W, X), Z \rangle_g &= 0
\end{aligned}$$

By adding them all together and applying property 3 to the underlined pairs noted below

$$\begin{aligned}
&\langle R(X, Y, Z), W \rangle_g + \langle R(Y, Z, X), W \rangle_g + \langle R(Z, X, Y), W \rangle_g \\
&+ \langle R(Y, Z, W), X \rangle_g + \langle R(Z, W, Y), X \rangle_g + \langle R(W, Y, Z), X \rangle_g \\
&+ \langle R(Z, W, X), Y \rangle_g + \langle R(W, X, Z), Y \rangle_g + \langle R(X, Z, W), Y \rangle_g \\
&+ \langle R(W, X, Y), Z \rangle_g + \langle R(X, Y, W), Z \rangle_g + \langle R(Y, W, X), Z \rangle_g = 0
\end{aligned}$$

One is left with

$$\langle R(Z, X, Y), W \rangle_g + \langle R(W, Y, Z), X \rangle_g + \langle R(X, Z, W), Y \rangle_g + \langle R(Y, W, X), Z \rangle_g = 0.$$

Applying property 3 and 1 several more times, the desired result is achieved:

$$\begin{aligned}
&\langle R(Z, X, Y), W \rangle_g - \langle R(W, Y, X), Z \rangle_g + \langle R(X, Z, W), Y \rangle_g - \langle R(Y, W, Z), X \rangle_g = 0 \\
\iff &\langle R(Z, X, Y), W \rangle_g - \langle R(W, Y, X), Z \rangle_g - \langle R(Z, X, W), Y \rangle_g + \langle R(W, Y, Z), X \rangle_g = 0 \\
\iff &\langle R(Z, X, Y), W \rangle_g - \langle R(W, Y, X), Z \rangle_g + \langle R(Z, X, Y), W \rangle_g - \langle R(W, Y, X), Z \rangle_g = 0 \\
\iff &2 \langle R(Z, X, Y), W \rangle_g - 2 \langle R(W, Y, X), Z \rangle_g = 0.
\end{aligned}$$

□

A.3. Geodesics and The Exponential Map

A.3.1. Existence and Uniqueness of Geodesics and The Exponential Map

Definition A.18. An admissible curve $\gamma : I \rightarrow M$ is called a *geodesic* if

$$\frac{D}{dt}\gamma' \equiv 0. \quad (\text{A.24})$$

In terms of a local coordinates $\{u^1, \dots, u^n\}$ with $\partial_i = \partial_{u^i}$, then locally $\gamma'(t) = \sum_{i=1}^n \frac{\partial u^i(t)}{\partial t} \partial_i$ and Equation A.5 for $\frac{D}{dt}\gamma'$ gives us the *geodesic equation*

$$\frac{D}{dt}\gamma' = \sum_k \left(\frac{d^2 u^k}{dt^2} + \left(\sum_{i,j} \frac{du^j}{dt} \frac{du^i}{dt} \Gamma_{ij}^k \right) \right) \partial_k = 0. \quad (\text{A.25})$$

Theorem A.19 (Existence and uniqueness). *Let M be a smooth manifold and ∇ a connection on M . For every $p \in M, v \in T_p M$, and $t_0 \in \mathbb{R}$ there exist an open interval $I \subseteq \mathbb{R}$ containing t_0 and a geodesic $\gamma_v : I \rightarrow M$ satisfying $\gamma_v(t_0) = p$ and $\gamma'_v(t_0) = v$. Any two such geodesics agree on their common domain.*

For the proof, we refer to [15, Theorem 4.27].

Remark A.3.1. The subscript v will be added when we want to make explicit what $\gamma'(t_0)$ is.

A geodesic $\gamma : I \rightarrow M$ is said to be *maximal* if it cannot be extended to a geodesic on a larger interval, that is, if there does not exist a geodesic $\tilde{\gamma} : \tilde{I} \rightarrow M$ defined on an interval $\tilde{I} \supset I$ and such that $\tilde{\gamma}|_I = \gamma$. A *geodesic segment* is a geodesic whose domain is a compact interval.

Corollary A.19.1. *Let M be a smooth manifold with connection ∇ . For each $p \in M$ and $v \in T_p M$, there is a unique maximal geodesic $\gamma : I \rightarrow M$ defined on some open interval I containing 0, such that $\gamma(t_0) = p$ and $\gamma'(t_0) = v$.*

For the proof, we refer to [15, Corollary 4.28].

The following lemma will help in calculations with geodesics and the exponential map:

Lemma A.20 (Rescaling Lemma). *For all $p \in M, v \in T_p M$ and $c, t \in \mathbb{R}$, then $\gamma_{cv}(t) = \gamma_v(ct)$, whenever either is defined.*

From the Existence and Uniqueness theorem, we can define the *exponential map*.

Definition A.21. Define the domain of the exponential map to be

$$\mathcal{E} := \{(p, v) \in TM \mid \gamma_v : I \rightarrow M \text{ is a geodesic such that } [0, 1] \subseteq I\} \quad (\text{A.26})$$

and the *exponential map*

$$\exp : \mathcal{E} \rightarrow M \quad (p, v) \mapsto \gamma_v(1). \quad (\text{A.27})$$

For each $p \in M$, the *restricted exponential map at p* is $\exp_p : \mathcal{E} \cap T_p M \rightarrow M$.

Theorem A.22 (Properties of the exponential map). *Let $\exp : \mathcal{E} \rightarrow M$ be the exponential map:*

- (a) $\mathcal{E}$ is an open subset of TM containing the image of the zero section, and each set $\mathcal{E}_p \subseteq T_pM$ is star-convex with respect to 0 (any line segment from 0 to another point of $\mathcal{E}_p$ is contained in $\mathcal{E}_p$).
- (b) For each $v \in TM$, the geodesic γ_v is given by $\gamma_v(t) = \exp(tv)$ for all t such that either side is defined.
- (c) The exponential map is smooth.
- (d) For any $p \in M$, the differential $d(\exp_p)_0 : T_0(T_pM) \cong T_pM \rightarrow T_pM$ is the identity map of T_pM .

A.3.2. Normal coordinates

For all $p \in M$, $\exp_p : \mathcal{E}_p \rightarrow M$ is smooth into M . Since $d(\exp_p)_0$ is invertible, the inverse function theorem guarantees that there exists an open neighborhood of 0, $V \overset{o}{\subseteq} T_pM$ and an open neighborhood of p , $U \overset{o}{\subseteq} M$ such that $\exp_p : V \rightarrow U$ is a diffeomorphism.

Definition A.23. • A neighborhood U of p in M that is the diffeomorphic image under $\exp_p$ of a star-convex neighborhood of 0 in T_pM is called a *normal neighborhood* of p .

- For $\epsilon > 0$ such that $\exp_p : B_\epsilon(0) \subseteq T_pM \rightarrow M$ is a diffeomorphism onto its image, the image set $\exp_p(B_\epsilon(0))$ is a normal neighborhood of p called the *geodesic ball* in M or *open geodesic ball*.
- If $\overline{B_\epsilon(0)} \subseteq V \overset{o}{\subseteq} T_pM$ with $\exp_p : V \rightarrow M$ a diffeomorphism onto its image, then $\exp_p(\overline{B_\epsilon(0)})$ is called a *closed geodesic ball* and $\exp_p(\partial B_\epsilon(0))$ is a *geodesic sphere*. By compactness, there exists a $\epsilon' > \epsilon$ such that $\overline{B_\epsilon(0)} \subset B_{\epsilon'}(0) \subseteq V$.

For simplicity, we will abuse notation and denote $B_\epsilon(p) := \exp_p(B_\epsilon(0))$, $\overline{B_\epsilon(p)} := \exp_p(\overline{B_\epsilon(0)})$, and $\partial B_\epsilon(p) := \exp_p(\partial B_\epsilon(0))$.

Theorem A.24. *Every orthonormal basis $\{b_i\}$ of T_pM determines a basis isomorphism $B : \mathbb{R}^n \rightarrow T_pM$ defined by $(x^1, \dots, x^n) \mapsto \sum_i x^i b_i$. If U is a normal neighborhood of p , via the diagram below, one obtains a smooth normal coordinate map $\varphi : U \rightarrow \mathbb{R}^n$:*

$$\begin{array}{ccc} T_pM & \xrightarrow{B^{-1}} & \mathbb{R}^n \\ (\exp_p|_V)^{-1} \uparrow & \varphi \nearrow & \\ U & & \end{array}$$

Theorem A.25 (Uniqueness of Normal coordinates). *Let $p \in (M, g)$ and U a normal neighborhood of p . For all normal coordinate charts on U centered at p , the coordinate basis is orthonormal at p , and for all orthonormal bases $\{b_i\}$ for T_pM , then there exists*

a unique normal coordinate chart $\{x^i\}$ on U such that $\partial_i|_p = b_i$ for $i = 1, \dots, n$. Furthermore, any two normal coordinates $\{x^i\}$ and $\{\tilde{x}^i\}$ are related by $\tilde{x}^i = A_i^j x^j$ for some matrix $(A_i^j) \in O(n)$.

Theorem A.26 (Properties of normal coordinates). *Consider a Riemannian manifold (M, g) and any normal coordinate chart $(U, \{u^i\})$ centered at $p \in M$.*

(a) *The coordinates of p are $(0, \dots, 0)$.*

(b) *The components of the metric at p are $g_{ij} = \left\langle \frac{\partial}{\partial u_i}, \frac{\partial}{\partial u_j} \right\rangle_g = \delta_i^j$.*

(c) *For all $v = \sum_i v^i \partial_i|_p \in T_p M$, the geodesic γ_v is represented in normal coordinates by the line*

$$\gamma_v(t) = (tv^1(t), \dots, tv^n(t)) \quad (\text{A.28})$$

as long as t is in some interval I containing 0 such that $\gamma_v(I) \subseteq U$.

(d) *The Christoffel symbols in these coordinates vanish at p .*

(e) *All the first partial derivatives of g_{ij} in these coordinates vanish at p .*

The geodesics starting at a point p and lying in a normal neighborhood of p are called *radial geodesics*.

Remark A.3.2. In normal coordinates centered at a point $p \in M$, the open and closed geodesic balls and geodesic spheres centered at p are simply coordinate balls and spheres.

A.4. Divergence, Gradient and Laplacian

A.4.1. The Gradient of a Function

Definition A.27. Let (M, g) be a Riemannian manifold with metric g . The *gradient* of $f : M \rightarrow \mathbb{R}$ is the vector field

$$\text{grad } f = \nabla f = \sum_{i=1}^n \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^i}. \quad (\text{A.29})$$

on M which is characterized by the identity

$$\langle X, \text{grad } f \rangle_g = X(f) \quad (\text{A.30})$$

Proposition A.28. *Let (M, g) be a Riemannian manifold, $f \in C^\infty(M, \mathbb{R})$ and $X \in \mathfrak{X}(M)$ is a nowhere-vanishing vector field. Then, $X = \text{grad } f$ if and only if $Xf = \|X\|_g^2$ and X is orthogonal to the level sets of f at all regular points.*

Proof. ‘ $\Rightarrow$ ’ Consider the gradient vector field of f , $\text{grad } f$. Since f is constant on the level sets $f^{-1}(c)$ (where $c \in \mathbb{R}$), for every curve $\gamma : (-\epsilon, \epsilon) \rightarrow f^{-1}(c)$, the function $h(t) = f(\gamma(t))$ is constant. Every tangent vector $Y = \gamma'(0)$ to the level set $f^{-1}(c)$ can be expressed in such a way, thus

$$0 = h'(0) = df(\gamma(0)) \gamma'(0) = Yf|_{\gamma(0)} = \langle Y, \text{grad } f \rangle_g|_{\gamma(0)}.$$

As Y was chosen arbitrarily, we see that the gradient of f is orthogonal to the level set.

Moreover, by the definition of the gradient vector field:

$$\text{grad } f(f) = \langle \text{grad } f, \text{grad } f \rangle_g = \|\text{grad } f\|_g^2$$

‘ $\Leftarrow$ ’ Consider $X \in \mathfrak{X}(M)$ such that $Xf = \|X\|_g^2$ and X is orthogonal to the level sets of f at all regular points. By the previous part, we have that $\text{grad } f$ is also orthogonal to the level sets of f , thus $\text{grad } f$ must be a multiple of X , so there exists a $\lambda \neq 0$ such that $\text{grad } f = \lambda X$. Since $Xf = \|X\|_g^2 = \langle X, X \rangle_g$, the definition of the gradient gives us $\lambda \langle X, X \rangle_g = \langle X, \lambda X \rangle_g = \langle X, \text{grad } f \rangle_g = X(f) = \langle X, X \rangle_g$. Therefore, $\lambda = 1$ and $X = \text{grad } f$. $\square$

From the gradient, we can give an equivalent definition of what the *Hessian* of a function is.

Definition A.29. Let (M, g) be a Riemannian manifold and $f : M \rightarrow \mathbb{R}$ a smooth function. If $p \in M$ is a critical point of f , we define the *Hessian*

$$\text{Hess}_p(f) : T_p M \rightarrow T_p^* M \quad \text{defined by} \quad (v) \mapsto \nabla_v(df). \quad (\text{A.31})$$

In local coordinates $\{x^i\}$ one has the local expression

$$\text{Hess}_p(f) = \sum_{i,j=1}^n \nabla_{\partial x^i} \partial x^j f dx^i \otimes dx^j = \sum_{i,j=1}^n \left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \Gamma_{ij}^k \frac{\partial f}{\partial x^k} \right) dx^i \otimes dx^j. \quad (\text{A.32})$$

A.4.2. Divergence and Laplacian

Suppose (M, g) is an orientable Riemannian n -manifold with or without boundary and

$$d\mu_g = \sqrt{\det g} dx^1 \wedge \cdots \wedge dx^n \quad (\text{A.33})$$

its volume form, for local coordinates $(x^1, \dots, x^n)$. For a smooth vector field $X \in \mathfrak{X}(M)$, the interior multiplication $\lrcorner_X : \Omega^n \rightarrow \Omega^{n-1}$ maps the volume form to an $(n-1)$ -form $\lrcorner_X(d\mu_g)$, and the exterior derivative $d : \Omega^{n-1} \rightarrow \Omega^n$ gives a top-form, $d(\lrcorner_X d\mu_g)$, so it can be expressed as multiple of the volume form by some smooth function, which we will call the *divergence of X* , $\text{div } X$:

$$d(\lrcorner_X(d\mu_g)) = (\text{div}_g X) d\mu_g \quad (\text{A.34})$$

Remark A.4.1. By Cartan’s formula, for $V \in \mathfrak{X}(M)$ and α any smooth differentiable form:

$$\mathcal{L}_V \alpha = \lrcorner_V(d\alpha) + d(\lrcorner_V \alpha) \quad (\text{A.35})$$

Equation A.34 is also of the form

$$\mathcal{L}_X(d\mu_g) = d(\lrcorner_X(d\mu_g)) = (\text{div}_g X) d\mu_g. \quad (\text{A.36})$$

Proposition A.30. *For $X \in \mathfrak{X}(M)$, the divergence of X can be described as the trace of the Levi-Civita connection ∇ of M : for $\{E_i\}$ an orthonormal basis for $T_p M$ for some $p \in M$ and $X \in \mathfrak{X}(M)$,*

$$\operatorname{div}_g X = \operatorname{Tr}(\nabla X) = \sum_i \langle \nabla_{E_i} X, E_i \rangle_g. \quad (\text{A.37})$$

Proof. Let $p \in M$ be arbitrary and consider a local oriented orthonormal frame $\{E^i\}$ of T^*M , then $dV_g = E^1 \wedge \cdots \wedge E^n$. Remark that the volume form is parallel to the Levi-Civita connection:

$$0 = \nabla_X d\mu_g = \sum_i E^1 \wedge \cdots \wedge \nabla_X E^i \wedge \cdots \wedge E^n = \sum_i \langle \nabla_X E^i, E^i \rangle_g d\mu_g. \quad (\text{A.38})$$

Thus

$$\begin{aligned} \operatorname{div}_g X d\mu_g &= \mathcal{L}_X d\mu_g = (\mathcal{L}_X - \nabla_X) d\mu_g \\ &= \sum_i E^i \wedge \cdots \wedge (\mathcal{L}_X E^i - \nabla_X E^i) \wedge \cdots \wedge E^n = \sum_i \langle \mathcal{L}_X E^i - \nabla_X E^i, E^i \rangle_g d\mu_g. \end{aligned}$$

Therefore,

$$\begin{aligned} \operatorname{div}_g X &= \sum_i \langle \mathcal{L}_X E^i - \nabla_X E^i, E^i \rangle_g \\ &= \sum_i (\mathcal{L}_X E^i)(E_i) - (\nabla_X E^i) E_i \\ &= \sum_i E^i (\nabla_X E_i - \mathcal{L}_X E_i) \\ &= \sum_i \langle \nabla_X E_i - [X, E_i], E_i \rangle_g = \sum_i \langle \nabla_{E_i} X, E_i \rangle_g \end{aligned}$$

the second to last equality is due to items 2 and 5 from Proposition A.2, which give us $(\nabla_X E^i)(E_i) = -E^i(\nabla_X E_i)$, and from Lie derivative on differential forms

$$\mathcal{L}_V (\eta(Y)) = (\mathcal{L}_V \eta)(Y) + \eta([V, Y]),$$

we have $(\mathcal{L}_X E^i)(E_i) = -E^i(\mathcal{L}_X E_i)$. The last equality is due to Lemma A.14. $\square$

Remark A.4.2. In the proof above, we used the fact that an inner product $\langle \cdot, \cdot \rangle$ on a finite-dimensional vector space V induces a linear isomorphism to its dual V^* : $V \rightarrow V^*$, where $v \in V \mapsto \nu \in V^*$ is such that

$$\nu(w) = \langle v, w \rangle \quad \forall w \in V.$$

Theorem A.31 (Divergence Theorem). *Suppose (M, g) is a compact Riemannian manifold with boundary. For $X \in \mathfrak{X}(M)$,*

$$\int_M (\operatorname{div}_g X) d\mu_g = \int_{\partial M} \langle X, \eta \rangle_{\hat{g}} d\mu_{\hat{g}} \quad (\text{A.39})$$

where η is the outward unit normal to ∂M and $\hat{g}$ is the induced metric on ∂M .

Proof. From the definition of divergence, we have the following equalities

$$\int_M (\operatorname{div}_g X) d\mu_g = \int_M d(\lrcorner_X(d\mu_g)) = \int_{\partial M} i^*(\lrcorner_X(d\mu_g))$$

where i^* is the pullback of the inclusion $i : \partial M \rightarrow M$. From [16, Lemma 16.30] one has $i^*(\lrcorner_X(d\mu_g)) = \langle X, \eta \rangle_{\hat{g}} d\mu_{\hat{g}}$ and we have the desired equality. $\square$

Corollary A.31.1. *For $f \in C^\infty(M)$ and $X \in \mathfrak{X}(M)$, the divergence satisfies the following Leibniz identity*

$$\operatorname{div}_g(fX) = f \operatorname{div}_g(X) + \langle \operatorname{grad} f, X \rangle_g, \quad (\text{A.40})$$

and

$$\int_{\partial M} f \langle X, \eta \rangle_{\hat{g}} d\mu_{\hat{g}} = \int_M f \operatorname{div}_g(X) d\mu_g + \int_M \langle \operatorname{grad} f, X \rangle_g d\mu_g \quad (\text{A.41})$$

Proof. We begin by showing the first equality. Using Equation A.37 and taking the orthonormal frame to be $\{\frac{\partial}{\partial x_i} = \partial_i\}$ for some local coordinates $\{x_i\}$, we have

$$\begin{aligned} \operatorname{div}_g(fX) &= \sum_i \langle \nabla_{\partial_i} X, \partial_i \rangle_g \\ &= \sum_i \langle f \nabla_{\partial_i} X + (\partial_i f) X, \partial_i \rangle_g \\ &= f \sum_i \langle \nabla_{\partial_i} X, \partial_i \rangle_g + \sum_i \langle (\partial_i f) X, \partial_i \rangle_g \\ &= f \operatorname{div}_g X + \langle \operatorname{grad} f, X \rangle_g. \end{aligned}$$

From this first equality and Theorem A.31, the second equality is plain to see:

$$\begin{aligned} \int_M \operatorname{div}_g(fX) d\mu_g &= \int_M f \operatorname{div}_g(X) d\mu_g + \int_M \langle \operatorname{grad} f, X \rangle_g d\mu_g \\ \iff \int_{\partial M} f \langle X, \eta \rangle_{\hat{g}} d\mu_{\hat{g}} &= \int_M f \operatorname{div}_g(X) d\mu_g + \int_M \langle \operatorname{grad} f, X \rangle_g d\mu_g. \end{aligned}$$

$\square$

The *Laplacian* is the linear operator $\Delta : C^\infty(M) \rightarrow C^\infty$ defined by

$$\Delta f = \operatorname{div}_g(\operatorname{grad} f). \quad (\text{A.42})$$

Proposition A.32. *Let (M, g) be a Riemannian manifold (with or without boundary) and let (x^i) be any smooth local coordinates on an open set $U \subseteq M$. The coordinate representations of the divergence and Laplacian are as follows :*

$$\operatorname{div}_g \left(\sum_i X^i \partial_i \right) = \sum_i \frac{1}{\sqrt{\det g}} \partial_i (X^i \sqrt{\det g}) \quad (\text{A.43})$$

$$\Delta f = \sum_i \frac{1}{\sqrt{\det g}} \partial_i (g^{ij} X^i \sqrt{\det g} \partial_j) \quad (\text{A.44})$$

where $X = \sum_i X^i \partial_i$ is a smooth vector field on M , f a smooth function on M , and $\det g = \det(g_{kl})$ is the determinant of the components of g in these coordinates ($g_{kl} = \langle \partial_k, \partial_l \rangle_g$). On $\mathbb{R}^n$ with Euclidean metric and standard coordinates, these reduce to

$$\operatorname{div}_g \left(\sum_i X^i \partial_i \right) = \sum_i \partial_i X^i \quad (\text{A.45})$$

$$\Delta f = \sum_i \partial_i^2 f. \quad (\text{A.46})$$

Proof. Let $\{x^i\}$ be local coordinates and $g_{ij} = \langle \partial_i, \partial_j \rangle_g$ and $\det g = \det(g_{ij})$. Remark that $\det g > 0$.

The inner product can be written as the 2-tensor $\frac{1}{2}g_{ij}dx^i \otimes dx^j$. Let $X = \sum_i X^i \partial_i$ be a smooth vector field in these coordinates, then

$$\lrcorner_X d\mu_g = \sum_j (-1)^{j-1} X^j \sqrt{\det(g)} dx^1 \wedge \cdots \wedge \widehat{dx^j} \wedge \cdots \wedge dx^n$$

and

$$d(\lrcorner_X d\mu_g) = \sum_j \partial_j (X^j \sqrt{\det(g)}) dx^1 \wedge \cdots \wedge dx^n,$$

as a result $\operatorname{div}_g(X) = \sum_i \frac{1}{\sqrt{\det g}} \partial_i (X^i \sqrt{\det g})$.

Now for the equation for the gradient, Consider $df = \sum_i \partial_i f dx^i$. Remark that the transformation of co-vector fields $\sum_i v_i dx^i$ to vector field $\sum_i w^i \partial_i$ is via the inverse of the metric: $w^i = g^{ij} v_j$. Thus

$$\langle g^{ij} v_j \partial_i, u^k \partial_k \rangle_g = g^{ij} u^k v_j g_{ik} = \delta_{jk} v_j u^k = v_j u^j$$

and $\operatorname{grad} f = \sum_i g^{ij} \partial_j f \partial_i$. Finally

$$\Delta f = \operatorname{div}_g(\operatorname{grad} f) = \operatorname{div}_g \left(g^{ij} \partial_j f \partial_i \right) = \frac{1}{\sqrt{\det g}} \partial_i (g^{ij} X^i \sqrt{\det g} \partial_j)$$

□

Corollary A.32.1. Let (M, g) be a Riemannian manifold, $f \in C^\infty(M)$ and $p \in M$, then

$$\Delta f = \operatorname{Tr}_g(\operatorname{Hess}_p(f)) = \sum_{i,j=1}^n g^{ij} \left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \Gamma_{ij}^k \frac{\partial f}{\partial x^k} \right) \quad (\text{A.47})$$

Proof. From the previous equations, we have $\Delta = \operatorname{div}_g \circ \operatorname{grad} = \operatorname{Tr}_g \circ \nabla \circ \operatorname{grad} = \operatorname{Tr}_g \circ \nabla \circ \nabla$, and from the definition of the Hessian, we have $\operatorname{Hess}_p(f) = \nabla_{\partial x^i} \partial x^j f dx^i \otimes dx^j = \nabla_{\partial x^i} \nabla_{\partial x^j} f dx^i \otimes dx^j$. □

Definition A.33. Let (M, g) be a Riemannian manifold. A function $f \in C^\infty(M)$ is said to be *harmonic* if $\Delta f = 0$.

It is not hard to see that the geodesic equation Equation A.25 is equivalent Equation A.32. It will be shown in this thesis that this connection exists not only when considering the shortest path in a manifold, but when considering minimal sub-manifolds of a Riemannian manifold.

Theorem A.34 (Green's identities). *If (M, g) is a compact Riemannian manifold with or without boundary, then*

$$\int_M f \Delta h \, d\mu_g = \int_{\partial M} f \eta(h) \, d\mu_{\hat{g}} - \int_M \langle \text{grad } f, \text{grad } h \rangle_g \, d\mu_g \quad (\text{A.48})$$

$$\int_M (f \Delta h - h \Delta f) \, d\mu_g = \int_{\partial M} (h \eta(f) - f \eta(h)) \, d\mu_{\hat{g}}. \quad (\text{A.49})$$

for $f, g \in C^\infty(M)$, N the outward unit normal to ∂M and $\hat{g}$ the induced metric on the boundary.

Proof. By substituting $\text{grad } h$ for X in Equation A.39, we have

$$\begin{aligned} \int_M f \Delta h \, d\mu_g + \int_M \langle \text{grad } f, \text{grad } h \rangle_g \, d\mu_g &= \int_{\partial M} f \langle \text{grad } h, \eta \rangle_{\hat{g}} \, d\mu_{\hat{g}} \\ \iff \int_M f \Delta h \, d\mu_g + \int_M \langle \text{grad } f, \text{grad } h \rangle_g \, d\mu_g &= \int_{\partial M} f \eta(h) \, d\mu_{\hat{g}}. \end{aligned}$$

Thus, we have the first identity, which is used to prove the second identity

$$\begin{aligned} \int_M (f \Delta h - h \Delta f) \, d\mu_g &= \int_M f \Delta h \, d\mu_g - \int_M h \Delta f \, d\mu_g \\ &= \int_{\partial M} f \eta(h) \, d\mu_{\hat{g}} - \int_M \langle \text{grad } f, \text{grad } h \rangle_g \, d\mu_g \\ &= \int_{\partial M} h \eta(f) \, d\mu_{\hat{g}} + \int_M \langle \text{grad } f, \text{grad } h \rangle_g \, d\mu_g \\ &= \int_{\partial M} f \eta(h) \, d\mu_{\hat{g}} - \int_{\partial M} h \eta(f) \, d\mu_{\hat{g}} \\ &= \int_{\partial M} (h \eta(f) - f \eta(h)) \, d\mu_{\hat{g}}. \end{aligned}$$

□

Corollary A.34.1. *If M is a compact connected manifold without boundary, the only harmonic functions on M are the constant function, and for all $f \in C^\infty(M)$ $\int_M \Delta f \, d\mu_g = 0$.*

Proof. By Stokes's theorem, for any compactly supported $(n-1)$ -form on M we have $\int_M d\eta = 0$. Furthermore, observe that for arbitrary $f, h \in C^\infty(M)$, we have

$$\begin{aligned} \Delta(fh) &= \text{div}_g(\text{grad}(fh)) = \text{div}_g(f \text{grad } h + h \text{grad } f) \\ &= f \text{div}_g(\text{grad } h) + h \text{div}_g(\text{grad } f) + 2 \langle \text{grad } f, \text{grad } h \rangle. \end{aligned}$$

Thus

$$0 = \int_M \Delta(fh) \, d\mu_g = \int_M f \Delta h \, d\mu_g + \int_M h \Delta f \, d\mu_g + \int_M \langle \text{grad } f, \text{grad } h \rangle_g \, d\mu_g.$$

Thus if $h = \frac{1}{2}f$ and f is harmonic, then

$$0 = \int_M \|\text{grad } f\|_g^2 \, d\mu_g.$$

Hence $\text{grad } f = 0$ so f is constant. □

Corollary A.34.2. *If M is a compact connected manifold with boundary and f, h are harmonic functions on M whose restrictions to ∂M agree, then $f = h$.*

Proof. By Equation A.49, the following holds

$$\begin{aligned} 0 &= \int_M (f\Delta h - h\Delta f) d\mu_g = \int_{\partial M} (h\eta(f) - f\eta(h)) d\mu_{\hat{g}} \\ &= \int_{\partial M} h\eta(f) - f\eta(f) d\mu_{\hat{g}} = \int_{\partial M} (h - f)\eta(f) d\mu_{\hat{g}}. \end{aligned}$$

Thus $h - f = 0$ and $h = f$. $\square$

Theorem A.35 (Dirichlet's Principle). *Let (M, g) be a compact connected Riemannian n -manifold with boundary. Then $f \in C^\infty(M)$ is harmonic if and only if it minimizes*

$$\int_M \|\text{grad } f\|_g^2 d\mu_g$$

among all smooth functions with the same boundary values.

Proof. ‘ $\Rightarrow$ ’ Let f be a harmonic function of M and consider $h \in C^\infty(M)$ such that h vanishes on the boundary. Then, by Equation A.48, we have

$$0 = \int_M h\Delta f d\mu_g = - \int_M \langle \text{grad } f, \text{grad } h \rangle_g d\mu_g.$$

Thus,

$$\begin{aligned} \int_M \|\text{grad}(f + h)\|_g^2 d\mu_g &= \int_M \|\text{grad } f\|_g^2 + \int_M \|\text{grad } h\|_g^2 d\mu_g \\ &\quad + 2 \int_M \langle \text{grad } f, \text{grad } h \rangle_g d\mu_g \\ &= \int_M \|\text{grad } f\|_g^2 d\mu_g + \int_M \|\text{grad } h\|_g^2 d\mu_g \\ &\geq \int_M \|\text{grad } f\|_g^2 d\mu_g. \end{aligned}$$

where the last line is because $\int_M \|\text{grad } h\|_g^2 d\mu_g \geq 0$. Hence, f minimizes $\int_M \|\text{grad } f\|_g^2 d\mu_g$.

‘ $\Leftarrow$ ’ If f is the minimizer of $\int_M \|\text{grad } f\|_g^2 d\mu_g$ then for all $h \in C^\infty(M)$ and $\epsilon > 0$,

$$0 = \lim_{\epsilon \rightarrow 0} \frac{\int_M \|\text{grad}(f + \epsilon h)\|_g^2 d\mu_g - \int_M \|\text{grad } f\|_g^2 d\mu_g}{\epsilon}.$$

Since

$$\begin{aligned} \int_M \|\text{grad}(f + \epsilon h)\|_g^2 d\mu_g &= \\ &= \int_M \|\text{grad } f\|_g^2 + \epsilon^2 \int_M \|\text{grad } h\|_g^2 d\mu_g + 2\epsilon \int_M \langle \text{grad } f, \text{grad } h \rangle_g d\mu_g, \end{aligned}$$

we have

$$\begin{aligned} 0 &= \lim_{\epsilon \rightarrow 0} \frac{\int_M \|\text{grad}(f + \epsilon h)\|_g^2 d\mu_g - \int_M \|\text{grad } f\|_g^2 d\mu_g}{\epsilon} \\ &= 2 \int_M \langle \text{grad } f, \text{grad } h \rangle_g d\mu_g = -2 \int_M h\Delta f d\mu_g \end{aligned}$$

so $\Delta f = 0$. $\square$

B. Hopf-Rinow Theorem

This chapter aims to show that metric and geodesic completeness on a manifold are equivalent.

Definition B.1. A subset of $U \subseteq M$ is said to be *geodesically convex* if for each $p, q \in U$, there is a unique minimizing geodesic segment from p to q in M whose image lies entirely in U .

Definition B.2. A Riemannian manifold is said to be *geodesically complete* if every maximal geodesic is defined for all $t \in \mathbb{R}$.

A connected Riemannian manifold is said to be *metrically complete* if it is complete as a metric space.

Lemma B.3. Consider a connected Riemannian manifold (M, g) and a point $p \in M$ such that $\exp_p$ is defined on the whole tangent space $T_p M$.

- (a) Given any other q in M , there is a minimizing geodesic segment from p to q
- (b) M is metrically complete.

Theorem B.4 (Hopf-Rinow). A connected Riemannian manifold is metrically complete if and only if it is geodesically complete

Proof. Let (M, g) be a connected Riemannian manifold

‘ $\Leftarrow$ ’ By the previous lemma, we see that if it is geodesically complete, it is then metrically complete.

‘ $\Rightarrow$ ’ Suppose M is metrically complete, and it is not geodesically complete. Then there is a unit-speed geodesic $\gamma : [0, b] \rightarrow M$ that has no extension. Let (t_i) be any increasing sequence in $[0, b)$ that approaches b and set $q_i = \gamma(t_i)$. Since γ is parametrized by arc-length, $L_g(\gamma|_{[t_i, t_j]}) = |t_j - t_i|$ do $d_g(q_i, q_j) \leq |t_j - t_i|$ and $\{q_i\}$ is a Cauchy sequence in M . By completeness, $\{q_i\}$ converges to some point $q \in M$. Let W be a uniformly δ -normal neighborhood of q for some $\delta > 0$. Choose j large enough that $t_j > b - \delta$ and $q_j \in W$. Remark that in $B_\delta(q_j)$ is a geodesic ball, every unit-speed geodesic starting at q_j is defined at least on $[0, \delta)$. Consider, a geodesic η such that $\eta(0) = q_j$ and $\eta'(0) = \gamma'(t_j)$, then it is in $B_\delta(q_j)$. Define $\tilde{\gamma} : [0, t_j + \delta) \rightarrow M$ by

$$\tilde{\gamma}(t) = \begin{cases} \gamma(t) & t \in [0, b) \\ \eta(t - t_j) & t \in (t_j - \delta, t_j + \delta) \end{cases}$$

Since both components are geodesics and they have the same initial position and velocity at t_j , by the uniqueness of geodesics, the two agree where they overlap. Since $t_j + \delta > b$, $\tilde{\gamma}$ is an extension of γ . Thus, we reach a contradiction.

□

C. Kähler Manifolds

This chapter references *Lectures on Kähler Geometry*[23].

Let (M, g, J) be a Holomorphic manifold, we define the Kähler form to be

$$\omega(X, Y) = \langle JX, Y \rangle_g \quad (\text{C.1})$$

for $X, Y \in TM$.

Definition C.1 (Kähler manifold (Part 1)). A holomorphic manifold (M, g, J) is called a Kähler manifold if the Kähler form ω is closed.

For M a holomorphic manifold with compatible Riemannian metric g and Kähler form ω , $h = g + i\omega$ defines a Hermitian metric on $T^{\mathbb{C}}M$. Conversely, given a Hermitian metric h on $T^{\mathbb{C}}M$, $\Re h$ defines a compatible Riemannian metric for M and $\Im h$ defines the associated Kähler form.

From this, we can define the Kähler form from associated to a hermitian metric.

Definition C.2. Let (M, J) be a holomorphic manifold with local coordinates $z = (z^1, \dots, z^n)$ with hermitian metric $h = h_{j\bar{k}}(z)dz^j \otimes d\bar{z}^k$. We refer to the 2-form related to the hermitian metric $\omega = \frac{i}{2}h_{j\bar{k}}(z)dz^j \wedge d\bar{z}^k$ as the *Kähler form*.

Definition C.3 (Kähler manifold (Part 2)). A hermitian metric $h = h_{j\bar{k}}(z)dz^j \otimes d\bar{z}^k$ is called *Kähler* if for all $z \in M$ there is a neighborhood U of z and a function $\Phi : U \rightarrow \mathbb{R}$ such that

$$\frac{i}{2}h_{j\bar{k}}(z)dz^j \wedge d\bar{z}^k = \partial\bar{\partial}\Phi,$$

in which case M is a Kähler manifold.

Theorem C.4. Any Riemann Surface, a complex 1 dimensional manifold, M is a Kähler manifold.

Proof. Since $\dim_{\mathbb{R}}(M) = 2$, we have that $d\omega = 0$, for ω the Kähler form of the hermitian metric. $\square$

As one can extend the Riemannian metric g of M to $T^{\mathbb{C}}M$, one can also extend the Levi-Civita connection to $T^{\mathbb{C}}M$: for X, Y, Z, W real vector fields, we have the following complex vector fields $U = X + iY$ and $V = Z + iW$ and

$$\nabla_U v = \nabla_{X+iY}(Z + iW) = \nabla_X Z - \nabla_Y W + i(\nabla_X W + \nabla_Y Z). \quad (\text{C.2})$$

Theorem C.5. *Let (M, g, J) be a Kähler manifold, then the complex structure J is parallel to ∇ , the Levi-Civita connection associated to g : $\nabla J = 0$, in more details for $X, Y \in TM$*

$$(\nabla_X J)Y = \nabla_X(JY) - J(\nabla_X Y) = 0 \quad (\text{C.3})$$

Proof. Let ω be the Kähler form of M and ∇ the Levi civita connection.

- We begin by showing that integrability of the complex structure implies $J(\nabla_X J)Y = (\nabla_{JX} J)Y$, for arbitrary vector fields.

$$\begin{aligned} N_J(X, Y) &= [X, Y] + J([JX, Y] + [X, JY]) - [JX, JY] \\ &= \nabla_X Y - \nabla_Y X + J\nabla_{JX} Y - J\nabla_Y JX \\ &\quad + J\nabla_X JY - J\nabla_{JY} X - \nabla_{JX} JY + \nabla_{JY} JX \\ &= J((\nabla_X J)Y) - J((\nabla_Y J)X) - (\nabla_{JX} J)Y + (\nabla_{JY} J)X \\ &= (J((\nabla_X J)Y) - (\nabla_{JX} J)Y) - (J((\nabla_Y J)X) - (\nabla_{JY} J)X) \end{aligned}$$

as $J((\nabla_U J)V) = J(\nabla_U(JV) - J\nabla_U V) = J\nabla_U(JV) + \nabla_U V$ and $(\nabla_{JU} J)V = \nabla_{JU}(JV) - J(\nabla_{JX} Y)$ vice versa. If we consider the tensor

$$A(X, Y, Z) = \langle J((\nabla_X J)Y) - (\nabla_{JX} J)Y, Z \rangle_g,$$

then we see that as $N_J = 0$, we have that $A(X, Y, Z) = A(Y, X, Z)$. Furthermore, as J and ∇J are anti-commutative skew-symmetric operators, $A(X, Y, Z) = -A(X, Y, Z)$, we thus have that $A(X, Y, Z) = -A(X, Y, Z)$ so $J((\nabla_X J)Y) - (\nabla_{JX} J)Y = 0$.

- Without loss of generality we can consider commuting vector fields X, Y, Z, JY, JZ , such that $d\omega(X, Y, Z) = X\omega(Y, Z) - Y\omega(X, Z) + Z\omega(X, Y)$.

By Koszul formula and the fact that g and ω are compatible, we have

$$\begin{aligned} 2\langle J(\nabla_X Y), Z \rangle_g &= -2\langle \nabla_X Y, JZ \rangle_g = -X\langle Y, JZ \rangle_g - Y\langle X, JZ \rangle_g + JZ\langle X, Y \rangle_g \\ &= -X\omega(JY, JZ) - Y\omega(Z, X) + JZ\omega(X, JY) \end{aligned}$$

and

$$\begin{aligned} 2\langle \nabla_X(JY), Z \rangle_g &= X\langle JY, Z \rangle_g + JY\langle X, Z \rangle_g - Z\langle X, JY \rangle_g \\ &= X\omega(Y, Z) - JY\omega(JZ, X) + Z\omega(X, Y) \end{aligned}$$

thus

$$\begin{aligned} 2\langle (\nabla_X J)Y, Z \rangle_g &= 2\langle \nabla_X(JY), Z \rangle_g + 2\langle J(\nabla_X Y), Z \rangle_g \\ &= X\omega(Y, Z) - Y\omega(Z, X) + Z\omega(X, Y) \\ &\quad - X\omega(JY, JZ) + JZ\omega(X, JY) - JY\omega(JZ, X) \\ &= d\omega(X, Y, Z) - d\omega(X, JZ, JY) = 0, \end{aligned}$$

and we have shown that $\nabla J = 0$.

□

Theorem C.6. *Let (M, g, J) be a Kähler manifold, then $\langle R(JX, JY, U)V \rangle_g = \langle R(X, Y, U)V \rangle_g$ for $X, Y, U, V \in T_{\mathbb{C}}M$.*

Proof. First, observe that since M is Kähler manifold $\nabla J = 0$, hence

$$\begin{aligned} R(X, Y, JZ) &= \nabla_X \nabla_Y JZ - \nabla_Y \nabla_X JZ - \nabla_{[X, Y]} JZ \\ &= J \nabla_X \nabla_Y Z - J \nabla_Y \nabla_X Z - J \nabla_{[X, Y]} Z = JR(X, Y, Z). \end{aligned}$$

Then from Properties of the Curvature Tensor

$$\begin{aligned} \langle R(JX, JY, U)V \rangle_g &= -\langle R(U, V, JX)JY \rangle_g = -\langle JR(U, V, X)JY \rangle_g \\ &= -\langle R(U, V, X)Y \rangle_g = \langle R(X, Y, U)V \rangle_g \end{aligned}$$

□

Corollary C.6.1. *Let (M, g, J) be a Kähler manifold, then $\langle R(X, Y, U)V \rangle_g = 0$ for $U, V \in T^{(1,0)}M$ or $U, V \in T^{(0,1)}M$ and it non-vanishing only on $(X, \bar{Y}, U, \bar{V}) \in T^{(1,0)}M \times T^{(1,0)}M \times T^{(1,0)}M \times T^{(1,0)}M$.*

Part V.

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